Techno-Economic Viability Study of Acquisition of LSAW and HSAW Pipe Manufacturing Plant at Dammam, Saudi Arabia

M/s Man Industries (India) Limited

Secant Consultants Private Limited

January 20, 2026

Certificate of Issuance

To whom so ever it may concern, this certificate is being issued on behalf of Secant Consultants Private Limited for Project titled "Techno-Economic Viability Study of Acquisition of LSAW and HSAW Pipe Manufacturing Plant at Dammam, Saudi Arabia" of M/s Man Industries (India) Limited. The record of issuance of reports at various stages, along with the date of issue, team of Consultants from Secant Consultants and official approving the report has been listed in table.

Date	Originators	Checker	Approver	Report Status
2nd December 2025	Vipul Sonkar	Amit Pandey	Vatsal Misra	Draft Report
10th December 2025	Vipul Sonkar	Amit Pandey	Vatsal Misra	Draft Final Report
20th January 2026	Vipul Sonkar	Amit Pandey	Vatsal Misra	Final Report

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For and On Behalf of
Secant Consultants Private Limited

Vatsal Misra

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Date: 2026.01.20 15:23:52 +05'30'

Vatsal Misra
Project Director

Abbreviations Used

Abbreviation	Full form
APAC	Asia Pacific
API	American Petroleum Institute
ARAMCO	Arabian American Oil Company
ASTM	American Society for Testing and Materials
Avg.	Average
BCB	Central Bank of Brazil
BEP	Break Even Point
BOO	Build-Own-Operate
BOOT	Build-Own-Operate-Transfer
CAGR	Compounded Annual Growth Rate
CHS	Chung Hung Steel Corporation
CIN	Corporate Identification Number
CMA	Credit Monitoring Arrangement
CPI	Consumer Price Index
CSC	China Steel Corporation
CY	Calendar Year
DER	Debt-Equity Ration
DI	Ductile Cast Iron
DIN	Director Identification Number
DPR	Detailed Project Reports
DSAW	Double Submerged Arc Welding
DSCR	Debt-Service Coverage Ratio
EBITDA	Earnings Before Interest, Taxes, Depreciation and Amortization
ECB	European Central Bank
EN	Europaische Norm (European standard)
ERW	Electric Resistance Welding
EU	European Union
FACR	Fixed Assets Coverage Ratio
FBE	Fusion-Bonded Epoxy
FCAW	Flux-Cored Arc Welding
FDI	Foreign Direct Investment
FG	Finished Goods
FW	Furnace Welding
FY	Financial Year
GCC	Gulf Cooperation Council
GDP	Gross Domestic Product
GEM	Global Energy Monitor
GMAW	Gas Metal-Arc Welding
GTAW	Gas Tungsten-Arc Welding
HIC	Hydrogen Induced Cracking
HR	Hot Rolled
HRC	Hot Rolled Coils
HSAW/ SSAW	Helical/Spiral submerged arc welded

Abbreviation	Full form
IMF	International Monitory Funds
IP	Industrial Production
IRR	Internal Rate of Return
ISO	International Organization for Standardization
ISTP	Independent Sewage Treatment Plant
KSA	Kingdom of Saudi Arabia
KV	Kilo Volt
KVA	Kilo Volt Ampere
LLC	Limited Liability Company
LNG	Liquefied Natural Gas
LPG	Liquefied Petroleum Gas
LSAW	Longitudinal Submerged Arc Welding
MA	Masters in Art
MBA	Masters in Business Administration
MEA	Middle East & Africa
MENA	Middle East & North Africa
MIIL	Man Industries India Limited
NACE	National Association of Corrosion Engineers
NDT	Non-Destructive Testing
NGL	Natural Gas Liquids
NISCO	Nanjing Iron and Steel Company
NPC	National Pipe Company
NPS	Nominal Pipe Size
NPV	Net Present Value
OCTG	Oil Country Tubular Goods
OPEC	Organization of the Petroleum Exporting Countries
PAT	Profit After Tax
PCE	Personal Consumption Expenditures
PE	Poly Ethylene
PG	Post Graduation
PGDBA	Post Graduation Diploma In Business Analytics
PMI	Purchasing Managers' Index
PPP	Public-Private Partnership
RO	Reverse Osmosis
SAAR	Seasonally Adjusted Annual Rate
SAGIA	Saudi Arabian General Investment Authority
SAR	Saudi Arabian Riyal
SAW	Submerged Arc Welding
SCECO/ SECO	Saudi Electric Company
SCPL	Secant Consultants Private Limited
SMAW	Shielded Metal-Arc Welding
SS	Stainless Steel
SSC	Sulphide Stress Cracking
SSTP	Small Sewage Treatment Plant
SWCC	Saline Water Conversion Corporation

Abbreviation	Full form
SWOT	Strength, Weakness, Opportunity and Threats
SWPC	Saudi Water Partnership Company
TISCO	Tata Iron and Steel Company
TNW	Tangible Net Worth
TOL	Total Outside Liability
TPA	Ton per Annum
UAE	United Arab Emirates
UK	United Kingdom
US/ USA	United State/ United State America
USD/ $	United States Dollar
UTS	Ultimate Tensile Strength
VP	Vice President
WDV	Written Down Value
WIP	Work In Process

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Executive Summary

Study Background

M/s Man Industries (India) Limited ("MIIL" or "Company") is a public limited company, incorporated on 19th May 1988 under the name of Man Aluminium Limited. Later, on 21st June 1995, the name was changed to its present name. The Company is head quartered in the financial capital of India i.e. Mumbai, with corporate identification number as L99999MH1988PLC047408. MIIL since its inception has been involved in manufacturing of Helical Submerged Arc Welded (HSAW or H-SAW) pipes and Longitudinal Submerged Arc Welded (LSAW or L-SAW) pipes and has manufacturing facilities located at Pithampur in Madhya Pradesh and Anjar in Gujarat. MIIL is considered amongst the pioneers is SAW Pipe segments in India and has over the years established itself as one of the leading players in the segment in India and Globally.

The current installed capacity of the Company includes –

1. 600,000 Tons of LSAW Pipes
2. 400,000 Tons of HSAW Pipes
3. Coating plant with installed capacity of 4 million Sq. M.
4. ERW Pipes plant with installed capacity of 100,000 Tons
5. 7 MW of Windmills capacity, for captive purpose at Gujarat

The Company has been observing an increase in demand for HSAW and LSAW pipes in the Kingdom of Saudi Arabia ('KSA'), due to rising demand from Oil & Gas and Infrastructure sector in KSA region. Further it is noted that with the change in the ruling authorities of KSA, there has been a focus on what is called as "Saudisation/ Saudization" in Government procurements, there by pushing for products expected to be procured by Government/ Government Departments/ Government Companies to be manufactured locally in KSA. In line with same vision, Aramco/Saudi Aramco, a state-owned petroleum and natural gas company that is the national oil company of Saudi Arabia, major oil and gas producer and supplier in KSA region, has been placing orders for the HSAW and LSAW pipes with local manufacturers. Similarly, the Government Departments of KSA like Saline Water Conversion Corporation ('SWCC') too has been following Saudisation and are placing orders for pipes requirement with local vendors/manufacturers.

Taking into consideration the announced pipeline (including replacement pipeline) by various Government Departments/ Companies, it is noted that the markets of KSA will remain buoyant during the next decade or so for the HSAW and LSAW pipe manufacturers.

Further, it is noted that most of the countries amongst Gulf Cooperation Council ('GCC') region and Middle East and North Africa ('MENA') region have been working towards reducing the dependence on oil and gas to power own economies and the region has been experiencing announcement of series of infrastructure projects including projects for oil and gas, water transport etc.

Considering these opportunities, MIIL has been evaluating various options to enter the KSA markets and through KSA, the GCC and MENA markets as well. The Company had been in advance stages of discussions with local authorities to set-up a greenfield HSAW pipe and coating plant facility in KSA and had registered a KSA entity by name of Man International Steel Industries Company ('MISICL') for the Project.

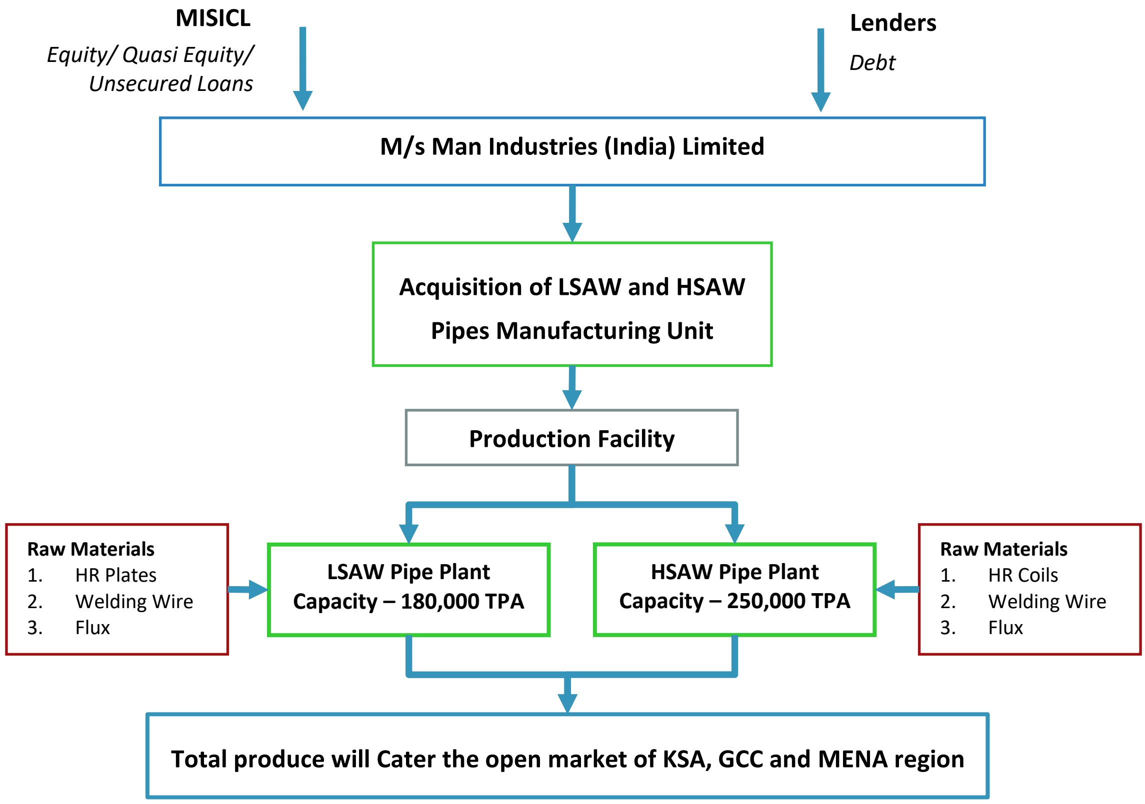
During the same period, an opportunity to acquire an existing local SAW Pipes manufacturing unit came to fore. The local entity is a well-established player by name of National Pipe Company Limited ('NPC'), which is located at Dhahran, near the city of Dammam. NPC was established in 1978 and the unit is promoted by Japanese Promoters (51.9%) including Nippon Steel and Sumitomo Corporation and Arab Promoters (48.1%) including Rezayat Group and other individuals/ companies. As per the confirmation provided by the Management/ Executives of NPC, the present installed capacity of NPC stands at –

- 250,000 Tons of HSAW Pipe facility based on 2 step technology.
- 180,000 Tons of LSAW Pipe facility based on 3 RB technology.

MIIL has undertaken technical, commercial, legal and financial due diligence of the assets of NPC and based on the same, MIIL is considering the acquisition of NPC. In this regard, MIIL has approached State Bank of India, Oversea Branch, Cuffe Parade, Mumbai ('SBI' or 'Bank' or 'Lender') to avail term loan facility for the proposed acquisition of NPC. The Bank in turn has appointed Secant Consultants Private Limited to undertake a Techno-Economic Viability Study of the proposed acquisition of NPC.

Project Configuration

The project configuration as envisaged by MIIL is presented in the exhibit below –



Source: MIIL and SCPL Estimates

Acquisition Cost

The total cost of acquisition has been finalised at SAR 382,500,000 or USD 102.00 Million plus USD 3.00 Million as capex towards HSAW Pipe mill upgrade, with total project cost being USD 105.00 Million and will be funded by promoters' equity of USD 26.25 Million (Rs. 234.68 Crores) and Long-Term Debt of USD 78.75 Million (Rs. 704.04 Crores) and the same has been exhibited in the table below –

All Figures in USD Million*	
Description	**Total**
Promoters Equity	26.25
Term Loan from Lenders	78.75
Total Means of Finance	**105.00**

Source: MIIL and SCPL Estimates

*** - Exchange rate considered is 1 USD = Rs. 89.4013 (exchange rate as on 29thj November 2025)**

Here it should be noted that the cost of acquisition of NPC has been considered at USD 102.00 Million and additional USD 3.00 Million has been considered for upgrade of HSAW Pipe mill, which has current capability to produce pipes of dimeter upto 84 inch. This is essential, as the newer orders from local customers in recent times indicate that the market trend for diameters upto 100 inches to 120 inches is gaining popularity. The upgrade project is expected to be completed within 12 months of acquisition of NPC by Man Group. This is the reason for considering the overall means of finance for USD 105.00 Million.

Promoters Equity

Based on the discussions with the Management of MIIL, it is understood that the Promoters of MIIL will be infusing funds to the tune of Rs. 26.25 Crores from own sources of funds and the balance of the equity (Rs. 209.43 Crores) will be raised from proceeds of sales of land own by the Company near Vashi in Navi Mumbai.

Term Loan

The Consultants understand that the Lenders will be issuing Stand-By Letter of Credit ('SBLC') of USD 78.75 Mn (Rs. 704.04 Crores), which will be utilised by the Company to fund the acquisition. The broad covenants of the term loan have been presented in the table below –

Parameter	Details
Term Loan Amount	USD 78.75 Million
Interest Rate	8.25% Per annum (SOFR at 4.05% + Hedging and Risk Spread of 4.20%)
Door to Door Tenure	7 Years
First Disbursement	January 2026
Date of Acquisition	1st January 2026
Post Acquisition Moratorium	6 Months
Commercial Operations Date	1st January 2026
Repayment Duration	26 Quarters
Installments	26 Equal Quarterly Instalments
First Instalment	3rd Quarter of CY 2026
Last Installment	4th Quarter of CY 2032

Source: SCPL Assumptions

Key Financial Parameters

Note – The proposal is to purchase the equity shares of NPC through MISICL (Man International Steel Industries Company), which will act as the holding company in the KSA. Hence the term loan will not reflect in the books of NPC post-acquisition, albeit the same will reflect in the books of the Holding Company. Hence, Consultants have considered dividend equivalent to the Interest + Principal amount issued by NPC, so as to pay the debt obligation of the Holding Company. The projected financial statements have been prepared considering the same, however the key financial parameters/ ratios have been estimated with the view the debt is on books of the Company, so as to provide the actual debt repayment capacity of the Project.

The key financial parameters of the project have been presented in the table below –

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Revenue	219.27	249.83	268.84	287.84	301.67	312.69	323.41
Total Operating Costs	179.87	206.21	221.90	236.47	246.92	255.21	263.27
EBDITA	39.40	43.63	46.94	51.38	54.75	57.48	60.14
EBDITA Margin	17.97%	17.46%	17.46%	17.85%	18.15%	18.38%	18.60%
Contribution	46.53	51.75	55.06	59.50	62.87	65.60	68.26
Contribution Margin	21.22%	20.71%	20.48%	20.67%	20.84%	20.98%	21.11%
BEP Sales	77.78	84.86	72.27	71.61	71.02	70.55	70.13
BEP Capacity Utilisation	35.47%	33.97%	26.88%	24.88%	23.54%	22.56%	21.68%
Cash Break Even	37.92	44.03	44.53	44.12	43.76	43.47	43.21
Cash Break Even Margin	17.29%	17.62%	16.56%	15.33%	14.50%	13.90%	13.36%
Net Profit	30.03	34.17	33.31	35.76	38.46	40.64	42.77
Net Profit Margin	13.69%	13.68%	12.39%	12.42%	12.75%	13.00%	13.22%
Equity Share Capital	53.27	53.27	53.27	53.27	53.27	53.27	53.27
Statutory Reserves	26.64	26.64	26.64	26.64	26.64	26.64	26.64
Reserves and Surplus	75.22	91.41	107.79	127.67	151.31	178.17	208.21
Tangible Net Worth (TNW)	155.12	171.32	187.70	207.58	231.21	258.07	288.11
Term Loan	72.69	60.58	48.46	36.35	24.23	12.12	-
Debt Equity Ratio	0.47	0.35	0.26	0.18	0.10	0.05	-
Total Outside Liability (TOL)	7.11	14.83	15.53	16.15	16.56	16.90	17.24
TOL/ TNW	0.05	0.09	0.08	0.08	0.07	0.07	0.06
Closing Cash Balance	31.13	45.69	69.91	85.59	100.32	125.02	149.24
DSCR	3.71	2.80	2.69	2.97	3.32	3.68	3.99
Minimum DSCR	2.69						
Average DSCR	3.25						
NPV	121.88						
IRR	49.99%						
Cost of Capital	9.19%						

Source: SCPL Analysis

Sensitivity Analysis

The Consultants had undertaken sensitivity analysis to assess the impact of various scenarios on the financial parameters and the result of the same is presented in the table below –

Description	**NPV**	**IRR**	**WACC**	**Min. DSCR**	**Avg DSCR**
	USD Mn	**%**	**%**	**Ratio**	**Ratio**
Base Case	121.88	49.99%	9.19%	2.69	3.25

Description	NPV	IRR	WACC	Min. DSCR	Avg DSCR
	USD Mn	%	%	Ratio	Ratio
5% decrease in capacity utilisation	111.54	46.80%	9.21%	2.62	3.10
10% decrease in capacity utilisation	101.26	43.64%	9.23%	2.50	2.94
15% decrease in capacity utilisation	106.28	44.09%	9.22%	2.58	3.02
1.50% decrease in selling price	69.90	31.35%	9.32%	2.09	2.50
2.50% decrease in selling price	16.89	14.62%	9.62%	1.38	1.74
5% decrease in selling price	-46.68	-3.55%	10.19%	0.67	0.82
1.50% increase in raw material cost	81.90	35.14%	9.29%	2.25	2.67
2.50% increase in raw material cost	41.46	21.77%	9.45%	1.70	2.09
5% increase in raw material cost	-1.18	9.45%	9.78%	1.16	1.50
1% increase in interest rates	118.72	50.91%	9.82%	2.64	3.20
2% increase in interest rates	115.71	51.85%	10.45%	2.59	3.15

Source: SCPL Estimates

It is noted from the exhibit above, that the Project will be susceptible to 5% decrease in selling price and 5% increase in raw material cost. However, it is also noted that the Project remains viable under all other adverse scenarios as NPV remains positive and IRR, remains above WACC under other all scenarios.

The average DSCR of the Project is 3.25, while the minimum DSCR of the Project 2.69 indicating the Project has fair repayment capacity.

Conclusion

Based on the study undertaken by the Consultants, the following is concluded about the proposed acquisition of MIIL –

- The manufacturing unit of NPC is located on 40-hectares, freehold land owned by the Company at Dhahran at the outskirts of city of Dammam in KSA.
- As per the confirmation provided by the Management/ Executives of NPC, the present installed capacity of NPC stands at –
  - 250,000 Tons of HSAW Pipe facility based on 2 step technology.
  - 180,000 Tons of LSAW Pipe facility based on 3 RB technology.
- The plant and machinery have been supplied and installed by renowned technology provider of the segment like (Nippon Steel Welding & Engineering, Sumitomo Heavy Industries, SMS Germany (PWS), Mitsubishi Heavy Industries, and Haeusler AG, Etc.)
- The Raw material will be procured from the open market from renowned supplier like (Nippon Steel Company, POSCO (Korea), Japan Future Enterprise, and NISCO – China, Etc.)
- The power requirement will be fulfilled by a permanent incoming supply of 17.5 KV issued from Saudi Electricity Company (SECO).
- Water requirement is fulfilled through 2 Nos. of Borewell available within the premise of company. Also, there is a RO water plant within the premises of the NPC factory.
- The company is operation and has received requisite approvals and clearances.
- The overall acquisition cost has been estimated at SAR 382,500,000 or USD 102.00 Million. In addition, the Group after acquisition of NPC will immediately undertake upgrade of HSAW Pipe mill, with an expected cost of USD 3.00 Million. Hence the overall project cost is estimated USD 105.00 Million.

- The proposed project is proposed to be funded in a Debt-Equity ratio of 3:1. The Promoters Contribution for the Project will be USD 26.25 Million, while the loan will be USD 78.75 Million.
- The Net Present Value (NPV) of the Project has been estimated at USD 121.88 Million, while Internal Rate of Return at 49.99% is higher than Post Tax Cost of Capita at 9.19%, indicating the Project is financially viable.
- The average DSCR of the Project has been estimated at 3.25, indicating fair capability of the Project in repaying the long-term debts.
- The peak manpower required for the proposed project is ascertained at 567 personnel for peak capacity utilisation.
- Based on the market assessment, technological assessment, SWOT and risk analysis and the financial evaluation as undertaken by the Consultants, the proposal of acquisition of NPC by MIIL is found to be techno-economically viable preposition.

Introduction

Study Background

M/s Man Industries (India) Limited ("MIIL" or "Company") is a public limited company, incorporated on 19th May 1988 under the name of Man Aluminium Limited. Later, on 21st June 1995, the name was changed to its present name. The company was formed by Dr. Ramesh Chandra Mansukhani and Family with a vision to "Obtaining market leadership position by providing innovation and quality products & services to improve quality of life & environment in quest for excellence". And with a mission to "Partnering with every Major Global Client in oil, Gas and water sector by providing quality solutions in committed timeline, enhancing value for stakeholders". The company's Corporate Identification Number (CIN) is L99999MH1988PLC047408. The corporate office address of the company is Man House, 101, S.V. Road, Vile Parle (W), Mumbai, Maharashtra 400056.

Initially, MIIL was engaged in manufacturing of aluminium sections. Later in 1995, the Company diversified into manufacturing of Submerged Arc Welded (SAW) line pipes, thereby operating two businesses viz. manufacturing of SAW pipes and manufacturing of aluminium sections. Further, in April 1999, the Company diversified into manufacturing of spiral welded pipes. Subsequently, MIIL set up Poly Ethylene (PE)/ Coal Tar Enamel (CTE) coating plant in its then existing works at Pithampur (Madhya Pradesh). The coating plant became operational since June 2001. Line Pipe and Coating Complex in Anjar at Gujarat were commissioned in FY 2005.

At present, MIIL engaged in manufacturing and export of LSAW and HSAW pipes with a total installed capacity of 10,00,000 Tons. MIIL has two plants i.e. one plant in Anjar, Kutch District of Gujarat and other in Pithampur, Madhya Pradesh. Anjar plant facilitates easy transportation to two major ports viz, Kandla and Mundra as well as provides good connectivity to the road network. Both facilities put together spread across about 150 acres of land. MIIL's facilities hold internationally accepted quality standards laid down by the American Petroleum Institute (API) which is a mandatory requirement to produce high pressure line pipes for hydrocarbon applications. The Company's products are used for various high-pressure transmission applications for Gas, Crude Oil, Petrochemical Products and Potable Water.

The Company has been observing an increase in demand for HSAW and LSAW pipes in the Kingdom of Saudi Arabia, due to rising demand from Oil & Gas and Infrastructure sector in KSA region. Further it is noted that with the change in the ruling authorities of KSA (Kingdom of Saudi Arabia), there has been a focus on what is called as "Saudisation/ Saudization" in Government procurements, there by pushing for products expected to be procured by Government/ Government Departments/ Government Companies to be manufactured locally in KSA. In line with same vision, Aramco/Saudi Aramco, a state-owned petroleum and natural gas company that is the national oil company of Saudi Arabia, major oil and gas producer and supplier in KSA region, has been placing orders for the HSAW and LSAW pipes with local manufacturers. Similarly, the Government Departments of KSA like Saline Water Conversion Corporation ('SWCC') too has been following Saudisation and are placing orders for pipes requirement with local vendors/manufacturers.

Taking into consideration the announced pipeline (including replacement pipeline) by various Government Departments/ Companies, it is noted that the markets of KSA will remain buoyant during the next decade or so for the HSAW and LSAW pipe manufacturers.

Further, it is noted that most of the countries amongst GCC region and MENA region have been working towards reducing the dependence on oil and gas to power own economies and the region has been experiencing announcement of series of infrastructure projects including projects for oil and gas, water transport etc.

Considering these opportunities, MIIL has been evaluating various options to enter the KSA markets and through KSA, the GCC and MENA markets as well. The Company had been in advance stages of discussions with local authorities to set-up a greenfield HSAW pipe and coating plant facility in KSA and had registered a KSA entity by name of Man International Steel Industries Company for the Project.

During the same period, an opportunity to acquire an existing local SAW Pipes manufacturing unit came to fore. The local entity is a well-established player by name of National Pipe Company Limited ('NPC'), which is located at Dhahran, near the city of Dammam. NPC was established in 1978 and the unit is jointly promoted by Japanese Promoters (51.9%) and Arab Promoters (48.1%). As per the confirmation provided by the Management/ Executives of NPC, the present installed capacity of NPC stands at –

- 250,000 Tons of HSAW Pipe facility based on 2 step technology.
- 180,000 Tons of LSAW Pipe facility based on 3 RB technology.

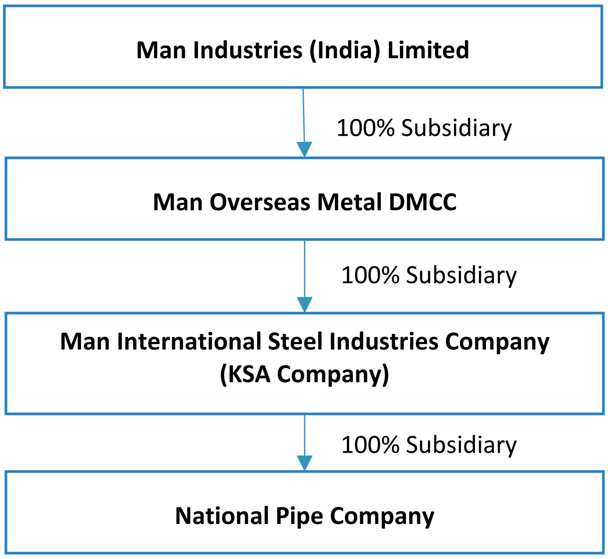
MIIL has undertaken technical, commercial, legal and financial due diligence of the assets of NPC and based on the same, is considering the acquisition of NPC. The company is now seeking term loan from the State Bank of India for the funding the acquisition of NPC. The bank in return has asked the company to undertake a techno-economic evaluation of the Project, by an empanelled Technical Consultancy Organisation. In this regard, SBI has approached and appointed Secant Consultants Private Limited for undertaking the Techno-Economic Viability Study of the said Project.

The key parameters of the Project have been summarised below –

- The combined installed capacity of the plant is 430,000 TPA, including LSAW pipe manufacturing capacity of 180,000 TPA and HSAW pipe manufacturing capacity of 250,000 TPA.
- The overall acquisition cost has been estimated at SAR 382,500,000 or USD 102.00 Million. In addition, the Group after acquisition of NPC will immediately undertake upgrade of HSAW Pipe mill, with an expected cost of USD 3.00 Million. Hence the overall project cost is estimated USD 105.00 Million.
- The proposed project is proposed to be funded in a Debt-Equity ratio of 3:1. The Promoters Contribution for the Project will be USD 26.25 Million, while the loan will be USD 78.75 Million.
- The Net Present Value (NPV) of the Project has been estimated at USD 121.88 Million, while Internal Rate of Return at 49.99% is higher than Post Tax Cost of Capita at 9.19%, indicating the Project is financially viable.
- The average DSCR of the Project has been estimated at 3.25, indicating fair capability of the Project in repaying the long-term debts.

Structure for Acquisition

Based on the discussions with Executives of the Company, the Consultants understand that the funds from MIIL will be acquiring NPC through its 100% subsidiary in UAE, by name of Man Overseas Metal DMCC ('MOMD'). The KSA based Man International Steel Industries Company LLC ('MISICL'), will be a 100% subsidiary of MOMD and MISICL will be holding 100% shares of NPC post-acquisition. The structure as proposed by MIIL has been presented in the exhibit below –



Source: MIIL

MIIL's Promoter Details

The brief about the Promoters of MIIL is provided in the exhibit below –

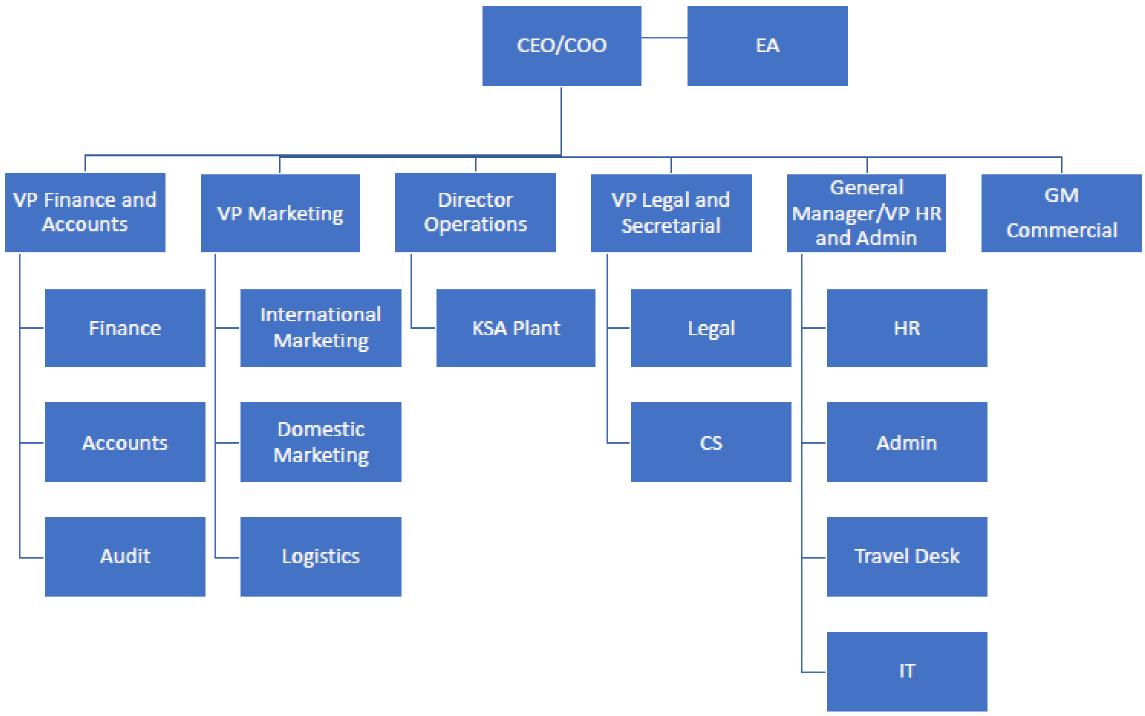
Name	Designation	Experience	Qualification
Dr. Ramesh Chandra Mansukhani	Promoter and Executive Chairman	Mr. Mansukhani is the First-Generation Entrepreneur and has over four decades of industrial experience. Also, has been instrumental in growth of the organization in domestic as well as international markets. Under his leadership, the Group has diversified in other business segments such as Real Estate, Infrastructure and Trading.	Ph.D. in International Economics and Finance by University in France, MA Economics {Gold Medallist from Vikram University, Ujjain (M.P.), Bachelor of Law
Mr. Nikhil Mansukhani	Promoter and Executive Director	Mr. Mansukhani is widely engaged in MIIL's day to day affairs and board of other group companies as well. Also, has an experience of over 14 years. He heads Group's real estate business particularly business development, land acquisition, designing & liasioning and	Graduate from King's College, UK, Bachelor of Engineering and Business

Name	Designation	Experience	Qualification
		also manages day to day affairs of group's realty companies.	
Mrs. Heena Vinay Kalantri	Non-Executive Director	Mrs. Kalantri has varied experience in the field of marketing, finance and specialization in Human Resource Management.	Postgraduate in Management from United Kingdom
Mr. Pramod Tondon	Independent Director	Mr. Tondon has industry experience of around 42 years. He has strong Knowledge on Company's businesses and policies; Business Strategy, Financial and Management Skills, Stakeholder Relationship.	Has Master's degree in Science.

Source: MIIL

Organisation Structure of MIIL

The organogram of the proposed company as shared by the management of the company is presented below –



Key Management of MIIL

Based on the information received from the executive of the company, the key management are as follows:

Name	Designation	Experience and qualification
Mr. Sandeep Kumar Garg	Chief Financial Officer	Mr. Garg is completed his LLB and is Chartered Accountant by profession and has 34 years of rich experience in the industry.
Mr. Rahul Rawat	Company Secretary	Mr. Rawat is Company Secretary of the Group and has completed is LLB and MBA as well. He has over 13 years of experience in the industry.
Mr. Jaspreet Singh Bhatia	Sr. Vice President – Operations	Mr. Bhatia is an Automobile Engineer and has 25 years of rich experience in the industry.
Mr. Hardik Desai	Jt. Vice President - Technical	Mr. Desai has completed his PG Diploma (Mechanical Engineering) and subsequently B. Sc. He has over 24 years of industry experience.
Mr. Gurinder Singh Sethi	Sr. Vice President – Marketing & Business Development	Mr. Sethi is an Electrical Engineer and has completed in MBA in marketing. He has over 32 years of experience in marketing of industrial products.
Mr. Hardik Shah	VP – Projects	Mr. Shah has bachelor's degree in mechanical Engineering followed by PG Operation and PG machine Line & AI and has 24 years of experience in the industry.
Mr. Rahul Sanghvi	Assistant Vice president – Finance	Mr. Sanghvi has bachelor's degree in commerce followed by MBA Finance and has 23 years of experience in the industry

Source: MIIL

About Target Company

National Pipe Company Limited (NPC)

Established in 1978, National Pipe Company Limited (NPC), a joint venture between Rezayat Group, Nippon Steel Sumitomo Metal Corporation and Sumitomo Corporation, is a leading manufacturer of large diameter steel pipes.

Located in Al Khobar, Saudi Arabia and established in 1978, NPC manufactures steel pipes for various applications such as oil, natural gas and water transmission, as well as for structural use.

NPC has 430,000 metric tons of annual pipe production capacity: 250,000 metric tons of helical (also known as spiral) Submerged Arc Welded (SAW) pipe and another 180,000 metric tons of longitudinal (also known as Straight Seam Submerged Arc Welded) pipe of various sizes.

The helical pipe production facility started commercial production in December 1980, producing and supplying high quality pipes to Saudi Aramco, the Saline Water Conversion Corporation and many others in Saudi Arabia and the GCC.

The longitudinal pipe mill started commercial production in September 2001 and mainly aims at meeting growing demand for heavier wall thickness pipes for natural gas transmission lines, associated with the development of non-associated gas fields and the construction of gas refinery plants in Saudi Arabia and other GCC countries.

NPC has technical and management support from Sumitomo Metal Industries Ltd of Japan, a major shareholder in the company and a leading manufacturer of steel and steel products.

Key Management of NPC

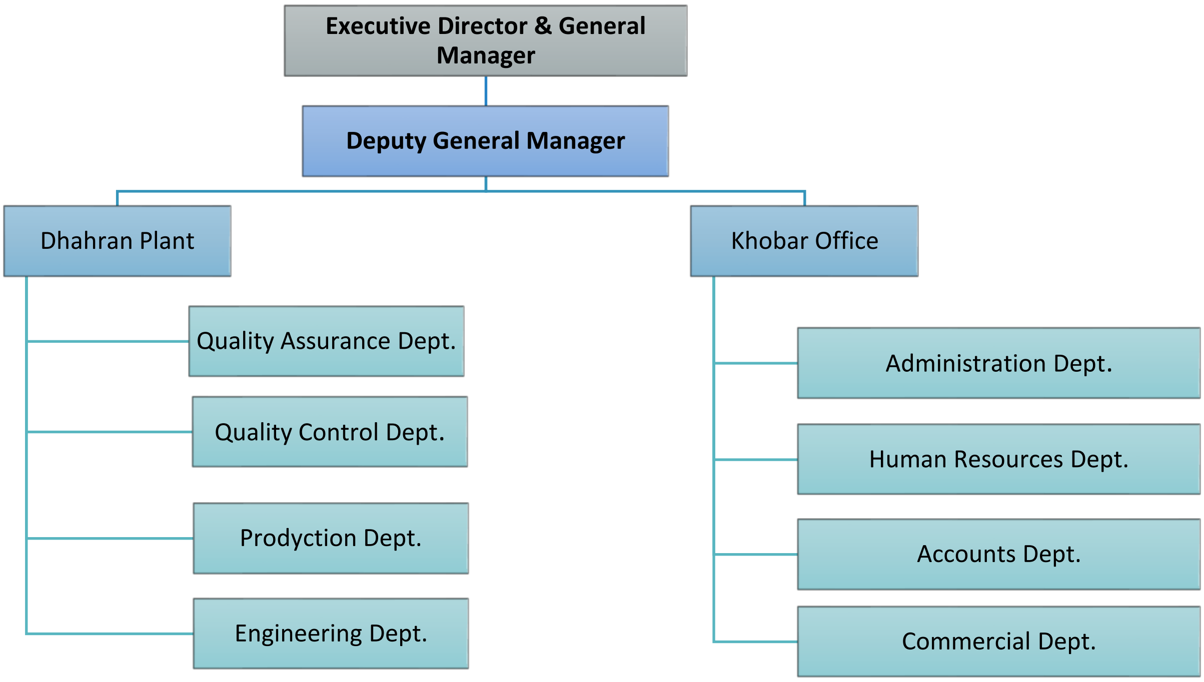
Based on the corporate meeting with executive of Target Company and data received from the Company, the key management of the Target Company are as listed below –

Name	Designation	Country
Mitsuru Miura	Deputy General Manager – Administration	Japan
Abdullah Al Tuwaijiri	Deputy General Manager – Commerce	Saudi Arabia
Shiroyama Takuya	Manager – Commercial	Japan
Tohru Fukushima	Sr. Manager – Production	Japan
Akihiro Mondel	Sr. Manager - Admin	Japan
George Mathews	Manager – Accounts/ Finance	India

Source: MIIL

Organisation Structure of NPC

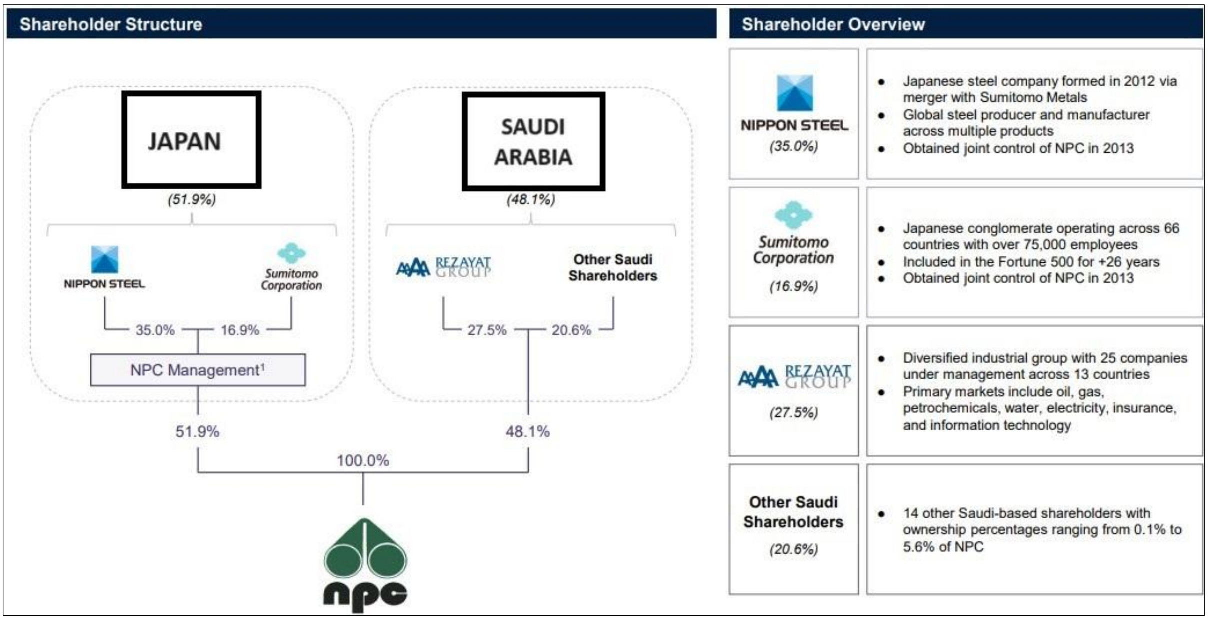
The existing organization structure of the Target Company has been exhibited below –



Source: MIIL

Shareholding Pattern of the Target Company

The shareholding pattern of NPC as provided to the Consultants has been presented in the exhibit below –



Source: MIIL

Historical Financials

The historical profit and loss account of NPC has been presented in the exhibit below –

All Figures in USD Million				
Description	Audited			Est
	31-Dec-22	31-Dec-23	31-Dec-24	31-Dec-25
Sales Realisation	34.48	29.32	119.90	186.77
Other Income	0.64	0.87	0.81	0.52
Total Revenue	**35.12**	**30.20**	**120.71**	**187.29**
Variable Cost				
Raw Material	32.24	21.73	80.09	156.55
Power Cost	0.35	0.46	0.74	1.74
Material Handling and Transport				0.45
Repairs and Maintenance	0.57	0.91	1.61	1.41
Other Direct Expenses	0.42	0.26	0.55	2.27
Manpower Cost	8.04	8.21	10.05	7.73
Total Variable Cost	**41.63**	**31.57**	**93.04**	**170.14**
Opening Balance - WIP	-	-	-	-
Sub-Total	**41.63**	**31.57**	**93.04**	**170.14**
Closing Balance - WIP	-	-	-	0.45
Opening Balance - FG	-	-	-	-
Sub-Total	**41.63**	**31.57**	**93.04**	**169.70**
Closing Balance - FG	-	-	-	21.73
Cost of Production	**41.63**	**31.57**	**93.04**	**147.97**
Fixed Costs				
Administrative Expenses	3.69	4.49	5.08	4.68
Selling and Distribution	1.14	1.09	0.96	1.40
Total Fixed Costs	**4.83**	**5.58**	**6.04**	**6.09**
Total Operating Expenses	**46.46**	**37.15**	**99.08**	**154.06**
EBITDA	**-11.35**	**-6.95**	**21.63**	**33.24**
EBITDA Margin	**-32.31%**	**-23.02%**	**17.92%**	**17.75%**
Other Expenses				
Depreciation	8.76	8.60	8.35	8.46
Interest on Term Loan	-	-	-	
Interest on Working Capital Loan	0.05	0.56	2.53	2.58
Write Offs	1.92	1.64	-	-
Non-Operating Expenses (Loss)	- 0.26	-	- 0.13	-
Total Other Expenses	**10.47**	**10.80**	**10.74**	**11.03**
Total Expenditure	**56.93**	**47.95**	**109.82**	**165.09**
Profit Before Tax	**- 21.82**	**- 17.76**	**10.88**	**22.20**
Applicable Tax and Adjustments	- 0.49	- 0.35	2.10	-
Profit After Tax	**- 22.31**	**- 18.11**	**8.78**	**22.20**
PAT Margin	-63.53%	-59.96%	7.28%	11.86%

All Figures in USD Million				
Description	**Audited**			**Est**
	31-Dec-22	**31-Dec-23**	**31-Dec-24**	**31-Dec-25**
Dividend				-
Retained Earning	**- 22.31**	**- 18.11**	**8.78**	**22.20**

Source: NPC Annual Reports

The historical balance sheet of the Company has been presented in the exhibit below –

All Figures in USD Million				
Description	**Audited**			**Est**
	31-Dec-22	**31-Dec-23**	**31-Dec-24**	**31-Dec-25**
Liabilities				
Shareholders Funds				
Equity Capital	53.27	53.27	53.27	53.27
Statutory Reserves	26.64	26.64	26.64	26.64
Reserves and Surplus	45.10	27.00	35.78	57.99
Total Shareholder Funds	**125.01**	**106.90**	**115.69**	**137.89**
Term Liabilities				
Term Loan	-	-	-	-
Unsecured Loan	-	-	-	-
Employee Terminal Benefits	5.77	5.95	5.89	5.95
Deferred Tax Liability (Net)	-	-	-	-
Total Term Liabilities	**5.77**	**5.95**	**5.89**	**5.95**
Current Liabilities				
Trade Payables	2.54	0.85	-	-
Other Current Liabilities	-	51.36	11.63	1.12
Working Capital Loan	-	25.40	-	-
Provision for Zakat	0.59	0.50	1.91	-
Total Current Liabilities	**3.13**	**78.11**	**13.53**	**1.12**
Total Liabilities	**133.91**	**190.96**	**135.12**	**144.96**
Assets				
Gross Block	264.87	264.62	265.09	266.06
Cumulative Depreciation	197.80	206.26	214.72	223.18
Net Block	**67.07**	**58.36**	**50.37**	**42.89**
Current Assets				
Inventories	21.77	72.37	61.47	47.91
Debtors/ Receivables	25.73	36.01	7.07	23.03
Other Current Assets	-	-	-	-
Cash and Cash Equivalent	19.34	24.21	16.20	31.13
Total Current Assets	**66.84**	**132.60**	**84.74**	**102.07**
Total Assets	**133.91**	**190.96**	**135.12**	**144.96**

Source: NPC Annual Reports

Secant Consultants Private Limited

Secant Consultants Private Limited is a growing multi-dimensional technical, development and management consultancy company with head office at Mumbai and regional office at Lucknow, Noida and Pune.

The Company's mission is to provide high end consulting services to corporate and government (both central and state) bodies by partnering to overcome their existing problems and to assist them on growth trajectory. SCPL combines industry specific managerial expertise along with consulting competence in all its services.

The team of experts at SCPL has a combined business advisory and operational experience of over 150 years. The Team is an optimum mix of Technocrats, Certified Asset Valuers, Market Research Analysts, Financial Analysts, Contract Specialist, Architects, Economists, Growth Strategists and Health Consultants.

Scope of Services

The Scope of Services, as envisaged by the Consultants is presented as below -–

Market Assessment

1. Demand and market
   a. Structure and characteristics of the H-SAW and L-SAW Pipes market in the Kingdom of Saudi Arabia ('KSA') and Gulf Co-operation Countries ('GCC') as a "region".
   b. Assess the historical demand of H-SAW and L-SAW Pipes in the region during the last few years, including the exports (if any)
   c. Supply of H-SAW and L-SAW Pipes in the area including the import trend during the last few years (if any)
   d. Commenting on the current, as well as expected the future demand-supply gap scenario
   e. Commenting on the historical price trend and expected future price trend
2. Review and comment on the marketing concept, sales forecast and marketing budget
   a. Description of the marketing concept, selected targets and strategies
   b. Estimated annual revenues from products, pricing of the product.
   c. Estimated annual operating cost and expenses towards costs of sales promotion and marketing etc.
3. Competition Analysis on qualitative basis.
4. Identify and analyse the company's existing customer base - type of clients, geographical reach, split by product categories and order booking for the next 12 months.
5. Commentary on import substitutes and duty structure for company's products.

Technology Assessment

Assessment of overall plant- performance by facility in terms of production volumes, capacity utilization, and key operational parameters of key modules and assessment of the plant in terms of capability to achieve the optimum level of production. This will be based on operating data provided by Company and benchmarking against comparable metrics. Analyse capital expenditure that may be needed towards immediate maintenance or repairs to ensure sustainable operations at the rated capacity.

1. Review the reports/ studies as submitted by the technical consultants identified by Client and comment on the same.
2. Review and comment on the existing infrastructure and manufacturing facilities of the target based in Saudi Arabia.
3. Review and comment on the production process expected to be followed at the unit and suggest variations/ changes (if any).
4. Review the land purchase/ sales deed and understand the availability of the land for the Project and possible future expansions (if any). (Land purchase/ sales deed to be provided by Client in or translated to English)
5. Review the layout of the facility and comment on sufficiency and efficiency in terms of free movement of manpower and material across the Unit.
6. Review and comment on the fixed asset register, from the perspective of capability of unit to produce the products i.e. H-SAW and L-SAW Pipes
7. Review and comment on the block acquisition cost as estimated by the Company/ technology providers. The block acquisition cost will be reviewed and commented on upon based on the following sub-heads –
    a. Hardware Cost
        i. Land and Land Development Cost
        ii. Building and Civil Works
        iii. Plant and Machinery Cost (including the civil works for plant and machinery foundation)
        iv. Miscellaneous Fixed Assets (including electrical works, utilities like water, fuel storage etc., firefighting system, furniture and fixture, computer and software etc.)
    b. Soft Costs
        i. Preliminary and Pre-operative Expenses
        ii. Interest During Construction Period
        iii. Contingency
    c. Margin Money for Working Capital
8. Raw Material Analysis
9. Review of the plant material, power, water and utility balance, and generation of an updated flow sheet.
10. Review/ Assess and/ or comment on -
    a. Power requirement
    b. Water requirement
    c. Secondary fuel requirement (if any)
    d. Utilities requirement etc.
11. Review the manpower requirement and manpower recruitment policy as proposed by the Company. Understand the organogram as proposed by the Company for operations of the Project and commenting on the suitability/ adequacy of the same. Assess the social impact in terms of jobs created directly or indirectly on account of the project.

12. Review of existing management setup and comment on expertise and experience of senior management officials, and adequacy of skilled manpower for carrying out manufacturing operations at projected levels.
13. Review the prevailing Environment Policies of the Local Government and comment on the same.
14. Review and comment on the Environmental Impact of the Project from the following perspective –
    a. Air pollution
    b. Land pollution
    c. Water pollution
    d. Noise pollution
15. Review and comment on the implementation (acquisition) schedule as proposed by the Company.
16. Based on the observation during site visit and the review of the data as discussed above, comment on the residual life of the Project.
17. Review the status of various statutory clearances, including renewal, for all the production facilities.

Financial Assessment

Based on review of all the above factors and review of the financial model as prepared by the Company/ its Consultants, undertake an independent financial evaluation/ assessment which would comprise of –

1. Project Financing
    a. Proposed capital structure and proposed financing.
    b. Cost/ Means of finance
2. Production cost (significantly large cost items to be classified by materials, personnel and overhead costs, as well as by fixed and variable costs)
3. Based on the above, estimate the EBDITA margin of the Project and based on EBDITA margin of the Project, comment on the interest-bearing capacity and repayment schedule of the Project
4. Estimate and comment on the working capital requirement of the Project and comment on the working capital limits required to achieve the projected level of sales/production.
5. Financial evaluation based on the above-mentioned estimated values
6. Validating operating and financial assumptions for the financial model.
7. Validation of cost assumptions in case of any capital expenditure/maintenance/refurbishment requirements.
8. Provide a complete financial model.
9. Run various scenarios/ sensitivities on key operating and financial assumptions.
10. Commenting on the Viability/ Sustainability of the Project, based on -
    a. Net Present Value
    b. Internal Rate of Return and Post Tax Cost of Capital
    c. Debt Service Coverage Ratio

Risk and SWOT Analysis

11. Based on market, technical and financial assessment, undertake a detailed risk assessment exercise
12. Suggest possible mitigation measures of the risks.
13. Undertake PESTLE (political, economic, social and technological) and SWOT (strengths, weaknesses, opportunities and threats) Analysis of the Project.

Approach and Methodology

SCPL adopted a consultative approach during the course of the assignment. The team from SCPL worked closely along with the team from MIIL and NPC for successful and early completion of the assignment. All the information required for successful completion of the assignment collated by team SCPL from the following sources –

- In house information available with the Consultants
- Information available with the Client Organization
- Published information available with various Government Departments
- Sector/ Industry/ Country reports published by reputed consultancy organizations
- Journals/ Publication of trade association
- Discussions with industry/ sector experts

Based on the secondary and primary analysis the Consultants estimated the facilities planning including service offerings. The Consultants have provided estimates of capital investments required and operating costs based on revenue and cost streams.

Industry Assessment

Introduction

The Oil and Gas sector is understandably classed as being the biggest industry in the world in terms of dollar value. Oil and Gas companies are usually significant contributors to National GDP worldwide and they provide employment to millions of people across the globe. By far, the largest volumes of products of the oil and gas industry are fuel oil and gasoline (petrol).

The sector has three key areas – Upstream, Midstream and Downstream.

- The Upstream market is exploration and production which involves the search for underwater and underground natural gas and crude oil fields, and the drilling of exploration wells.
- The Midstream market covers the transportation, storage and processing of oil and gas. Transportation can involve anything for oil tankers to pipelines and even fleets of trucks.
- The Downstream market incorporates the filtering of the raw materials that were attained during the Upstream phase. In includes the refining of crude oil and the purification of natural gas.

The oil and gas industry widely operates in demanding environments that sees machinery working in very high or low temperatures in potential toxic substances. Because of this, special grades of metal have been developed to ensure that it can resist corrosion and withstand extreme temperatures.

As the oil and gas market is increasingly focusing its search on even more hazardous environments, the demand for corrosive resistant duplex steel products becomes higher. The further the depth of exploration for offshore oil, the greater pressure is put on stainless-steel pipes installed in harsh corrosive environments.

Stainless steel is often used in the oil and gas industry due to its many benefits –

- High strength that therefore enables lighter constructions
- Is relatively maintenance free, reducing life-cycle costs
- Has very good corrosion resistance, including pitting corrosion resistance and intergranular corrosion
- Has higher tensile and yield strength
- Has good weldability

Steel Pipes and Tubes

Globally, 10% of the steel produced is estimated to be converted to tubes. Higher demand for oil & gas and chemical & petrochemical industry – two of the largest consumers of steel pipes and tubes – is driving the demand across the world. Steel pipes and tubular products are broadly classified in following six type –

1. **Standard Pipe:** There are three different types of standard metal pipes - welded (ERW Pipe), seamless pipe, and galvanized pipe.
2. **Line Pipe:** Used primarily in Oil and Gas Applications. Line pipe includes ERW, FW, SAW and DSAW Pipe. They are manufactured to API 5L Specification and are available in X42, X50, X60 etc. grades.
3. **Oil Country Tubular Goods (OCTG):** This includes drill pipes, tubing and casing and are primarily used in drilling and completion of Oil and Gas wells. OCTG are produced by ERW and Seamless manufacturing.

4. **Pressure Tubing:** Pressure tubings are produced using seamless manufacturing and are used for industrial and pressure application.
5. **Mechanical tubing:** These are used for mechanical and structural application and is produced by ERW and seamless manufacturing. They conform to ASTM specification.
6. **Structural Tubing:** These are used for support or retention purpose and are produced by ERW manufacturing. These are available in round or square and are largely used in construction, for fencing and other misc. support needs.

The two most common types of pipes are welded pipe and seamless pipe, both of which are available in carbon steel and stainless steel. They are part of tubular goods, which are manufactured to different specifications and standards. Pipes are sold by "nominal pipe size" in sizes from 1/8" to 72".

- **Welded Pipes:** Welded pipe is manufactured using following three processes i.e. ERW (Electric Resistance Welded) pipe, Furnace Weld (FW) also called as Continuous Weld, and Submerged Arc Weld (SAW), also DSAW. Most common specification for welded carbon steel pipe is A53. Welded SS Pipe is made to specification ASTM A312 and A358. A312 is the most common specification for SS pipe. Welded stainless pipe is made from 1/8" to 24" NPS (Nominal Pipe Size).
- **Seamless pipes:** Seamless pipes is manufactured using process that requires no welding. They are made from steel billets have uniform structure and strength across the entire pipe body because of which it can withstand high pressure, temperature and stress as compared to welded pipes wherein the strength of the pipe is somewhat limited to the strength of the weld joint. These are used in applications which require properties of high anti corrosion and ability to withstand high pressure. Most common specification for seamless carbon steel pipe is A106B and for seamless stainless-steel pipe is A312. Seamless Stainless-Steel Pipe is made to specification ASTM A312 and A376. A312 is also the most common spec for seamless SS pipe. Seamless SS pipe is made from size 1/8 to 14" nominal.

Based on manufacturing process and differentiated by their end-usage, several types of steel pipes & tubes manufactured are listed below

Pipe Type	Size	Manufacturing	Key Applications
Seamless	1/2” –14”	Piercing ingots/ billets of steel at high temperatures	It finds wide application in High Pressure condition Oil & Gas Exploration & Drilling, Boiler, Automobiles, Process, Pipelines, Refineries.
Spiral HSAW	18” –100”	Spirally Welding HR Coils	Low Pressure Application Cross-Country Line Pipes for Oil & Gas and Water Transportation
Spiral LSAW	16” –50”	Longitudinally submerged arc welding of steel plates	High Pressure Application Cross Country Line Pipes for Oil & Gas Transport

Pipe Type	Size	Manufacturing	Key Applications
ERW	1/2" –20"	Hot Rolled steel coils using electrical resistance welding process	Low/Medium Pressure Application, Application in urban and rural infrastructure (scaffolding), industrial application in engineering, automobile and process industry such as chemical, food processing, fertilizers, dairy, amongst other; pharmaceutical, power plants and water & sewage transport
Black Steel Pipe	Dia: ½" to 20"Thickness: 1.00 mm – 12.7 mm	Are Forged and threaded	Water, Gas, Air, Steam, Sewage, Water Wells, Mechanical, Hot water circulation in Boilers system, General Engineering purpose.
Galvanized Iron Pipes	15 - 200 mm.	Coated with Zinc layers. Generally screwed & socketed plain bevelled cut ends in pipe are used	It is primarily used in the carrying water in homes and commercial building, Structural Application, Scaffolding
DI Pipes	10 – 300 mm and above	DI also known as ductile cast iron is advanced form of cast iron that can be manufactured in multiple grades to achieve high ductility and tensile strength. Aus tempered ductile iron has even better mechanical properties and resistance to wear	It finds large application in transporting water for drinking water application, sewage treatment and for industrial water supply.

Source: SCPL Analysis

Product Description

As per the discussion with the Executives of Company and data received, it is understood that HSAW and LSAW pipe will be manufactured at National Pipe Company. The detailed description of HSAW and LSAW pipe has been given below –

HSAW Pipe

Helical submerged arc welded (HSAW) pipes has spiral welding joints and the pipes are welded and formed at the same time. The strip steel with the same specification can produce HSAW pipes with various diameters. The welding joints can avoid major stress and have better force resistance. Spiral welded pipe is formed by forming the material strip spiral molding, through internal and external submerged arc welding, thus producing different specifications of the spiral welded pipe. Range and Grade Dimensions

HSAW pipes can be produced of any diameter as desired by customer by adjusting dimensions. HSAW pipes are available in a variety of thickness and diameters, diameters ranging from 219mm to 914 mm and thickness ranging from 6 mm to 22 mm. HSAW pipe's prices do not move significantly with change in diameter and thickness. It remains almost stable over various diameters and thicknesses.

The dimensions of spiral weld pipes are continuously adjustable so that any diameter, as desired by the costumer can be produced from a base material of the same width. Unlike LSAW it is not necessary to use a large number of forming tools for various separate sizes. Spiral welded pipes are manufactured in steel grades from API 5L Grade B to X80 and to various other national and international standards.

The manufacturing of HSAW pipes involves various tests such as hydrostatic tests, automatic non-destructive ultrasonic tests, and X-ray tests. Depending upon requirements and intended use pipes can be produced from various steel grades. Pipes are generally manufactured in diameter range from 323,9 mm to 1219 mm, in length from 4 m to 24 m (line pipes) and from 4 m to 30 m (pipes for construction purposes), with plain or bevelled ends. The manufacturing of Pipes also involves hydrostatic tests and depending on requests: automatic non-destructive ultrasonic tests and/or X-ray test (pipelines).

Advantages of HSAW Pipe

Spiral submerged arc welding (SSAW) steel pipe has many advantages over other welded pipes, the same has pointed below –

- There are two overlapping spiral welds inside and outside the pipeline, which can increase the stiffness and the bearing pressure.
- Submerged arc welding technology has good slag control effect. Therefore, it has good impact toughness and low temperature performance.
- Steel strips of the same width can produce pipes of different diameters and lengths with good straightness. Easy to adjust, precise size, no need for sizing and straightening after welding.
- Spiral steel pipes have strong habitual standards, which means that any standard can be produced by spiral steel pipes.
- The length of spiral steel pipes can also be arbitrary, and the longest can be up to 30 meters.
- It can also be made into a conjoined pillar, which has better function than channel steel and is cheaper.

Application

HSAW steel pipe is mainly used in public waterworks, petrochemical industry, chemical industry, electric power industry, agricultural irrigation, and urban construction etc. The strength and flexible manufacturing of spiral welded pipe make it the product of choice for a variety of applications. They are also used in the design of multiple structures such as piling, bridge, wharf, road, and architectural structures etc

- **Liquid Transmission -** HSAW pipes are used in transportation of liquids in various ways. They are used in supply of fresh water, drainage of sewage water and transportation of petroleum products.
- **Gas Transmission -** HSAW pipes also have ability to transmit various kinds of gases such as coal gas, steam, natural gas, LPG etc.
- **Building Structure -** They are also used in the design of multiple structures such as piling, bridge, wharf, road, and architectural structures etc

LSAW Pipe

Longitudinally Submerged Arc Welding (LSAW) pipes use single plate steel as raw material, applied double-sided submerged arc welding. The finished products have larger specifications and the welding joints have excellent ductility, plasticity and good sealing. LSAW pipes have large diameter, thick wall and can withstand high pressure and low temperature. LSAW pipes play a major role in oil and gas pipelines with features of high strength, high quality and long distance.

The production of Longitudinally Submerged Arc Welding (LSAW) pipes involves the utilization of the JCOE or UOE method. In this process, bevelled steel plates are shaped into a U configuration using a U-press, followed by the application of an O-press to transform them into an O shape. Due to UOE method of production LSAW pipes are sometimes also called UOE pipes.

Range and Grade Dimensions

LSAW pipes are used in sizes (diameter) from 16” (406 mm) to 60” (1520 mm) with wall thickness upto 60 mm. Pipe size and wall thickness depend mainly on the available plates but the processes that are used have limitations with respect to forming forces and these also limit pipe size and wall thickness.

In terms of grades the main trigger is given by the pre-material. Usually grades with strength classes up to L 485 (X70) are available, but also grade L555 (X80) are frequently used for pipelines worldwide as standard grades. Even higher grades, L 690 (X100) and L830 (X120) have been developed, but the pipeline market is reluctant to use these very high strength pipe grades at this time.

Higher grades allow reduction of pipe weight with smaller wall thickness at the same diameter and operational pressure, which is beneficial for pipe transport and laying activities.

Advantages of LSAW Pipe

- Spiral submerged arc welding (SSAW) steel pipe has many advantages over other welded pipes, the same has pointed below –
- LSAW pipes are known for their high strength and structural integrity, making them suitable for applications where the pipes must withstand significant loads and pressures.
- LSAW pipes can be manufactured in large diameters, making them ideal for applications that require the transportation of large volumes of fluids or gases. The large diameter also contributes to their strength and load-bearing capacity.
- LSAW pipes are often more cost-effective than seamless pipes for large-diameter applications. The manufacturing process allows for efficient production of pipes with substantial dimensions.
- LSAW pipes find applications in diverse industries, including oil and gas transportation, water pipelines, structural construction, piling, and more.
- The submerged arc welding (SAW) process used in LSAW pipes is an efficient and automated welding method. It ensures a high-quality weld with minimal defects, contributing to the overall reliability of the pipes.
- LSAW pipes are well-suited for long-distance pipelines due to their strength and the ability to produce pipes in lengths that minimize the number of welding joints. This is crucial for pipeline integrity over extended distances.

- LSAW pipes can be manufactured to meet specific industry standards, such as API 5L for the oil and gas industry. Adherence to standards ensures that the pipes meet the required mechanical and chemical properties.
- LSAW pipes can be coated with protective layers such as fusion-bonded epoxy (FBE) or three-layer polyethylene (3LPE) to enhance their resistance to corrosion, especially in corrosive environments.

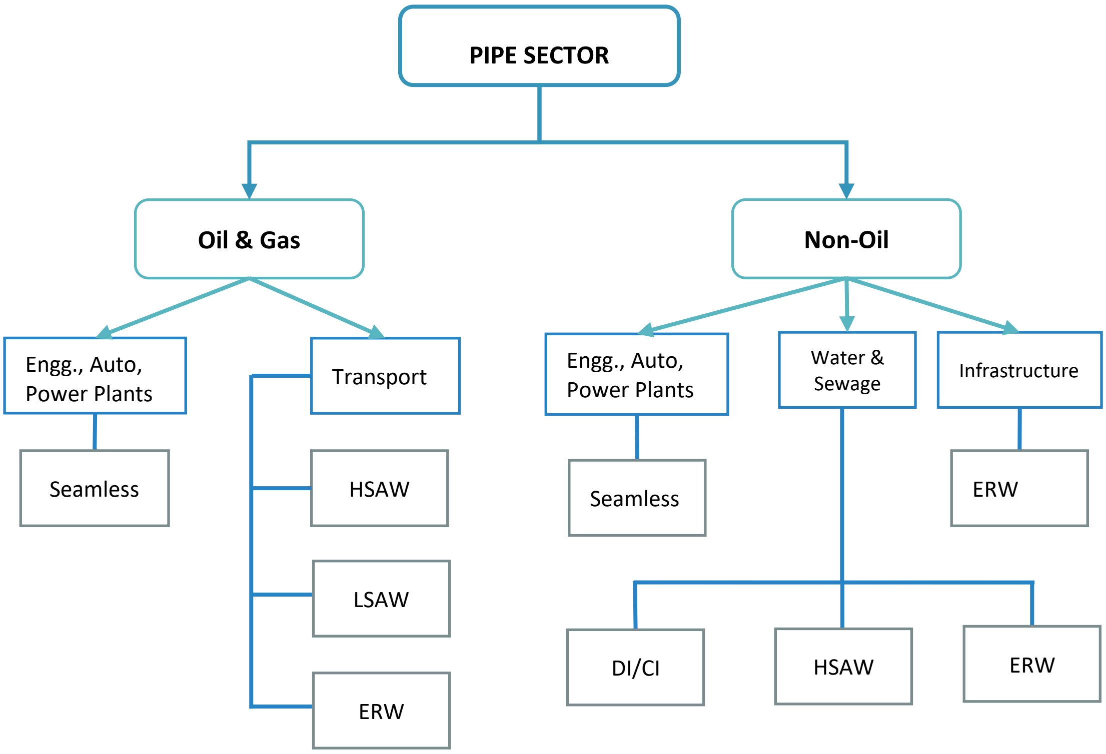
Application

Longitudinally Submerged Arc Welding (LSAW) pipes find applications across various industries due to their structural integrity, strength, and suitability for transporting fluids and gases over long distances. Some common applications include –

- **Oil and Gas Industry -** LSAW pipes are extensively used for the transportation of oil and natural gas over long distances. They are employed in both onshore and offshore pipelines, providing a reliable and durable solution for energy transmission.
- **Water and Sewage Pipelines -** LSAW pipes are utilized for the construction of water and sewage pipelines, delivering a robust solution for the efficient conveyance of water resources.
- **Infrastructure and Construction -** In construction projects, LSAW pipes are employed for structural purposes, such as piling, bridge construction, and building foundations. Its strength and large-diameter capabilities make them suitable for supporting heavy loads.
- **Piling and Foundation Projects -** LSAW pipes are commonly used in piling applications where a strong foundation is required, such as in the construction of bridges, buildings, and other structures.
- **Power Generation -** In the power generation sector, LSAW pipes may be used for conveying cooling water or other fluids in thermal power plants.
- **Transportation of Chemicals -** LSAW pipes can be employed for the safe and reliable transportation of chemicals in industrial applications.
- **Infrastructure Development -** Large-diameter LSAW pipes are suitable for infrastructure development projects, including the construction of tunnels and underground structures.
- **Mining Industry -** LSAW pipes may be used in the mining industry for transporting materials, slurries, or water in a reliable and efficient manner.
- **Structural Engineering -** LSAW pipes contribute to the structural integrity of various engineering projects, including the construction of stadiums, high-rise buildings, and other large-scale structures.
- **Offshore Construction -** LSAW pipes are employed in the fabrication of offshore structures, including platforms and pipelines, where they can withstand challenging environmental conditions.

Structure of Industry

The steel pipes industry comprises of series/ variety of pipes, which are utilised as per the end-use/ application and these have been depicted in the structure of the industry illustration as provided below –



Source: SCPL Database

Standards

Standards prescribed by institutions like ASTM, BS, DIN, BIS, GOST, JIS & EN governs the global pipe industry. Some of these standards include –

- Outside diameter
- Wall thickness
- Nominal weight per unit length
- Hydrostatic test pressures
- Tolerance preferences
- Tensile Strength
- Yield Strength

American Petroleum Institute (API) prescribes the pipe specifications for application in crude, petroleum products, gas and other applications requiring operation under high pressure.

The Consultants API approved SAW pipes are sold in the open market at a premium as compared to the non-API SAW pipes. The non-API SAW pipes compete with other pipes like ERW, DI, CI, concrete and plastic pipes for non-critical applications like water and sewerage transport.

As discussed previously the project is proposed to manufacture SAW pipes and hence the Consultants have focused on SAW pipes only in subsequent sections.

Installed Capacity and Operational Capacity

Discussion with industry experts indicate that there is marked variation in the installed capacity (rated capacity) and the actual capacity (operational capacity) across the global SAW pipes industry. The Consultants understand that the installed capacity of the plants is only a theoretical value, which is based on the amount of pipes (of various diameters) manufactured during the pilot run of similar plants by the technology provider. However the market conditions and the specific order from the customers forces the pipe manufacturers to vary the amount of specific diameter pipes from the one used by the technology provider hence the variation in the actual capacity from the installed capacity.

The Consultants understand that the operational capacity will reduce significantly if the amount of lower diameter pipes produced is increased. On the other hand the operational capacity of the plant will be higher if more numbers of higher diameter pipes are manufactured by the same plant.

The other reasons for the variation in the installed and actual capacity have been provided below –

Plant Availability

The SAW pipe industry is old, with many plants operating for more than 20-25 years. Availability and reliability of these plants are poor as compared to new plants.

Financial Problem

Global pipe industry is very competitive. Hence, players with higher operating costs find it difficult to operate their facilities as they do not get sufficient orders at their price. This makes the company financially weak and they find it increasingly difficult to get additional banking support both for renovation and working capital. This further affects their performance.

Raw Material Sourcing

Global steel industry manufactures wide variety of HR Plates and HR Coils, which can be used to manufacture LSAW pipes and HSAW pipes respectively. However, it is noted all the HR Plate and Coils manufacturers do not have the capability to manufacturer the entire range of HR Plates and Coils. This in turn becomes a disadvantage for SAW pipe manufacturers in vicinity. As they have to source the raw material from far off places, thus increasing the overall raw material cost. This in turn affects their performance.

Labour Problem

The global pipe industry is cost competitive. It is noted that developed countries in Europe and North America face a disadvantage of higher labour cost. This has lead to steps being taken by management to rationalise labour cost leading to labour problems.

Set-up time and Productivity

Globally all the SAW manufacturers try to optimise utilisation by balancing changeovers and size of pipes. It can be noted that change in the size of pipes can have about 10% impact on the utilisation of the plant.

Gulf Cooperation Council – Regional Scenario

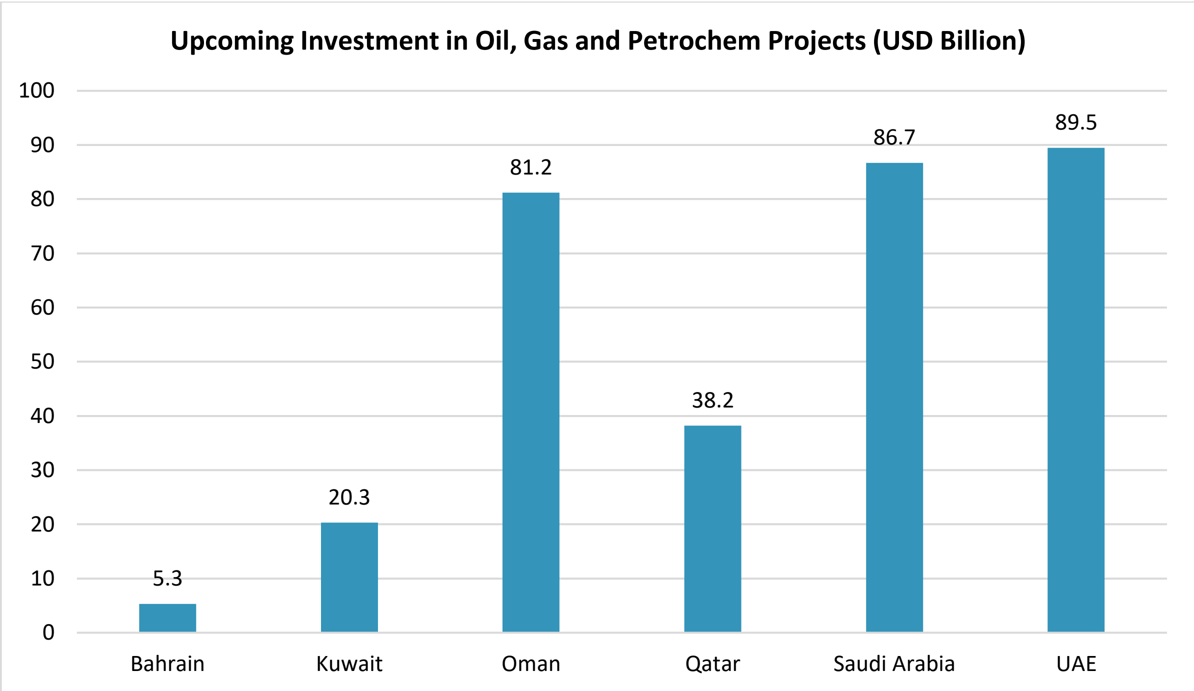
Introduction

The Cooperation Council for the Arab States of the Gulf, also known as Gulf Cooperation Council (GCC) is regional intergovernmental, political and economic Union comprising of Bahrain, Kuwait, Oman, Qatar, Saudi Arbia and

United Arab Emirates (UAE). The charter of GCC was formally signed during the year 1981 and its head office is located at Riyadh, the capital city of KSA.

The region has been experiencing higher than global average GDP growth in period prior to CoVId-19 pandemic hit the global markets/ economies. The economies have been driven by the Oil and Gas supplies found in the region, with the region being rich in these resources. However, as recent push has been seen amongst the countries in GCC to move towards developing the other areas like manufacturing sector, IT, financial services etc. However, under the prevailing scenario, oil and gas sector is what drives the GCC region.

The exhibit below provides as idea of the upcoming/ planned investment in oil, gas and petrochemical projects in the region. A part of these investments will be for the purpose of oil and gas pipelines, whether inter region, intra city or intercity pipelines.



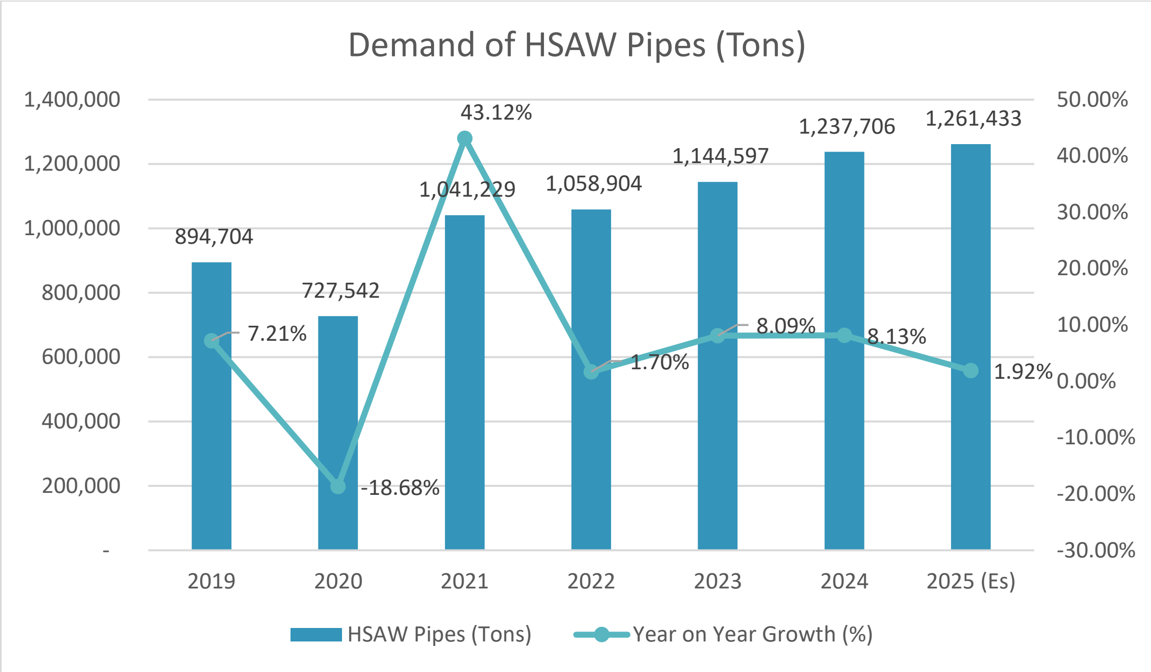
Source: World Bank Gulf Economic Update 2025

HSAW Pipes – Demand Supply Scenario

Demand

As discussed in the section above, the oil, gas and petrochemical industry are the prime drivers of economies in the region and the demand for HSAW Pipes in the region is primarily driven by these industries and also demand from the water transportation industry in the region. There are a series of oil and gas pipeline projects and water transportation pipeline project which have been declared in the region, between 2015 to 2018 and many of these have been delayed by 2 years on account of the pandemic scenario prevailing in the international markets during the period 2019-21. These projects have kept the demand buoyant for HSAW Pipes in the region.

The demand for HSAW Pipes in the region stood at 894,704 Tons during the year 2019. Since then, the demand for HSAW Pipes in the GCC region has grown at a CAGR of 5.89% to reach an estimated 1,261,433 Tons during the current CY 2025. The growth in demand for the HSAW Pipes in the region during the period 2019 to 2025 has been illustrated in the exhibit below –



Source: Sector Reports and SCPL Analysis

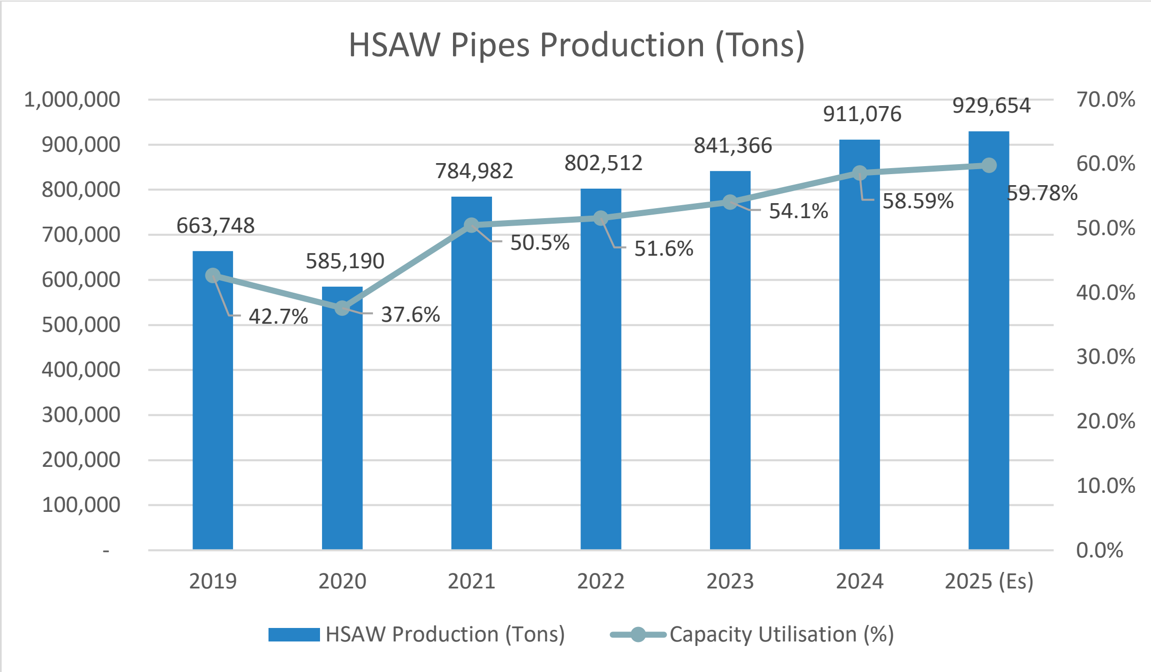
It is noted from the exhibit above, that the demand for HSAW Pipes saw a dip during the year 2020, which was impacted on account of the global pandemic scenario. The demand of HSAW Pipes in the region has since then recovered and showing an increasing trend, which is on account of recommencement of stalled projects and commencement of newer projects.

Installed Capacities and Production

Based on the data available in secondary domain, limited primary market assessment undertaken by the Consultants and reports published by reputed consultancy firms, the Consultants understand that there are total of 6 operational players of HSAW Pipes in the region and these include the following –

1. East Pipes (KSA) – 450,000 TPA
2. Group 5 (KSA) – 570,000 TPA
3. National Pipe Company (KSA) – 250,000 TPA
4. KPIOS (Kuwait) – 150,000 TPA
5. PSL Limited (UAE) – 75,000 TPA
6. Sharjah Steel Pipe Company (UAE) – 60,000 TPA

Considering the above, the total installed capacity of HSAW Pipes in the region stands at 1.56 million Tons. The production of HSAW Pipes in the region stood at 663,748 Tons during the year 2019 and since then the production of HSAW Pipes in the region has grown at a CAGR of 6.41% to reach and estimated 929,654 Tons during the current CY 2025. The production levels as experienced by the HSAW Pipes industry in the region and capacity utilisation levels during the period 2019-2025 has been presented in the exhibit below –



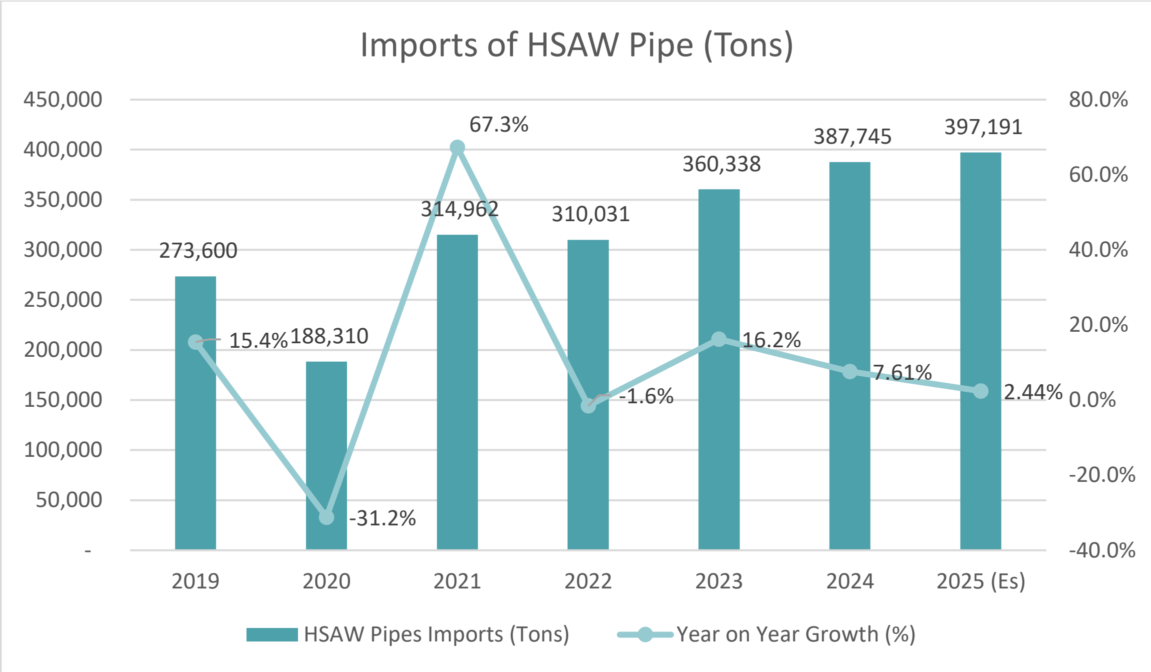
Source: Trade Map, Sector Reports and SCPL Research

It is noted from the exhibit above, that the production levels in the regional HSAW Pipes industry saw a dip during the year 2020, which was on account of CoVid-19 pandemic. Further it is noted from the figure, that the production levels have revived since then and have reach the pre-pandemic level of capacity utilisation level of around 50%-55% level.

Imports and Exports

From the discussions above, it is noted that the demand for HSAW Pipes in the region is much higher than the production levels achieved by the HSAW Pipe manufacturers in the region. The difference in the demand and production at the present moment is being fulfilled by imports of HSAW Pipes from countries like China, India, Korea and Turkey etc.

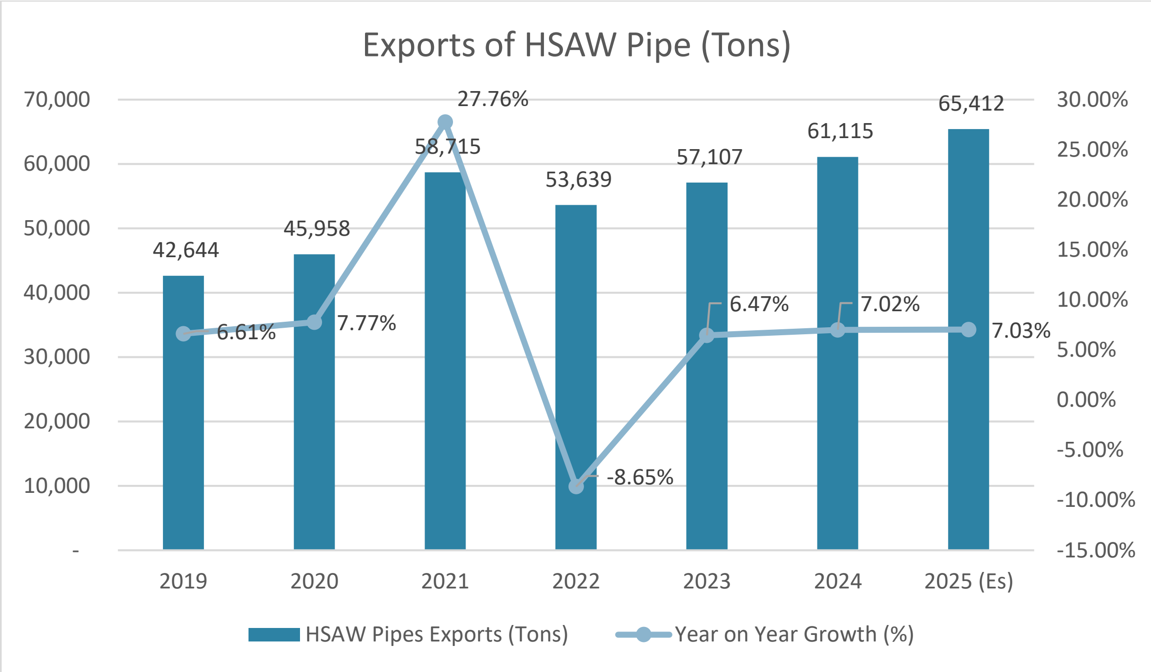
The imports of HSAW Pipes in the region stood at 273,600 Tons during the year 2019 and since then the imports have grown at a CAGR of 6.41% to reach an estimated 397,191 Tons during the current CY 2025. The growth in imports of HSAW pipes in the region during the period 2019-25 has been presented in the exhibit below –



Source: Trade Maps and SCPL Research

Here it should be noted that the imports had seen a decreasing trend in the region prior to CoVid-19 pandemic, which was on account of the fact that the local Governments are pushing for local production/ manufacturing capacities. The demand in the region remained unmet and hence the imports saw an increasing trend post CoVid-19 pandemic, which reflect in the exhibit above. However, based on the discussion with industry experts, it is noted that the trend is expected to reverse in near future with the focus of local manufacturing coming to fore again.

There have been smaller amounts of exports also from the region, but mainly to GCC countries and to some extent the other middle eastern countries like Iraq, Jordan, Lebanon etc. The exports of HSAW Pipes from the region stood at 42,644 Tons during the year 2019. Since then, the exports of HSAW Pipes from the region have grown at a CAGR of 7.39% to reach an estimated 65,412 Tons during the current CY 2025. The exhibit below illustrates the growth of exports of HSAW pipes from the region during the period 2019-2025.



Source: Trade Maps and SCPL Research

Demand Supply Gap Scenario

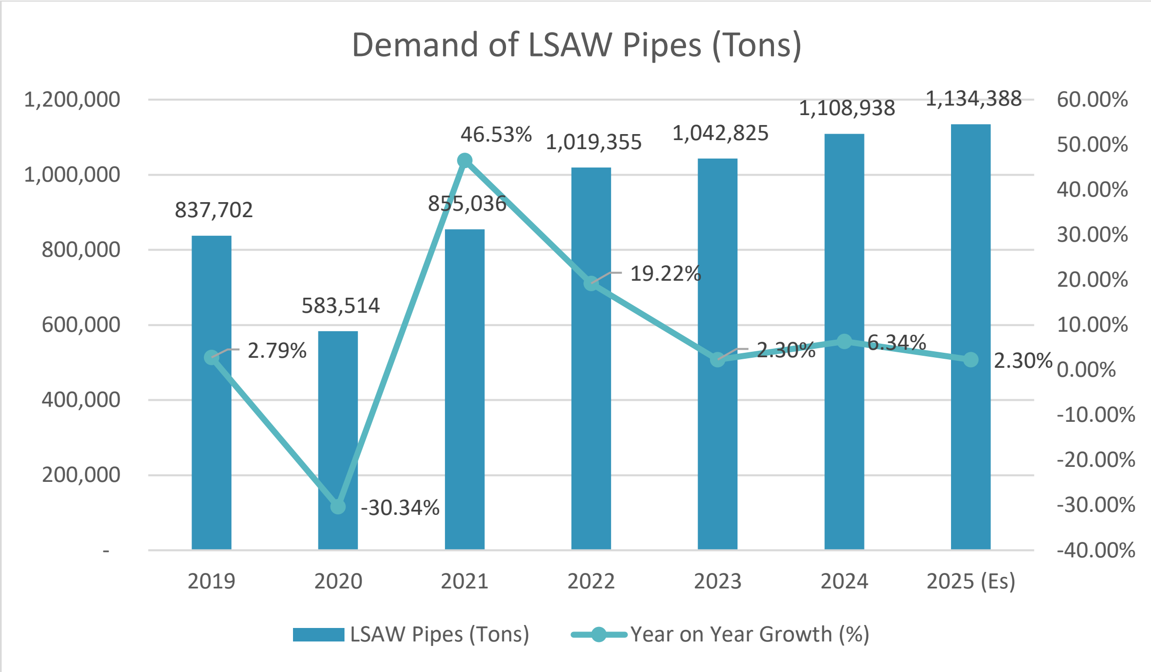
Based on the discussions in the section above, it is noted that the demand for HSAW Pipes in the region under the prevailing circumstances outruns the production level in the region and hence there is a demand supply gap in the markets. At the moment, this gap is being filled in by the imports of HSAW Pipes from various locations including China, India, Korea, Turkey etc. and the region remain a net importer of HSAW Pipes.

LSAW Pipes – Demand Supply Scenario

Demand

Unlike the HSAW Pipes, which find non oil and gas, in water transport sector and to a small extent in construction industry as well, the LSAW Pipes find application primarily in oil and gas sector, with a small demand from petrochemical industry. The demand for LSAW Pipes in the region was primarily from the onshore and offshore oil and gas pipelines announced by various Government Departments/ Companies.

The demand for LSAW Pipes in the region stood at 837,702 Tons during the year 2019. Since then, the demand for LSAW Pipes in the GCC region has grown at a CAGR of 5.18% to reach an estimated 1,134,388 Tons during the current CY 2025. The growth in demand for the LSAW Pipes in the region during the period 2019 to 2025 has been illustrated in the exhibit below –



Source: Sector Reports and SCPL Analysis

It is noted from the exhibit above, that the demand for LSAW Pipes saw a dip during the year 2020, which was impacted on account of the global pandemic scenario. Since then, the demand has seen an upward trend, as the stalled oil and gas pipeline projects have recommenced construction/ implementation, which had stalled/ slowed down during pandemic.

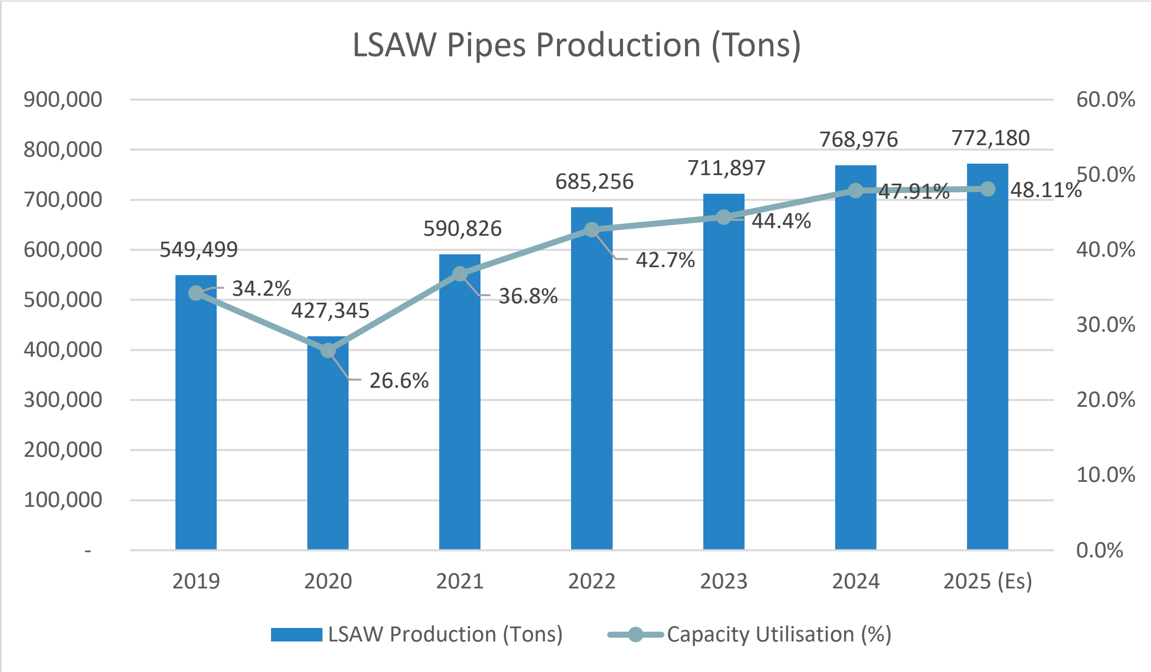
Installed Capacities and Production

Based on the data available in secondary domain, limited primary market assessment undertaken by the Consultants and reports published by reputed consultancy firms, the Consultants understand that there are a total of 6 operational players of LSAW Pipes in the region and these include the following –

1. National Pipe Company (KSA) – 180,000 TPA
2. Global Pipe Company (KSA) – 375,000 TPA
3. Arabian Pipe Company (KSA) – 300,000 TPA
4. PCK (KSA) – 300,000 TPA
5. JPIOS (UAE) – 300,000 TPA
6. SeAH Steel Pipe Co (UAE) – 150,000 TPA

Considering the above, the total installed capacity of LSAW Pipes in the region stands at 1,605,000 Tons. However, here it should be noted that a consignment of Global Pipe Company was rejected by Aramco on quality standard reasons and hence the plant of GPC is currently not operational. The GPC plant is expected to become operational only by end of year 2025, so effectively the current operational capacity of the region is 1,230,000 Tons, but will become 1,605,000 Tons from CY 2026 onwards.

The production of LSAW pipes in the region stood at 549,499 Tons during the year 2019. Since then, the production of LSAW Pipes in the region has grown at CAGR of 6.69% to reach an estimated 772,180 Tons during the current CY 2025. The production levels as experienced by the LSAW Pipes industry in the region and capacity utilisation levels during the period 2019-2025 has been presented in the exhibit below –



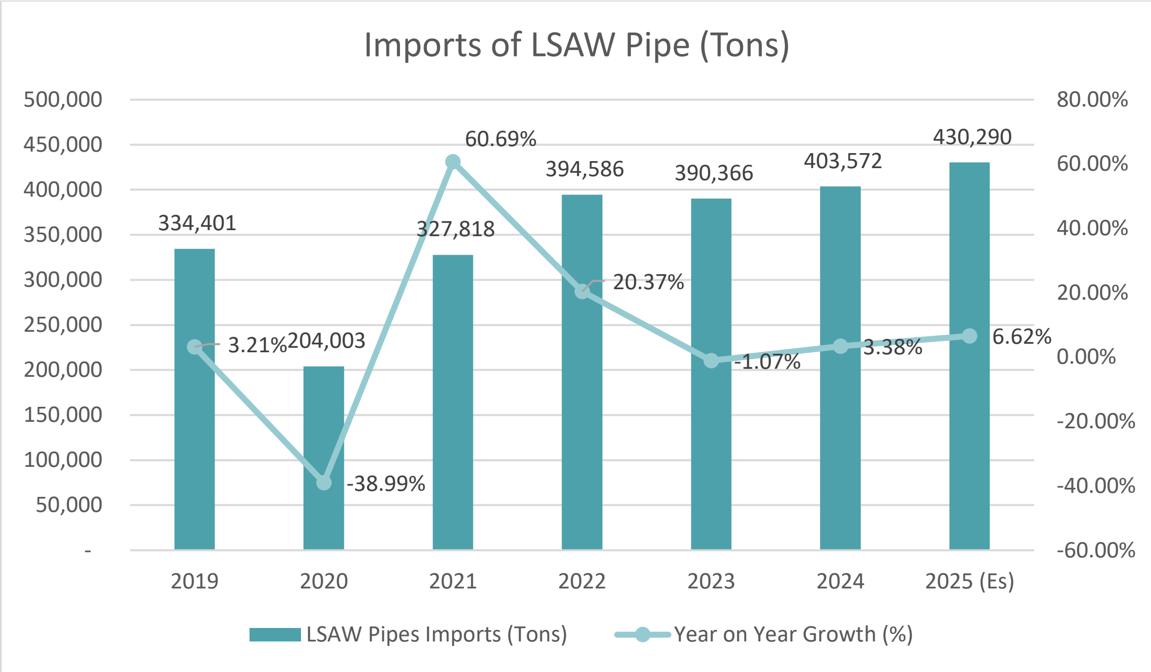
Source: Trade Map, Sector Reports and SCPL Research

It is noted from the exhibit above, that the production levels in the regional LSAW Pipes industry saw a dip during the year 2020, which was on account of CoVid-19 pandemic. Further it is noted from the figure, that the production levels have revived since then and have reach the pre-pandemic level of capacity utilisation level of just below 50% level since last 2 years.

Imports and Exports

From the discussions above, it is noted that the demand for LSAW Pipes in the region has been leading the production of LSAW Pipes in the region. The gap between the demand and production is at the present moment being bridged by imports from various countries like China, India, Korea and Turkey.

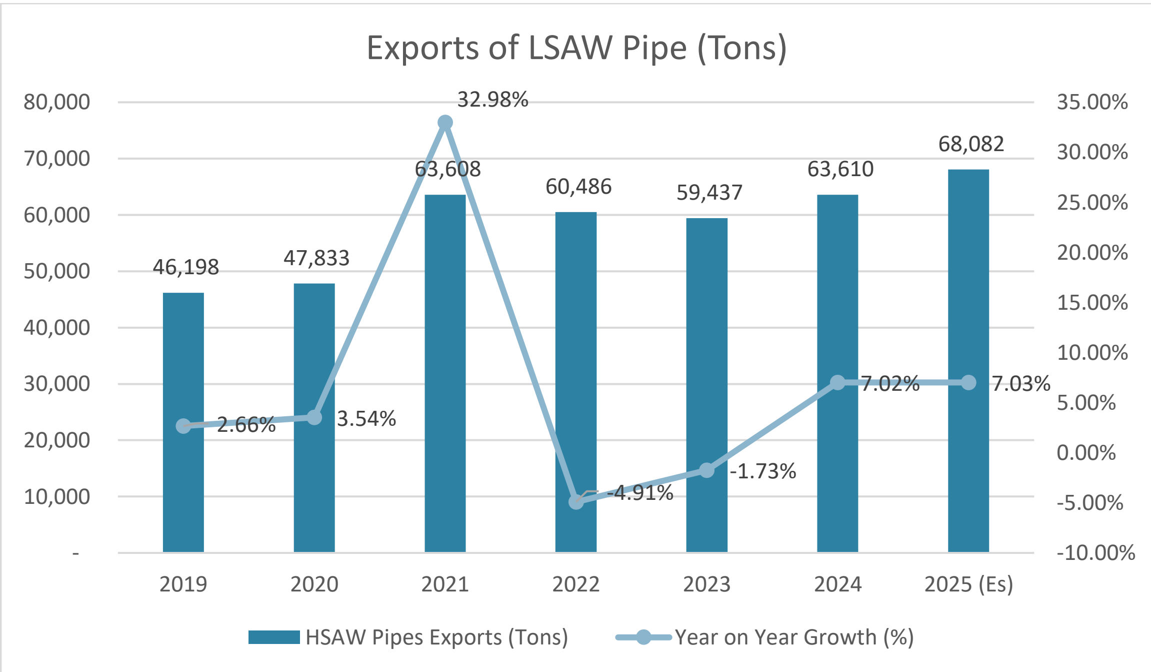
The imports of LSAW Pipes in the region stood at 334,401 Tons during the year 2019 and since then the imports have grown at a CAGR of 4.29% to reach an estimated 430,290 Tons during the current CY 2025. The growth in imports of LSAW pipes in the region during the period 2019-25 has been presented in the exhibit below –



Source: Trade Maps and SCPL Research

In line with the imports of HSAW Pipes, the imports of LSAW Pipes also saw a slow-down during the year 2020, on account of the pandemic scenario in the global markets. However, it is noted that the imports have since then picked up and have reached pre-pandemic level during CY 2021 and have further grown during CY 2023 and recently concluded CY 2024.

The Consultants note that there have been minor number of exports of LSAW Pipes, from the region within the region to other GCC countries and to middle eastern countries like Jordon, Iraq, Lebanon etc. The exports of LSAW Pipes from the region stood at 46,198 Tons during the year 2019. Since then, the exports of LSAW Pipes from the region have grown at a CAGR of 6.68% to reach an estimated 68,082 Tons during the current CY 2025. The exhibit below illustrates the growth of exports of LSAW pipes from the region during the period 2019-2025.



Source: Trade Maps and SCPL Research

Demand Supply Gap Scenario

Based on the discussions in the section above, it is noted that the demand for LSAW Pipes in the region under the prevailing circumstances outruns the production level in the region and hence there is a demand supply gap in the markets. At the moment, this gap is being filled in by the imports of HSAW Pipes from various locations including China, India, Korea, Turkey etc. and the region remain a net importer of HSAW Pipes.

Saudi Arabia – Market Overview

The Kingdom of Saudi Arabia is a member of the Gulf Cooperation Council and the Organization of Petroleum Exporting Countries (OPEC). The economy of Saudi Arabia is the largest of all the GCC countries. It is dominated by petroleum, of which the country is by far the largest producer within OPEC. The country is among the top three petroleum producers in the world (the other two being the US and the Russian Federation) and is the largest exporter of crude oil in the world. It exports mainly to non-EU countries (China, Japan, India, and the US). Saudi Arabia's petroleum sector accounts for roughly 87 % of budget revenues, 42 % of Gross Domestic Product (GDP), and 90 % of export earnings.

Saudi Arabia produced 12.1 million b/d in total liquid fuels in 2022, up 12% from 10.8 million b/d in 2021.

Saudi Arabia produced 10.4 million b/d of crude oil in 2022, 14% higher than the 9.1 million b/d in 2021. This increase drives Saudi Arabia’s increased total liquid fuels production, reflecting a gradual reversal of OPEC+ production cuts from 2020.

Saudi Arabia accounts for 43% (0.5 million b/d) of agreed on OPEC+ production cuts that began in May 2023 and extends through the end of 2024. Saudi Arabia had taken an additional oil production cut of 1.0 million b/d from July through December 2023. The Consultants expect these oil production cuts to result in decreased crude oil production in Saudi Arabia for 2023 compared with 2022.

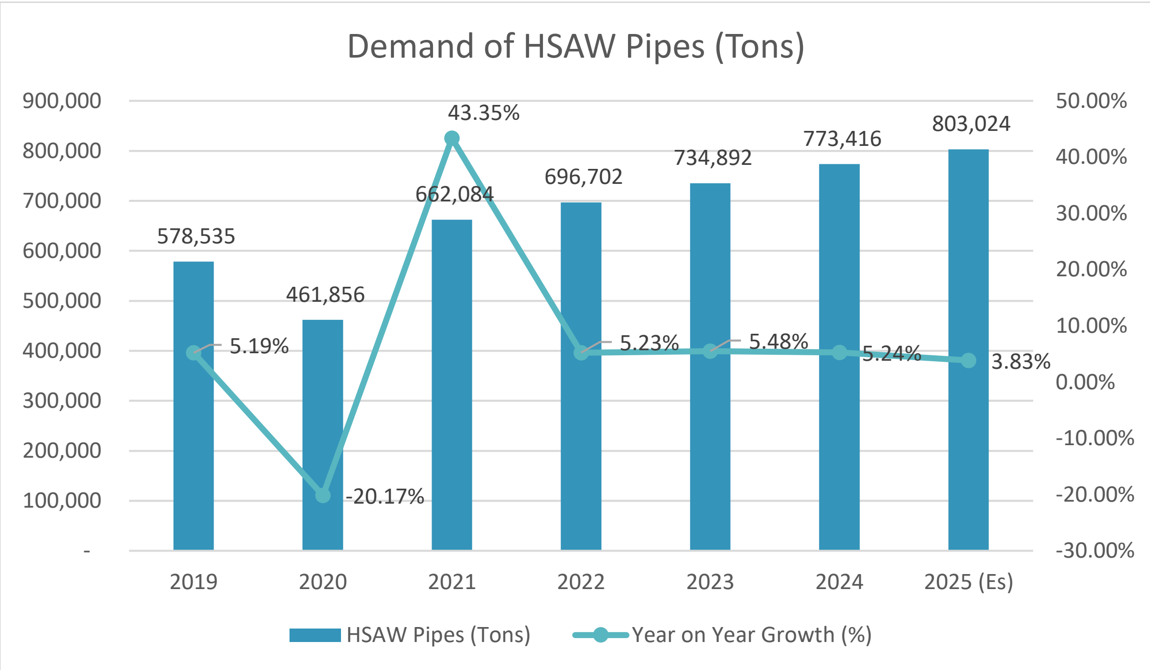
Apart from the oil and gas-based demand, the change in regime and taking over of the designation of crown prince by Mohammed-Bin-Salman-Bin-Saud (Prince Salman) has also resulted in modification of long term vision of the country, whereby focus on development of infrastructure, development of new cities like Neom city etc. has commenced in the KSA. There are series of road projects, water transportation projects, which have been announced by the various Government Departments and the focus has been on what is called as “Saudization”. Saudization is the term given to focus or push for procurement of goods and services from the local manufacturers and suppliers, rather than trying to import the products/ services. Saudization of products and services has resulted in development of non-oil and gas sectors in the country and these include manufacturing, engineering, IT, Telecom, financial services to name a few. In simple word Saudization is Saudi Arabian version of “Make in India” policy of India.

HSAW Pipes Market

Demand

As discussed above, after the change in regime in Saudi Arabia, there has been a series of infrastructure projects announcements both in oil and gas sectors and in non-oil and gas sectors in Saudi Arabia. These include projects for oil and gas pipeline to the tune of 5,354 Km and water transportation pipeline projects to the tune of 4,797 Km. These are the main demand drivers of the HSAW Pipes and LSAW Pipes in the country.

The demand for HSAW Pipes in KSA stood at around the demand for HSAW Pipes in the region stood at 578,535 Tons during the year 2019. Since then, the demand for HSAW Pipes in KSA has grown at a CAGR of 5.62% to reach an estimated 803,024 Tons during the current CY 2025. The growth in demand for the HSAW Pipes in the country during the period 2019 to 2025 has been illustrated in the exhibit below –



Source: Sector Reports and SCPL Analysis

It is noted from the exhibit above, that the demand for HSAW Pipes saw a dip during the year 2020, which was impacted on account of the global pandemic scenario. Since then, the demand for HSAW Pipes in KSA has recovered and is expected to further increase with the recently announced oil and gas pipeline projects and the water supply and transportation projects as announced in the country.

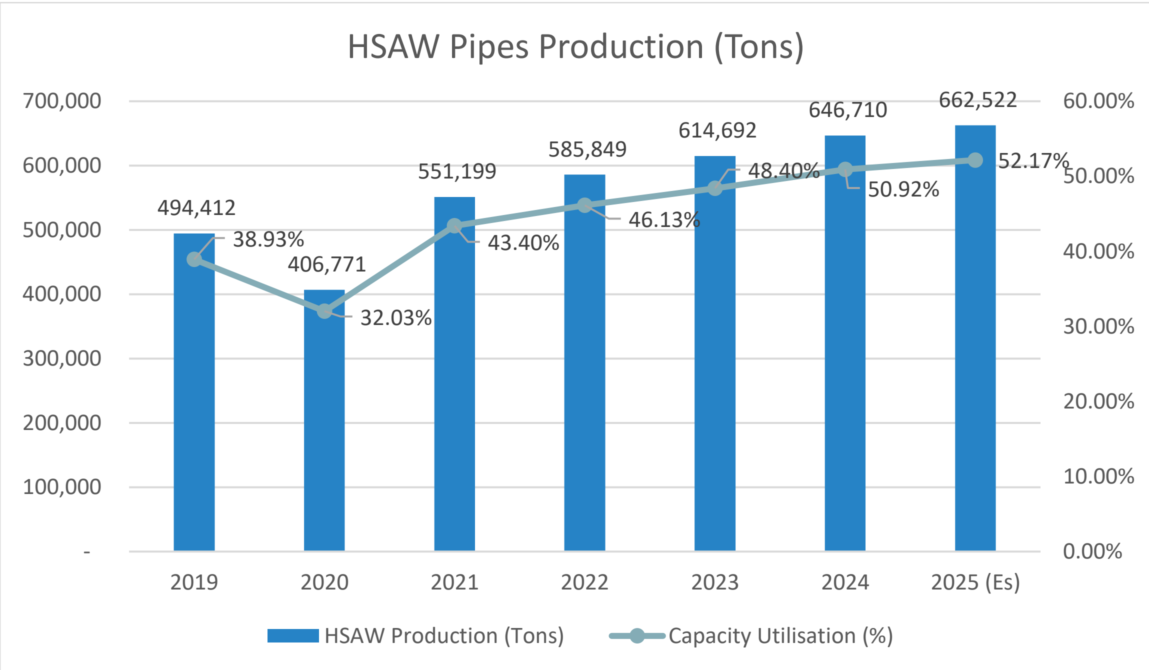
Installed Capacities and Production

Based on the data available in secondary domain, limited primary market assessment undertaken by the Consultants and reports published by reputed consultancy firms, the Consultants understand that there are a total of 3 operational players of HSAW Pipes in KSA and these include the following –

1. East Pipes (KSA) – 450,000 TPA
2. Group 5 (KSA) – 570,000 TPA
3. National Pipe Company (KSA) – 250,000 TPA

Considering the above, the total installed capacity of HSAW Pipes in the region stands at 1.27 million Tons.

The production of HSAW Pipes in KSA stood at 494,412 Tons during the year 2019. The production of HSAW Pipes saw a slight dip during the year 2020, which was on account of the global pandemic scenario. The production of HSAW Pipes in the country saw a growth of CAGR 5.00% during the period 2019-25, to reach an estimated 662,552 Tons during the current CY 2025. The growth in production of HSAW Pipes in the country during this period and corresponding utilisation level has been presented in the exhibit below –



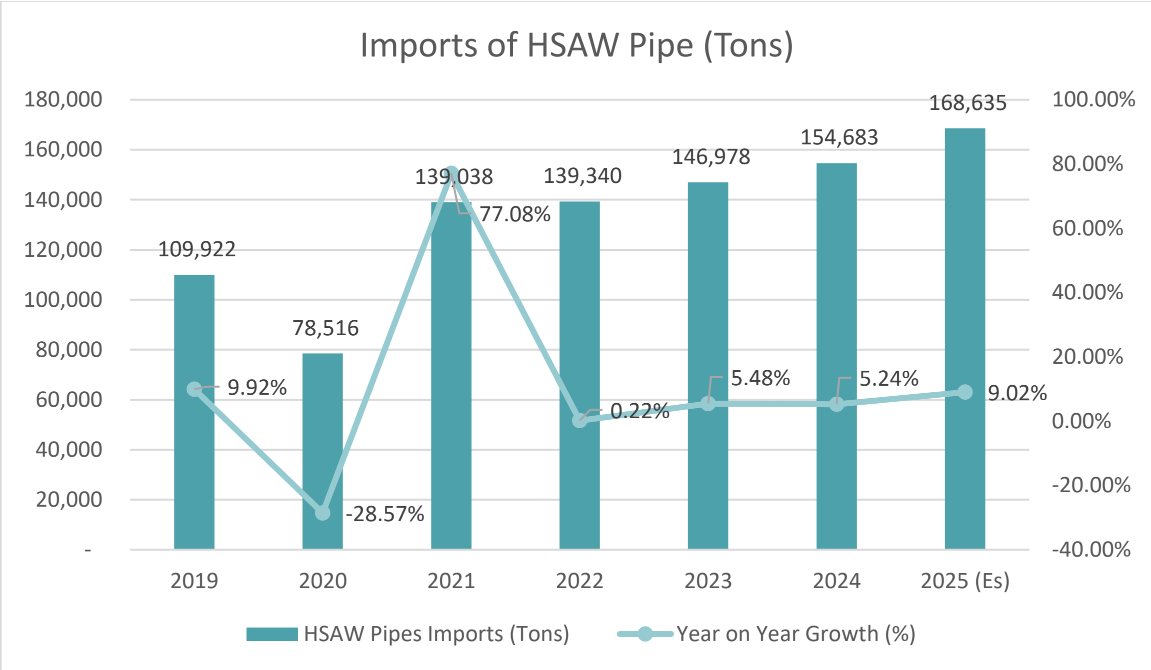
Source: Trade Map, Sector Reports and SCPL Research

It is noted from the exhibit above, that the production levels in HSAW Pipes industry saw a dip during the year 2020, which was on account of CoVid-19 pandemic. Since then, there has been a revival in the production of HSAW Pipes in the KSA, which is evident from the exhibit above, with capacity utilisation reaching above 45% level during CY 2022 and crossing 50% level during CY 2024.

Imports and Exports

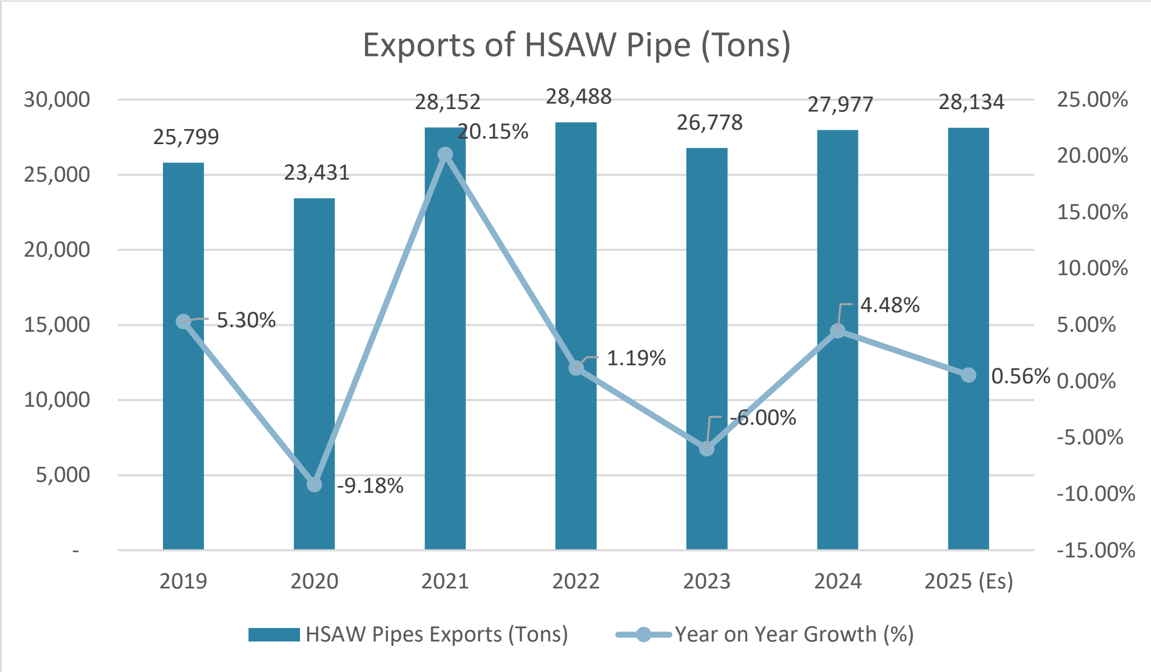
Even though there has been a marked move towards Saudization, there have been imports of HSAW Pipes in the Kingdom of Saudi Arabia and the same saw an upward trend post pandemic. This is on account of the fact that the declared projects as announced faced delays on account of slowdown in the global markets, during the pandemic season and in KSA as well. To make up for the time lost, the customers like Aramco and SWCC have placed some order abroad. The difference in the demand and production at the present moment is being fulfilled by imports of HSAW Pipes from countries like China, India, Korea and Turkey etc.

The imports of HSAW Pipes in KSA stood at 109,922 Tons during the year 2019 and since then the imports have grown at a CAGR of 7.53% to reach an estimated 168,635 Tons during the current CY 2025. The growth in imports of HSAW pipes in the region during the period 2019-25 has been presented in the exhibit below –



Source: Trade Maps and SCPL Research

The exports of HSAW Pipes from KSA stood at 25,799 Tons during the year 2019. Since then, the exports of HSAW Pipes from the region have grown at a CAGR of 1.45% to reach an estimated 28,124 Tons during the current CY 2025. The exhibit below illustrates the growth of exports of HSAW pipes from the region during the period 2019-2025.



Source: Trade Maps and SCPL Research

Demand Supply Gap Scenario

Based on the discussion in the sections above, it is noted that a series of oil and gas and non-oil and gas pipeline project had been announced in the country almost immediately after the regime change. The projects as perceived at that point of time got delayed by over 2 years on account of the pandemic scenario, followed by the Ukraine conflict. The Projects have recommenced and the gap in the demand has been fulfilled by imports of HSAW Pipes from various destination.

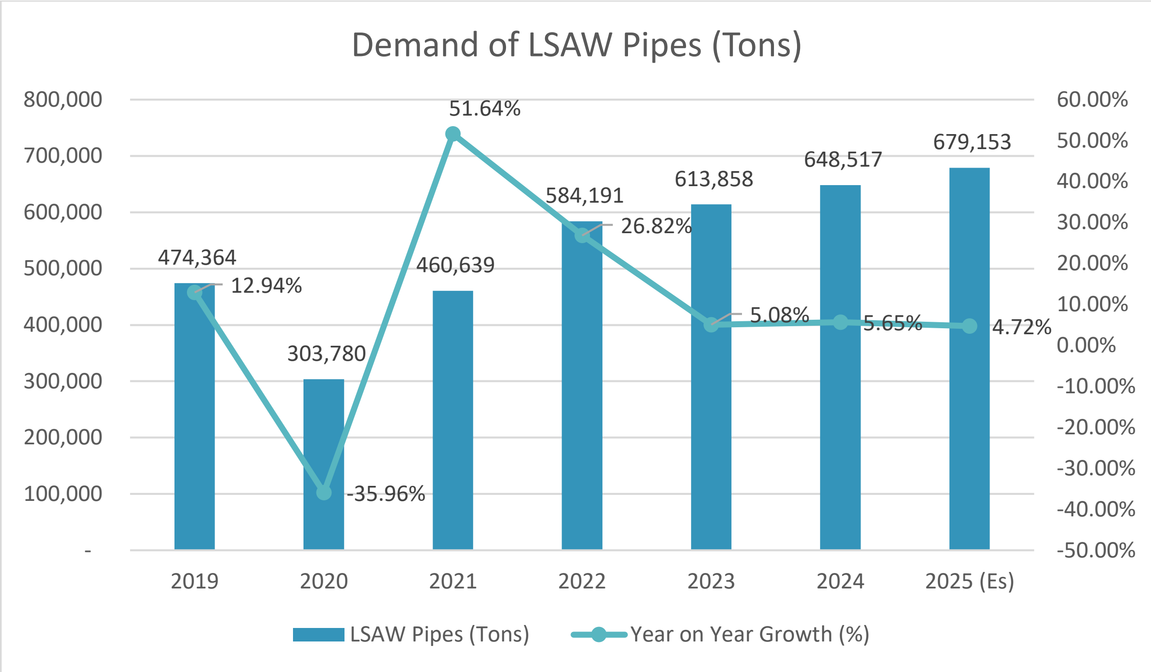
Further it is noted that the focus on Saudization recommenced from the CY 2024 and the imports are expected to reduce, thereby increasing the demand supply gap and this gap will have to be fulfilled by increasing production levels. Hence local players are expected to receive more order from players like Aramco, SWCC, Water Transportation Department etc.

LSAW Pipes Market

Demand

As discussed above, after the change in regime in Saudi Arabia, there has been a series of infrastructure projects announcements both in oil and gas sectors and in non-oil and gas sectors in Saudi Arabia. These include projects for oil and gas pipeline to the tune of 5,354 Km and these are the main demand drivers of the LSAW Pipes in the country.

The demand for LSAW Pipes in KSA stood at around 474,364 Tons during the year 2019. Since then, the demand for HSAW Pipes in KSA has grown at a CAGR of 6.16% to reach an estimated 679,153 Tons during the current CY 2025. The growth in demand for the LSAW Pipes in the country during the period 2019 to 2025 has been illustrated in the exhibit below –



Source: Sector Reports and SCPL Analysis

In line with the demand for HSAW Pipes in the country, the demand for LSAW Pipes in KSA saw a slow down during the pandemic year 2020. Since then, the demand for LSAW Pipes in KSA has recovered and is expected to further increase with the recently announced oil and gas pipeline projects as announced in the country.

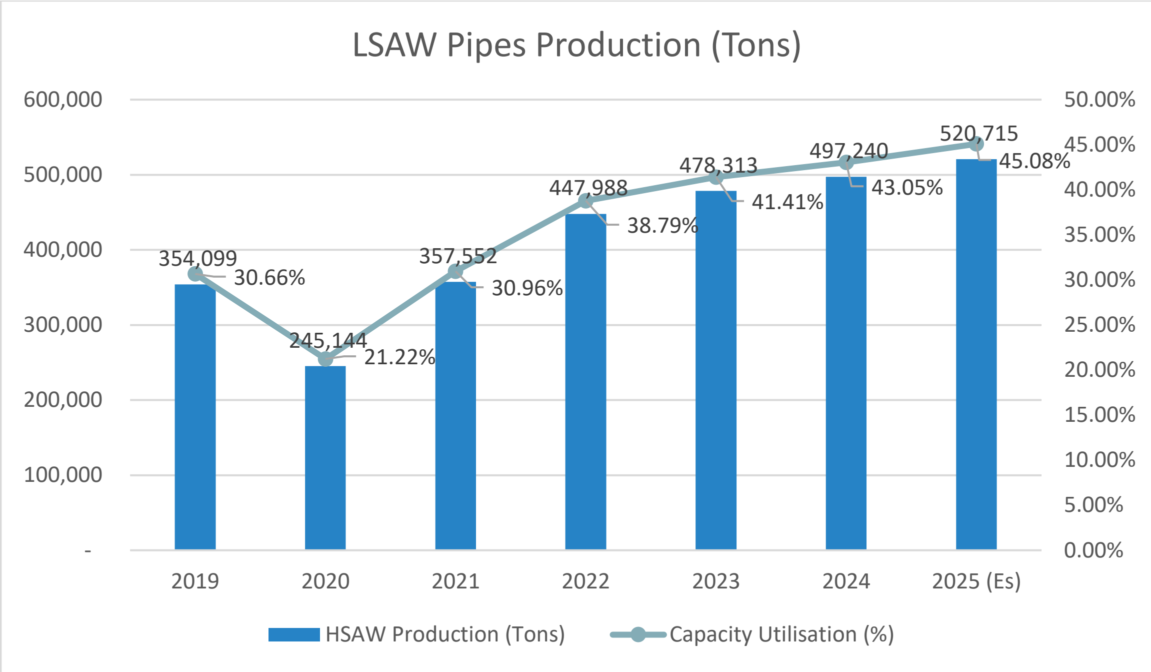
Installed Capacities and Production

Based on the data available in secondary domain, limited primary market assessment undertaken by the Consultants and reports published by reputed consultancy firms, the Consultants understand that there are a total of 4 operational players of LSAW Pipes in the region and these include the following –

1. National Pipe Company (KSA) – 180,000 TPA
2. Global Pipe Company (KSA) – 375,000 TPA
3. Arabian Pipe Company (KSA) – 300,000 TPA
4. PCK (KSA) – 300,000 TPA

Considering the above, the total installed capacity of HSAW Pipes in the region stands at 1.16 million Tons. However, here it should also be noted that one of the consignments of Global Pipes Company was rejected by Aramco during the year 2023, because of non-compliance as the per standards. The GPC plant is expected become operational again by end of CY 2025 and till then the operational capacity of the country in LSAW Pipes segment will remain at 0.78 million Tons, subsequent to which the capacity in the country will again reach 1.16 million Tons.

The production of LSAW Pipes in KSA stood at 354,099 Tons during the year 2019. The production of LSAW Pipes in the country saw a growth of CAGR 6.64% during the period 2019-25, to reach an estimated 520,715 Tons during the current CY 2025. The growth in production of LSAW Pipes in the country during this period and corresponding utilisation level has been presented in the exhibit below –

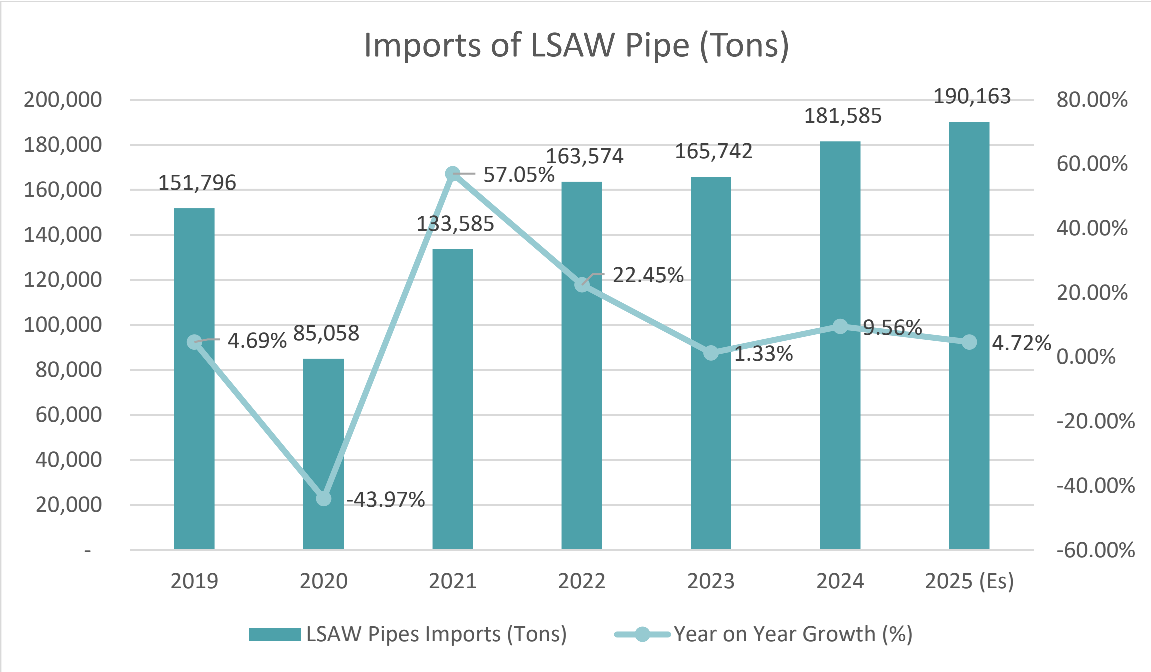


Source: Trade Map, Sector Reports and SCPL Research

It is noted from the exhibit above, that the production levels in HSAW Pipes industry saw a dip during the year 2020, which was on account of CoVid-19 pandemic. As evident from the from the exhibit above, the production of LSAW Pipes in the country has picked up from CY 2022, with utilisation level reaching about 40% level, further improving to 45% utilisation level in current year 2025.

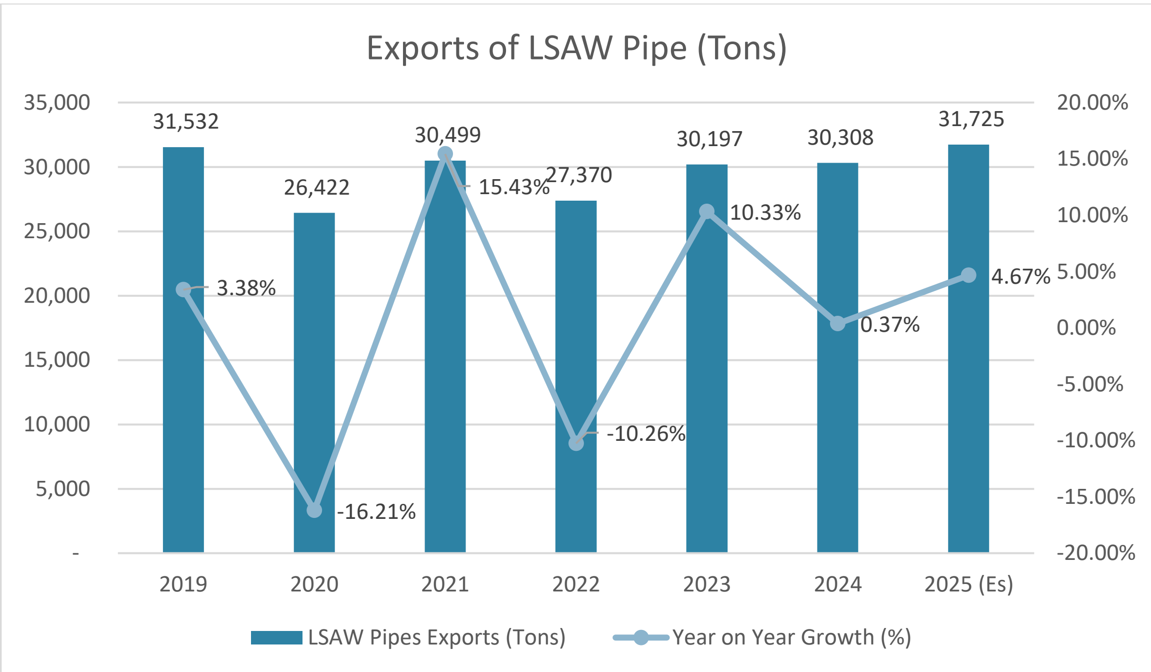
Imports and Exports

The imports of LSAW Pipes in the kingdom stood at 151,796 Tons during the year 2019 and since then the imports have grown at a CAGR of 3.82% to reach an estimated 190,163 Tons during the current CY 2025. The growth in imports of HSAW pipes in the region during the period 2019-25 has been presented in the exhibit below –



Source: Trade Maps and SCPL Research

The exports of LSAW Pipes from KSA stood at 31,532 Tons during the year 2019. Since then, the exports of LSAW Pipes from the region have grown at a CAGR of 0.10% to reach an estimated 31,725 Tons during the current CY 2025. The exhibit below illustrates the growth of exports of HSAW pipes from the region during the period 2019-2025.



Source: Trade Maps and SCPL Research

Demand Supply Gap Scenario

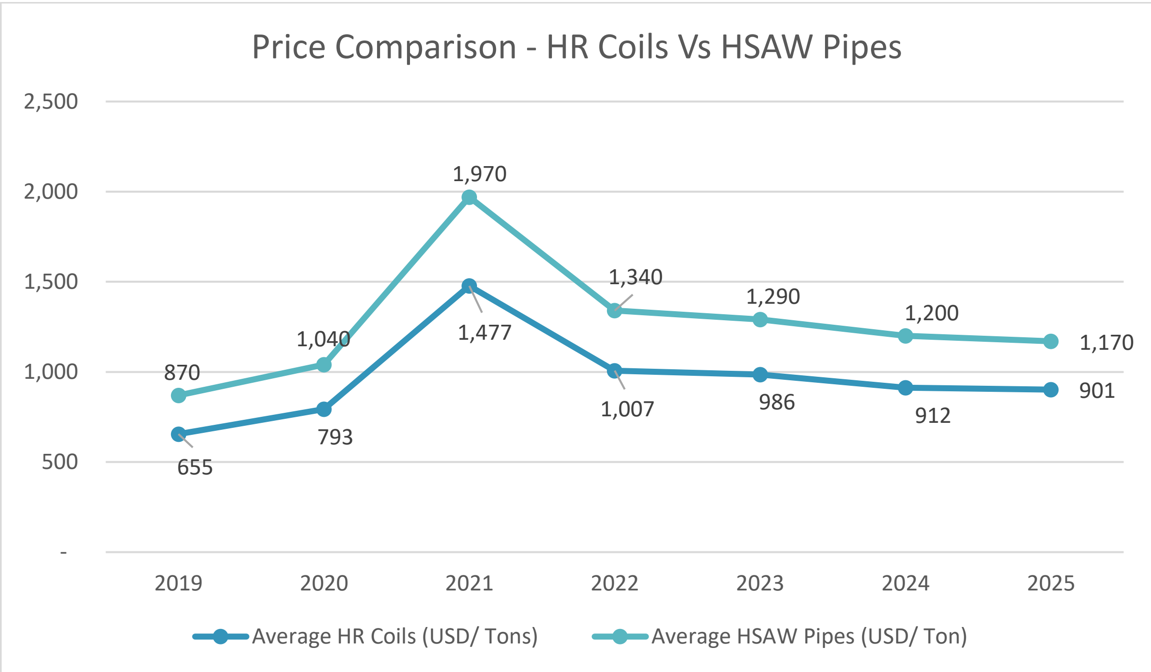
Based on the discussion in the sections above, it is noted that a series of oil and gas had been announced in the country almost immediately after the regime change. The projects as perceived at that point of time got delayed by over 2 years on account of the pandemic scenario, followed by the Ukraine conflict. The Projects have recommenced and the gap in the demand has been fulfilled by imports of LSAW Pipes from various destination.

Further it is noted that the focus on Saudization recommenced from the CY 2024 and the imports are expected to reduce, thereby increasing the demand supply gap and this gap will have to be fulfilled by increasing production levels. Hence local players are expected to receive more order from players like Aramco, SWCC, Water Transportation Department etc.

Prices

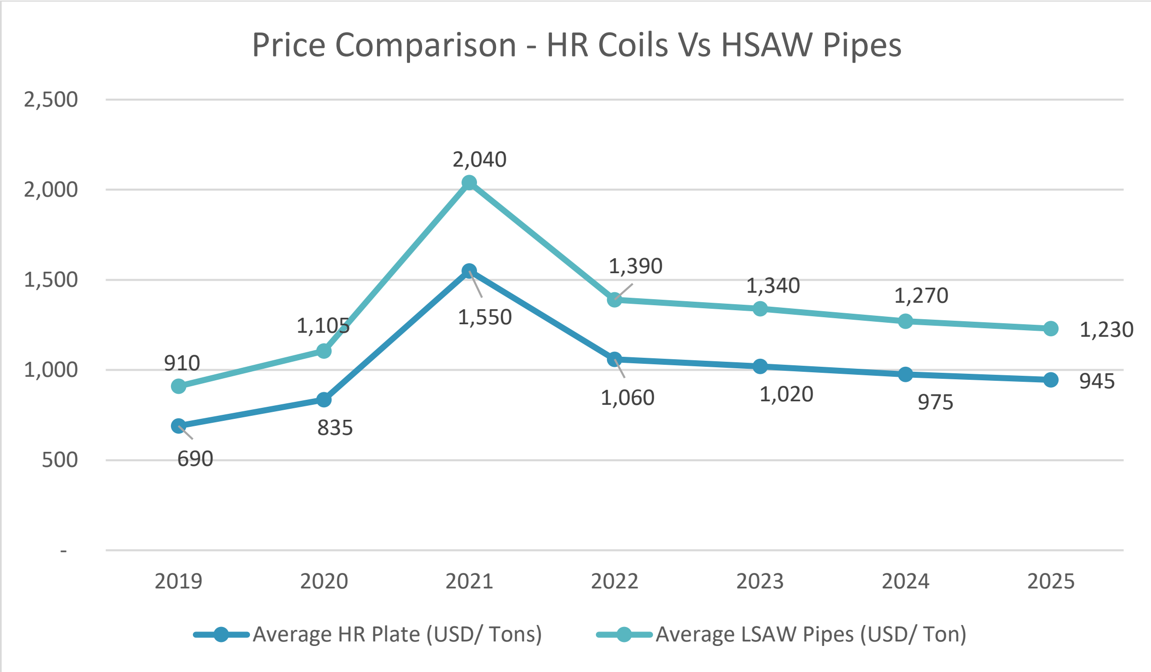
The Consultants note that the major cost component for manufacturing of HSAW and LSAW Pipes are the raw material prices i.e. HR Coils and HR Plates respectively. The raw material prices account for almost 80-82% of the finished goods selling prices. Further, it is noted that most of the contracts are fixed price contracts, hence the SAW Pipes manufacturers on receiving a contract, enter in bulk purchase contracts or long term contracts with raw material suppliers.

The exhibit below provides the historical trend in prices of HR Coils vis-a-vis HSAW Pipes.



Source: Trading Economics and SCPL Research

Similarly, the prices of HR Plate vis-s-vis the prices of LSAW Pipes in the international markets has been presented in the exhibit below –



Source: Trading Economics, Steel Mint and SCPL Research

Government Policies and Initiatives

The Government of Kingdom of Saudi Arabia has announced the National Industrial Strategy (NIS) at the end of 2022, which is roadmap for accelerating the diversification of its industrial base to increase non-oil exports, encourage privatization, attract more foreign investment, increase investment in innovation and research, and create local jobs. The strategy prioritizes 118 segments within 12 main industrial sub-sectors chosen for the country's ability to compete in these sectors at the regional and global levels.

The NIS seeks to achieve three main strategic objectives –

1. Build a flexible national industrial economy capable of adapting to changes.
2. Form an integrated regional industrial center to meet demand.
3. Achieve global leadership in manufacturing a group of selected industrial commodities.

The NIS identified four objectives to achieve the 3 strategic objectives:

1. Build and strengthen supply chains that are characterized by reliability, high productivity, advanced manufacturing, and the flow of goods and services at a competitive cost.
2. Develop the industrial business ecosystem by improving its legislative and financing environment, increasing contribution of small and medium enterprises (SMEs), and encouraging local content.
3. Enhancing international trade by opening access to new markets for industrial investors and facilitating the export process for local companies.
4. Develop and promote a culture of innovation and knowledge by attracting and retaining talent and fostering a culture of innovation in the industrial sector.

Local Content Development

A pivotal aspect of the National Industry Strategy is to increase the level of locally manufactured products and services. The Kingdom has set an ambitious target of raising local content in goods and services to 60% by 2025.

This would not only create job opportunities but also enhance economic resilience. Moreover, the strategy emphasizes attracting and retaining foreign investment through incentives and localized production requirements.

Promoting Saudi Industries on the Global Stage

Under the National Industry Strategy, Saudi Arabia aims to position its industries as global leaders. To achieve this, the Kingdom is investing in research and development, innovation, and advanced technologies. The strategy also encourages collaboration among academia, research centers, and industrial enterprises to promote knowledge transfer and develop cutting-edge solutions. By fostering a culture of innovation, Saudi industries are primed to compete on an international level and create high-value job and business opportunities.

Investment Initiatives: Driving Industrial Growth

To stimulate industrial growth, the Saudi government has launched several investment initiatives. For instance, the Industrial Development Fund (IDF) supports local industries by providing financing, while the Saudi Industrial Property Authority (MODON) develops industrial cities and provides infrastructure. Additionally, specialized economic cities, such as King Abdullah Economic City (KAEC) and NEOM, are being established, creating opportunities for both local and foreign investors.

The NIS seeks to achieve the following economic impacts:

1. Industrial sector contribution to GDP: increase from USD 88.26 billion in 2020 to USD 377.06 billion by 2035.
2. Industrial sector job opportunities: increase from 900,000 in 2020 to 3.3 million by 2035.
3. Value of industrial exports: increase from USD 45.06 billion in 2020 to USD 237.86 billion by 2035.
4. Competitiveness Industrial Performance Index and Economic Complexity Index: be in the top 15 countries by 2035.
5. Localization rate: increase from 41% to 64% in 2035.

The sectors primarily targeted by the NIS are chemicals/specialty chemicals, plastics and rubber conversion, food/agriculture, renewable energy, aviation (civil and military), automobiles (including electric vehicles), maritime manufacturing, pharmaceuticals, medical devices and supplies, military manufacturing, construction materials, machinery/equipment, and mining/minerals.

The NIS will be overseen by several authorities, including the Council of Economic and Development Affairs, the Supreme Committee of Industry (under direct supervision of the Crown Prince), the National Industrial Development and Logistics Program, and the Ministry of Industry and Mineral Resources.

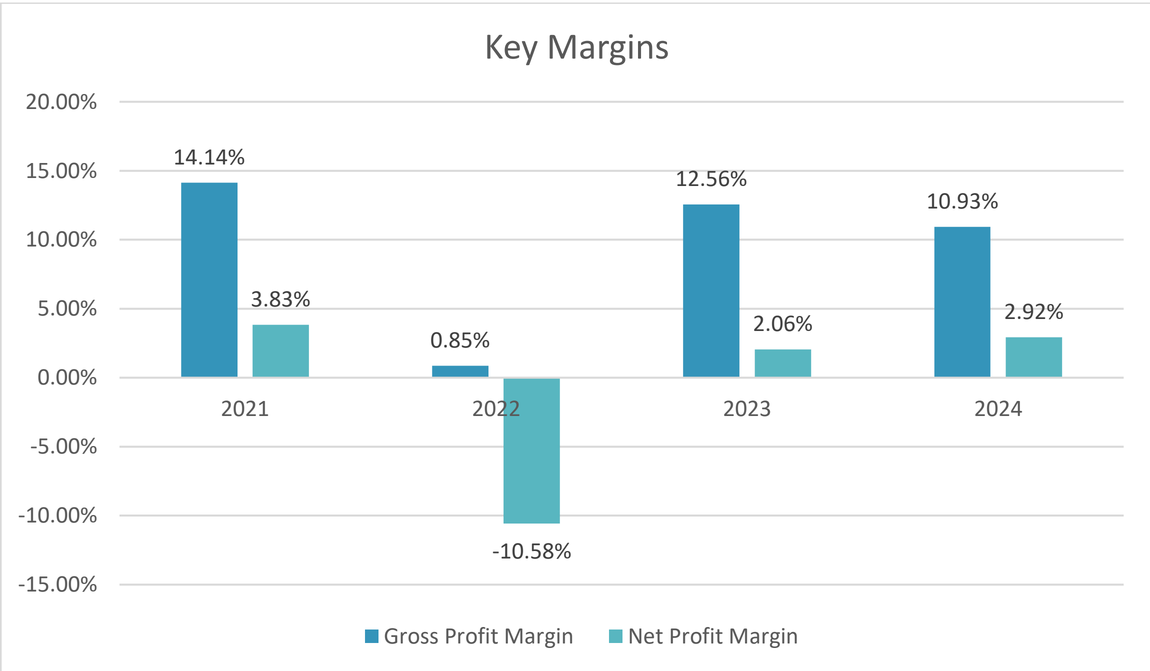
Competitive Landscape

There are some competitive players in the same line of business, the details of which have been penned below –

Group Five Pipe Company

In 2001 Group Five Pipe Saudi (G5PS) was established as a Joint between South-African companies Group Five Construction, Marine Civil and Al-Qahtani Group of Companies. The G5PS is located at Dammam, Second Industrial City, Kingdom of Saudi Arabia. G5PS is the leading HSAW pipe providers in Saudi Arabia and Middle East. It has five (5) production lines that has an annual capacity of 570,000 metric tons of H-SAW pipe productions. G5PS has 2 decades of experience in providing major projects in the Middle East particularly in the Kingdom of Saudi Arabia.

- The unit of G5PS is located at Dammam on a plot area of over 350,000 Sq. M.
- There are more the 536 employee from a diverse nationalities.
- The pipe making technologies for HSAW Pipe is "Conventional Forming Method"
- Range of pipes manufactured –
    - OD (Inches) – 20" to 132"
    - WT (mm) – 6.35 to 25.40
    - Length – Max 24 Meters
- The key margin of the Company as per data available in secondary domain has been presented in the exhibit below –



Source: Bloomberg, Wall Street Journal, www.investing.com, www.barron.com and SCPL Research

Quality Certificates

- API certification.

- ISO 9002.
- API Spec Q1: Pipe

Clients

- Aramco
- SWCC
- Transcanada

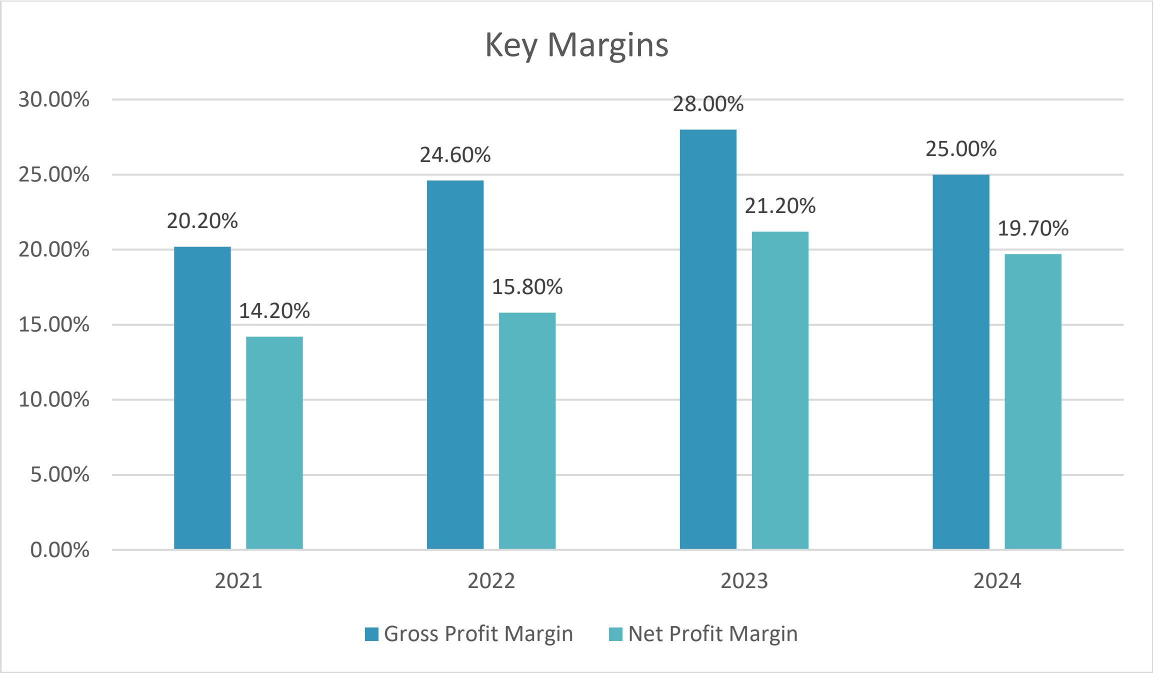
East Pipes Integrated Company ('East Pipes') or Welspun

The Company's journey commenced with the establishment of "Pipe Development Company for Pipe Production" in Dammam, Saudi Arabia, on May 3, 2010, evolving into a shareholding limited liability entity. In a transformative move in November 2010, the partners increased the Company's capital and renamed it "Welspun Middle East Pipes Company." Over time, the Company underwent significant development, marked by the merger of "Welspun Pipes Coating Company" in July 2020, leading to its current identity as "East Pipes Integrated Company for Industry." This strategic shift, from a limited liability to a joint stock company in September 2020, solidified its prominent status in the Kingdom and the region, specializing in the production and coating of spiral-welded steel pipes (18 to 120 inches in diameter) and cement-coated pipes and fittings. The Company excels in manufacturing HSAW pipes, particularly valued for their large diameters, catering to water, oil, and gas sectors.

The HSAW Plant is one of the largest spiral pipe manufacturing facilities in the region equipped with 2-Step/ Conventional forming process, with an installed capacity of 450,000 MT per annum with capability to manufacture large outer diameter ranging from 20" to 100" inches of H-SAW pipe.

Along with the H-SAW Plant it has a coating plant as well, The Coating Plant was set up in 2011 as a standalone unit; it was then combined with the HSAW Plant to provide external anti-corrosion coating through a triple layer of polyethylene, polypropylene, or high-performance epoxy primer. The nominal production capacity of the Coating Plant is 4,000,000 sqm per annum. The capability of the Coating Plant ranges from an outer diameter of 2 inches to 120 inches and a length of up to 26 meters pipes.

- The manufacturing facility of the Company is located in Dammam on a plot area of about 400,000 Sq. M.
- There are more the 490-employee working under law of the company.
- The pipe making technologies for HSAW Pipe is "2 Step Process and Conventional Forming Method"
- Range of pipes manufactured –
    - OD (Inches) – 20" to 100"
    - WT (mm) – 6.35 to 25.40
    - Length – Max 18 Meters
- The Key Margin of the Company as available in secondary domain have been summarised below –



Source: Bloomberg, Wall Street Journal, www.investing.com, www.barron.com and SCPL Research

Quality Certificates

- API 5 L: Pipe.
- API Spec Q1: Pipe.
- ISO 9001: Pipe.
- ISO 14001: Pipe.
- ISO 18001: Pipe.
- ISO 14001: Coating.
- ISO 18001: Coating.
- DAC coating & Spiral Coating.

Clients

- Transcanada.
- Kindermorgan.
- Texas Gas.
- Hunt Oil.
- Saudi Aramco.
- Exxon Mobil.
- Qatar Petroleum.

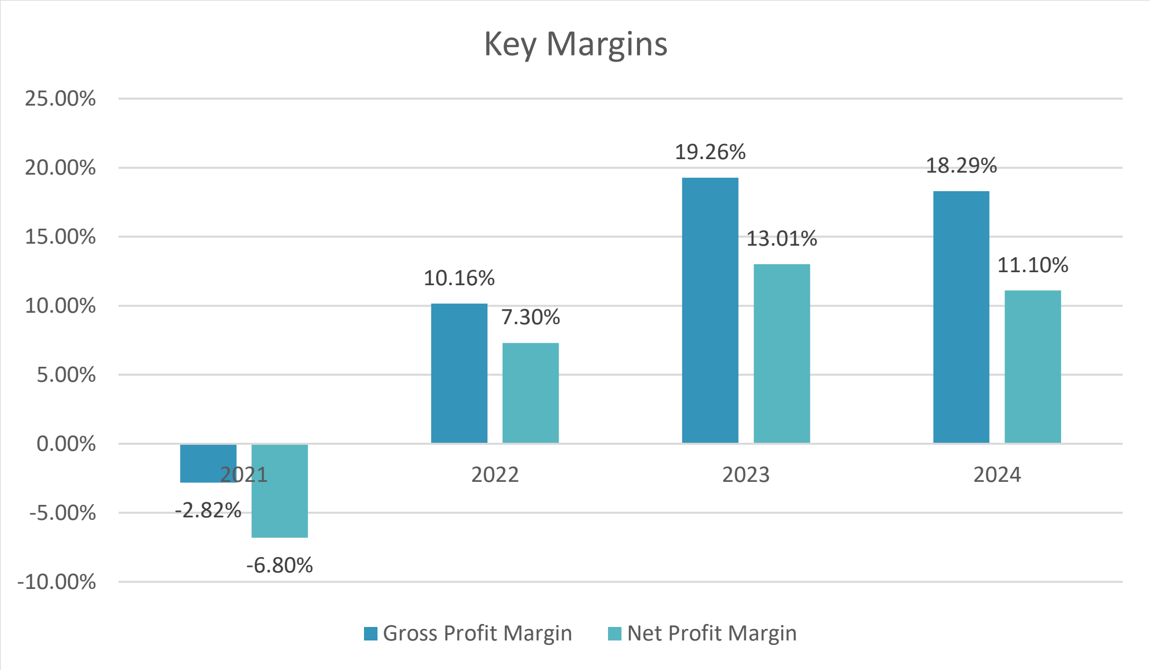
Global Pipe Company ('GPC')

GPC is a Saudi-German joint venture with a total investment exceeding SAR 70 million, established in 2010 at PO Box No. 24, Jubail Industrial City, Kingdom of Saudi Arabia. GPC is a leading manufacturer of LSAW pipes in MENA region for oil, gas, petrochemical, OCTG, power and civil engineering industries. It is the first manufacturer of heavy walled L-SAW pipe with wall thickness of upto 50.80 mm in the Middle East Region. Also, it has the State of Art manufacturing equipment as well as forming process like (JCOE, 3RB/ 4RB) with fully equipped laboratories. The combined installed capacity of the GPC through all forming process for manufacturing L-SAW pipe is about 375,000 MT per annum.

***The Consultants note that one large consignment of LSAW Pipes manufactured by GPC was rejected on quality basis by Aramco during the year 2023. Subsequent to that the plant has been under minimum production and in semi closed down state. The plant is expected to reattain its full operational levels by end of 2025, by which it would have reverified all quality standards and regained approval from Aramco*.**

Further, it is noted that Saudi Steel Pipe Company Limited, acquired around 57.37% stake in Global Pipe Company, during the early CY 2024 and the updated financial information of the Company is now available as consolidated financials of Saudi Steel Pipe Company Limited and the same has been presented in sections below.

- The manufacturing facility of GPC is located at Jubail Industrial Area in Saudi Arabia, where the Company has acquired 120,000 Sq. M. of plot Area
- There are around 250 employee from a diverse nationalities.
- The pipe making technologies for LSAW Pipe are "JCOE and 4RB and 3RB"
- Range of pipes manufactured –
    - OD (Inches) – 18" to 62"
    - WT (mm) – 6.35 to 50.80
    - Length – Max 12 Meters
- The Key Margin of the Company as available in secondary domain have been summarised below –



Source: Bloomberg, Wall Street Journal, www.investing.com, www.barron.com and SCPL Research

Quality Certificates

- API certification.
- ISO 9002.
- API Spec Q1: Pipe

Clients

- Aramco

Arabian Pipes Company ('APC')

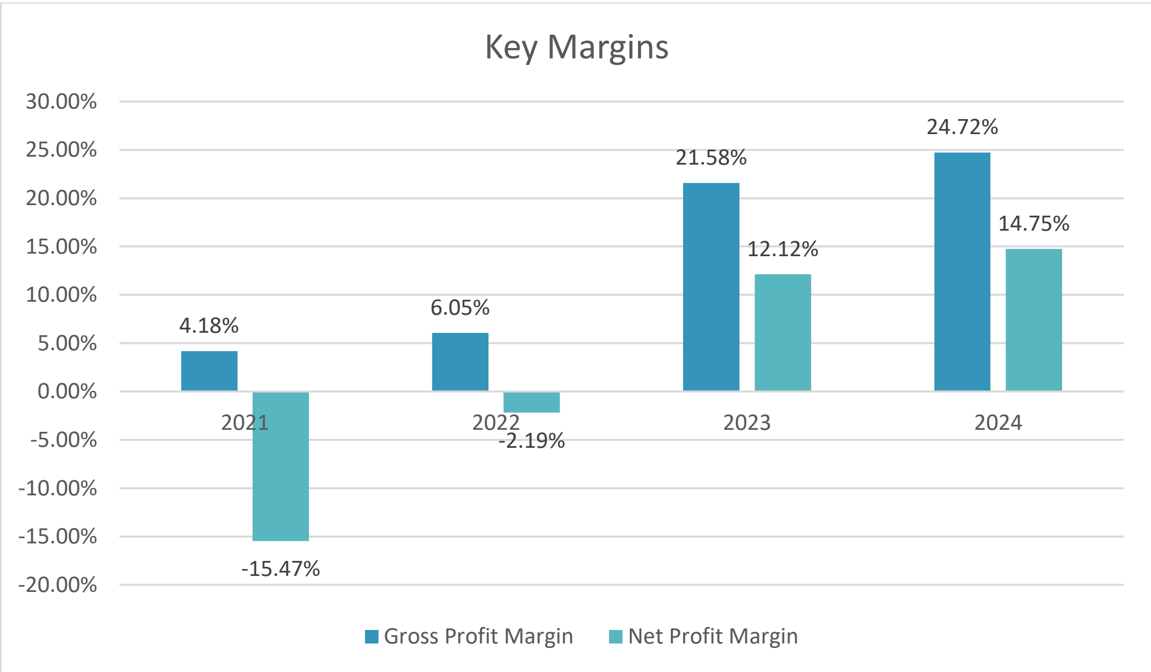
Arabian Pipes Company ("the Company") is a Saudi Joint Stock Company formed in accordance with the Companies Regulation and is registered in the Kingdom of Saudi Arabia under the Commercial Registration No. 1010085734 dated 25 August 1991. Arabian pipes co. is manufacturing and marketing of welded steel pipes for Oil and Gas, Structural and commercial utilization with sizes range from 6 to 48 inch. The company owns and operates two factories, the first plant in Riyadh with average capacity of 160,000 MT per annum of ERW pipes with sizes range from 6 to 20 inch and the second plant in Jubail industrial city with average capacity of 300,000 MT per annum of L-SAW pipes with sizes range from 16 to 48 inch to reach a total average capacity to 460,000 MT per annum. Also the company has threading line for casing pipes, a coating facility and slitting line for steel coils.

APC is capable of manufacturing L-SAW pipe with a wall thickness 6.35 mm to 25.40 mm, with a length of upto 12 meters by UOE forming process. Also to ensure good quality result of APC product, use advanced, and effective laboratory testing equipments (calibrated by third party) and conform to international standards such as API, ASTM, IOS, NACE, SAMSS, Shell, and the most valuable client specification.

Mechanical testing done by - metal analysis spectrometer, Zwick universal testing machine, Vickers hardness testing, Leica microscope, struers micro polishing and mounting machine, corrosion test - HIC and SSCC test.

Key financial of ACP are as follows –

- The ERW Pipes division of APC is located at Riyadh on a plot of size 124,000 Sq. M., while the LSAW Pipe facility of APC is located at Jubail on a plot of size 220,000 Sq. M.
- The Company employees a total manpower of close to 410 personnel of diverse nationality
- The pipe making technologies for LSAW Pipe is "UOE"
- Range of pipes manufactured –
  - OD (Inches) – 16" to 48"
  - WT (mm) – 6.35 to 25.40
  - Length – Max 12 Meters
- The Key Margin of the Company as available in secondary domain have been summarised below –



Source: Bloomberg, Wall Street Journal, www.investing.com, www.barron.com and SCPL Research

Quality Certification

- API Certification
- ISO9001,
- ISO14001,
- OHSAS18001 and
- ILAC

Clients

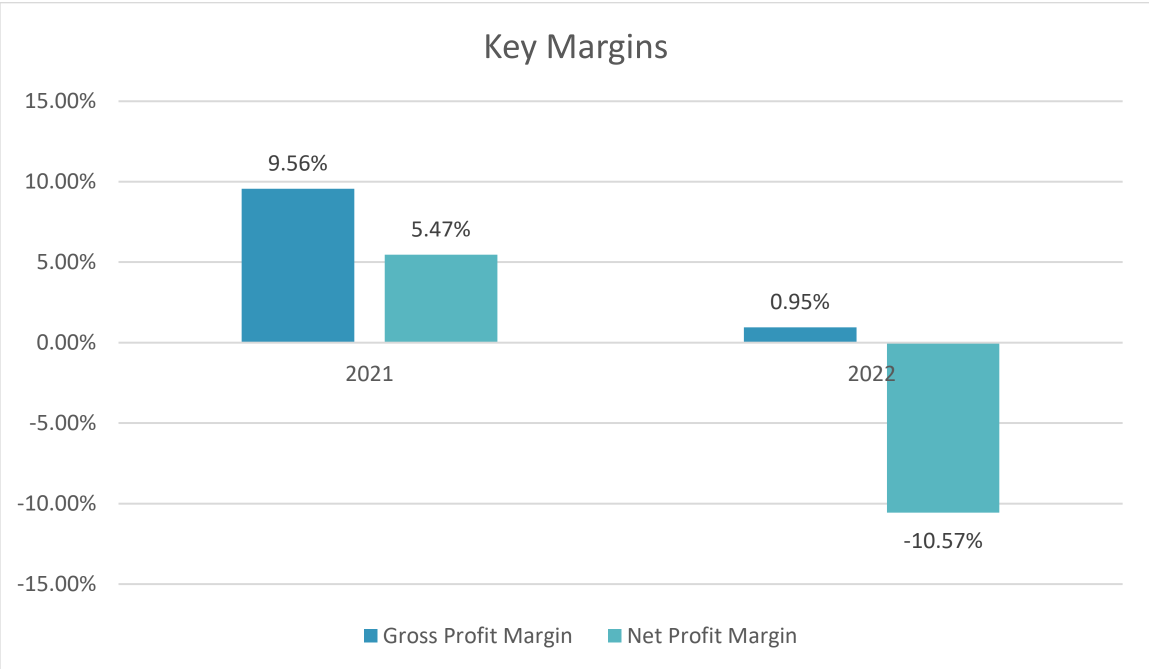
- Aramco

Al Qahtani PCK Pipe Company

Al Qahtani PCK Pipe Company (AQPCK) is a Saudi-Chinese Joint Venture Company formed in 2023 between Abdel Hadi A. Al Qahtani & Sons Group (AHQ), Saudi Arabia & Panyu Chu Kong Steel Pipe Co., Ltd (PCK) China, located at P.O. Box 7465, Dammam 31462 Kingdom of Saudi Arabia. AQPCK aims to become LEADING manufacturer in producing longitudinal submerged arc welded (LSAW) pipes and high frequency welded (HFW) pipes for oil, gas, water, petrochemical, power generating and structure industries. The LSAW Pipe manufacturing nominal capacity of the Company is 300,000 TPA.

Key financial of ACP are as follows –

- The plant of PCK is located at Dammam on a plot area of around 250,000 Sq. M.
- The Company employee around 375 permanent employees of diverse nationality
- The pipe making technologies for LSAW Pipe is “JCOE”
- Range of pipes manufactured –
    - OD (Inches) – 16” to 48”
    - WT (mm) – 6.35 to 25.40
    - Length – Max 12 Meters
- The Key Margin of the Company as available in secondary domain have been summarised below –



Source: Bloomberg, Wall Street Journal, www.investing.com, www.barron.com and SCPL Research

Since PCK is not a listed company, the data available on PCK is limited and the most updated data has been presented in the exhibit above.

Quality Certification

- API Certification
- ISO9001,
- ISO14001,
- DNV,
- ASTM

Clients

- Aramco

Conclusions

The Consultants conclude the following from the discussions in the chapter –

- The demand for HSAW Pipes in the region stood at 894,704 Tons during the year 2019. Since then, the demand for HSAW Pipes in the GCC region has grown at a CAGR of 5.89% to reach an estimated 1,261,433 Tons during the current CY 2025.
- The total installed capacity of HSAW Pipes in the GCC region stands at 1.56 million Tons.
- The production of HSAW Pipes in the region stood at 663,748 Tons during the year 2019 and since then the production of HSAW Pipes in the region has grown at a CAGR of 6.41% to reach and estimated 929,654 Tons during the current CY 2025.
- The imports of HSAW Pipes in the region stood at 273,600 Tons during the year 2019 and since then the imports have grown at a CAGR of 6.41% to reach an estimated 397,191 Tons during the current CY 2025.
- The exports of HSAW Pipes from the region stood at 42,644 Tons during the year 2019. Since then, the exports of HSAW Pipes from the region have grown at a CAGR of 7.39% to reach an estimated 65,412 Tons during the current CY 2025.
- Based on the review of the data available in secondary domain and limited primary market assessment as undertaken by the Consultants, it is understood that the demand leads the production in the GCC region and the GCC region is a net importer of HSAW Pipes.
- The demand for LSAW Pipes in the region stood at 837,702 Tons during the year 2019. Since then, the demand for LSAW Pipes in the GCC region has grown at a CAGR of 5.18% to reach an estimated 1,134,388 Tons during the current CY 2025.
- The total installed capacity of LSAW Pipes in the region stands at 1,605,000 million Tons.
- The production of LSAW pipes in the region stood at 549,499 Tons during the year 2019. Since then, the production of LSAW Pipes in the region has grown at CAGR of 6.69% to reach an estimated 772,180 Tons during the current CY 2025.
- The imports of LSAW Pipes in the region stood at 334,401 Tons during the year 2019 and since then the imports have grown at a CAGR of 4.29% to reach an estimated 430,290 Tons during the current CY 2025.
- The exports of LSAW Pipes from the region stood at 46,198 Tons during the year 2019. Since then, the exports of LSAW Pipes from the region have grown at a CAGR of 6.68% to reach an estimated 68,082 Tons during the current CY 2025.

- Based on the market assessment undertaken by the Consultants, it is understood that the demand for LSAW Pipes in the GCC is outruns the production of LSAW Pipes in the region and that the region is a net importer of LSAW Pipes.
- The demand for HSAW Pipes in KSA stood at around the demand for HSAW Pipes in the region stood at 578,535 Tons during the year 2019. Since then, the demand for HSAW Pipes in KSA has grown at a CAGR of 5.62% to reach an estimated 803,024 Tons during the current CY 2025.
- There are 3 operational players in the HSAW Pipes segment in KSA with total installed capacity of 1.27 million Tons per annum.
- The production of HSAW Pipes in KSA stood at 494,412 Tons during the year 2019. The production of HSAW Pipes saw a slight dip during the year 2020, which was on account of the global pandemic scenario. The production of HSAW Pipes in the country saw a growth of CAGR 5.00% during the period 2019-25, to reach an estimated 662,552 Tons during the current CY 2025.
- The imports of HSAW Pipes in KSA stood at 109,922 Tons during the year 2019 and since then the imports have grown at a CAGR of 7.53% to reach an estimated 168,635 Tons during the current CY 2025.
- The exports of HSAW Pipes from KSA stood at 25,799 Tons during the year 2019. Since then, the exports of HSAW Pipes from the region have grown at a CAGR of 1.45% to reach an estimated 28,124 Tons during the current CY 2025.
- The demand for HSAW Pipes in KSA Lead the production of HSAW Pipes in the Kingdom. The country remains a net importer of HSAW Pipes as off now.
- The demand for LSAW Pipes in KSA stood at around 474,364 Tons during the year 2019. Since then, the demand for HSAW Pipes in KSA has grown at a CAGR of 6.16% to reach an estimated 679,153 Tons during the current CY 2025.
- There are 4 LSAW Pipe manufacturers located in KSA, with total installed capacity of 1.16 million TPA. However, with semi-closure of GPC's plant, the actual operational capacity of the kingdom stands at 0.78 million TPA.
- The production of LSAW Pipes in KSA stood at 354,099 Tons during the year 2019. The production of LSAW Pipes in the country saw a growth of CAGR 6.64% during the period 2019-25, to reach an estimated 520,715 Tons during the current CY 2025.
- The imports of LSAW Pipes in the kingdom stood at 151,796 Tons during the year 2019 and since then the imports have grown at a CAGR of 3.82% to reach an estimated 190,163 Tons during the current CY 2025.
- The exports of LSAW Pipes from KSA stood at 31,532 Tons during the year 2019. Since then, the exports of LSAW Pipes from the region have grown at a CAGR of 0.10% to reach an estimated 31,725 Tons during the current CY 2025.
- Based on the market assessment undertaken by the Consultants, it is understood that the demand leads the production of LSAW Pipes in the Kingdom and that KSA is a net importer of LSAW pipes at the moment.
- The Consultants note that the major cost component for manufacturing of HSAW and LSAW Pipes are the raw material prices i.e. HR Coils and HR Plates respectively. The raw material prices account for almost 80-82% of the finished goods selling prices.

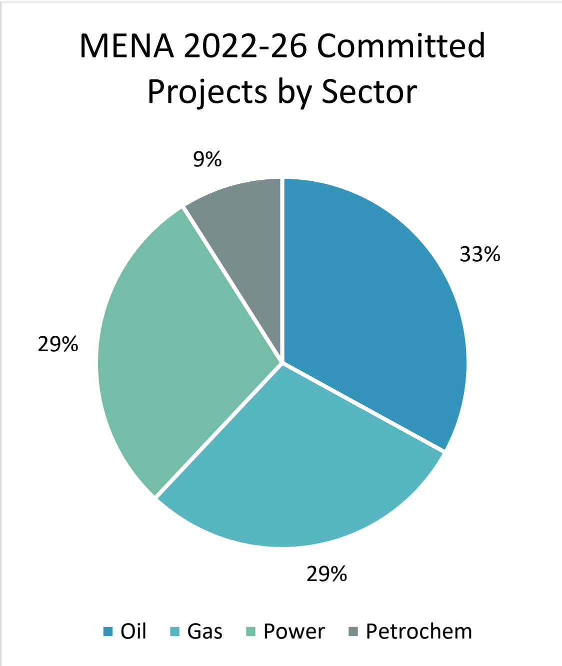
Future Demand Supply Scenario

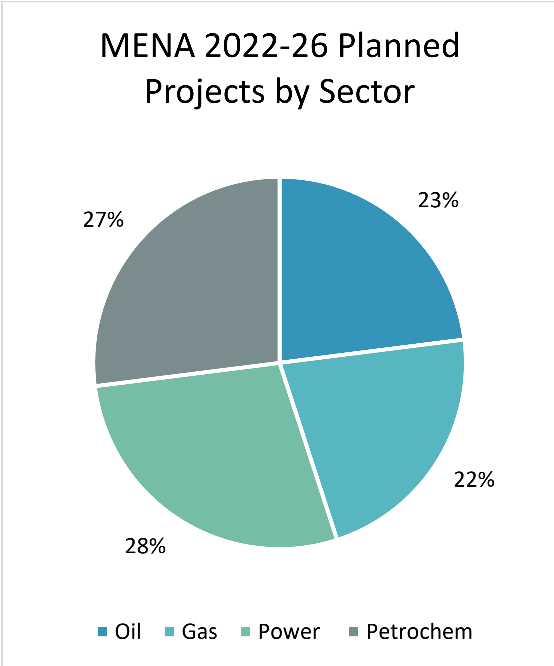
MENA/ GCC Region

Demand Drivers

As per the revised 5 year outlook data issued by the MENA region during the year 2022, there is a total anticipated investment of USD 879 billion in the region, up from USD 805 billion as per data released in 2021. Of this total, committed projects (i.e. projects which have entered the execution stage) make up around 30%, while planned projects (i.e. projects still in the planning phase) constitute the remaining 70%.

The increase in project expenditure is spearheaded by the GCC, with committed projects making up more than 45% of the Gulf states' total energy investments thanks to the windfall from oil and gas export revenues. The exhibit below provides a sector wise snapshot of committed and planned project in the entire MENA region.



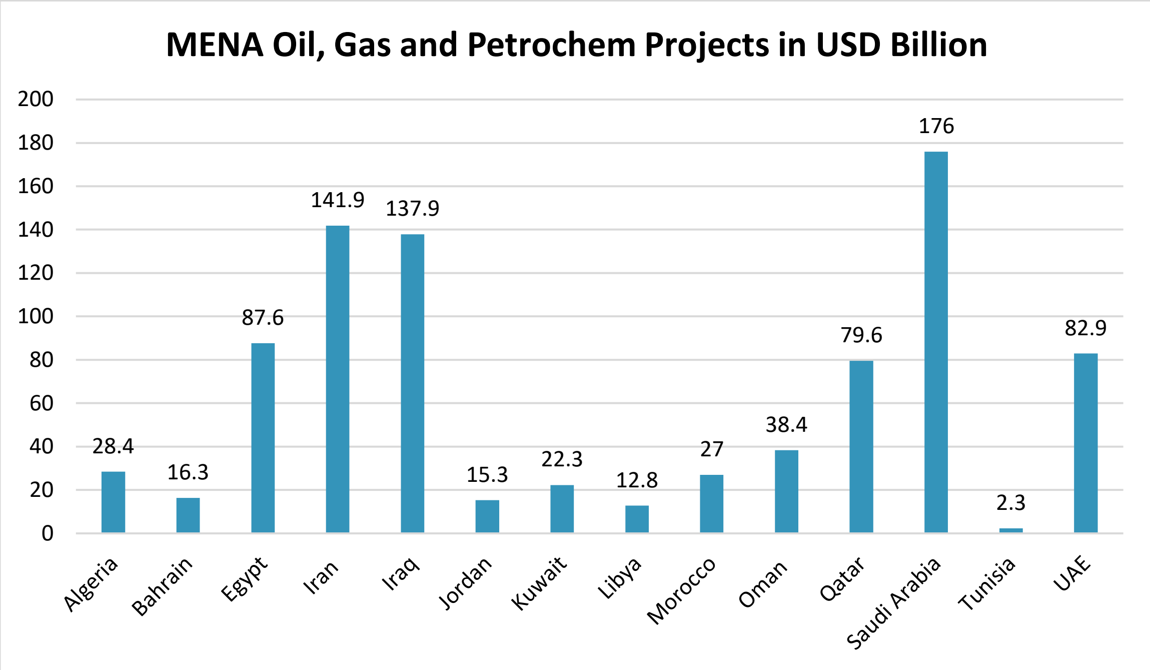


Natural gas investments are forecast to grow by 5% owed to Qatar LNG expansion projects, UAE sour gas developments (plus a newly planned LNG terminal on the Indian ocean straddling the Hormuz Strait chokepoint), and Iraq GTP projects. Committed projects in North Africa register a growth –especially in Algeria and Egypt– to stem gas exports

in light of recent MOUs with EU countries to substitute part of the Russian gas imports. Petrochemicals investments show the highest increase at 45% YoY —as several projects were put on hold in the last two years due to the combined crises where the focus was more on power sector and maximizing oil & gas export revenue to support the squeezed state budgets.

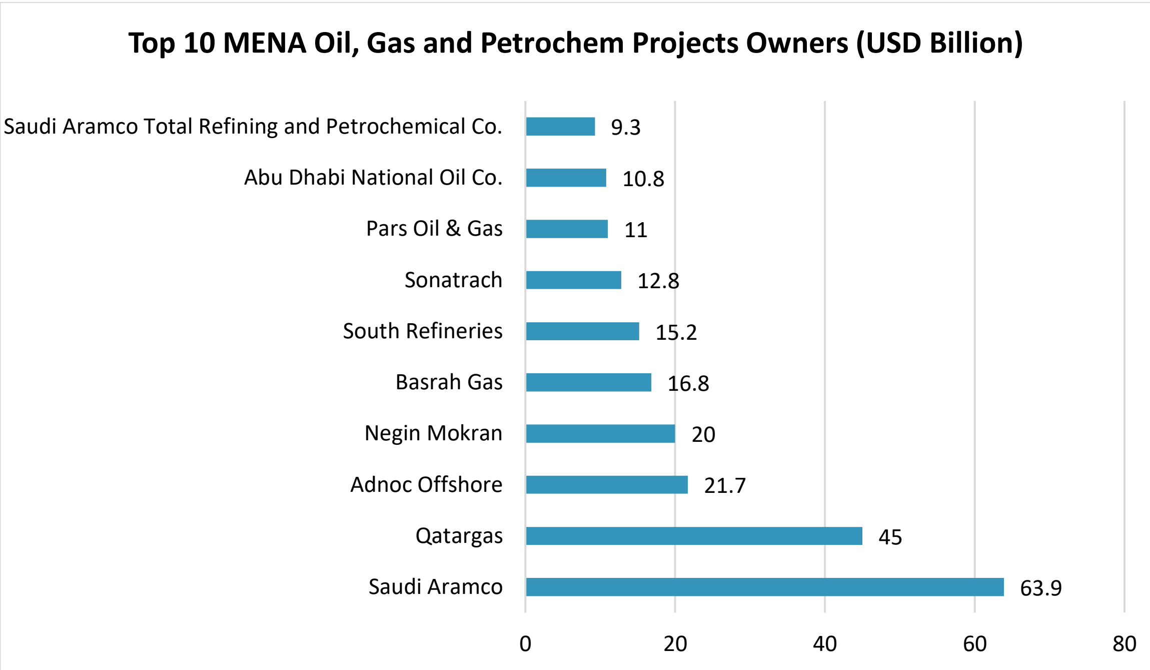
The power sector investments remain almost unchanged as MENA countries carry on with investments across the power value chain —in generation and T&D. On the supply power mix, natural gas and renewables drive the power decarbonization momentum via switching to natural gas as a cleaner fuel and leveraging sizeable renewables capacities. Although few MENA countries have already pledged their net-zero targets by 2050 (UAE) and 2060 (KSA, Bahrain), electrification via renewable energy sources will be a key driver to reach those targets.

However, due to the intermittency of renewable energy sources and the lack of utility-scale grid storage solutions to date, fossil fuels and nuclear will remain indispensable in the power supply mix in the foreseeable future.



Source: Sector Reports and SCPL Estimates

Saudi Arabia has the largest share of all active oil, gas and chemical projects, followed by Iran and Iraq. Egypt is the largest market in North Africa. The UAE and Qatar are the second and third-largest markets in the GCC.



Source: Sector Reports and SCPL Estimates

The list of top clients is led by national oil companies with Saudi Aramco ranked first. Qatargas is second due to its mammoth liquefied natural gas expansion programme. European contractors continue to dominate the regional engineering, procurement and construction market, with Japanese and South Korean firms making a strong comeback.

There are multiple tenders at the bidding stage across the region's upstream and downstream sectors. The UAE, Iraq and Saudi Arabia have the largest short-term pipeline.

Project	Owner	Country	Segment	Value (USD Billion)
Hail and Ghasha sour gas development: onshore package	Adnoc	UAE	Gas	5.4
Hail and Ghasha sour gas development: offshore package	Adnoc	UAE	Gas	4.9
Fujairah LNG project	Adnoc	UAE	Gas	4.5
North Field Production Sustainability: phase 2: scope D	Qatargas	Qatar	Gas	4.0
Al-Kut Refinery	Ministry of Oil	Iraq	Oil	3.0
Iraq strategic crude oil export pipeline: Najaf-Aqaba pipeline	Ministry of Oil	Iraq	Oil	2.8
Tail gas treatment desulphurization project	Saudi Aramco	Saudi Arabia	Gas	2.0
Riyas NGL fractionation plant: processing plant	Saudi Aramco	Saudi Arabia	Gas	2.0
Zuluf oil field: upgrade of offshore infrastructure (CRPO 100)	Saudi Aramco	Saudi Arabia	Oil	1.2
Jafurah: utilities & offsite facilities: phase 2: package 2	Saudi Aramco	Saudi Arabia	Gas	1.0
Jafurah: gas treatment facility: phase 2: package 1	Saudi Aramco	Saudi Arabia	Gas	1.0
Riyas NGL fractionation plant: offsite facilities	Saudi Aramco	Saudi Arabia	Gas	1.0
Marsa LNG terminal	Marsa LNG	Oman	Gas	1.0
Fadhili gas plant expansion: gas processing plant	Saudi Aramco	Saudi Arabia	Gas	0.8
Fadhili gas plant expansion: sulphur recovery units	Saudi Aramco	Saudi Arabia	Gas	0.8

Source: Sector Reports and SCPL Estimates

Future Demand Supply Gap Scenario

From the discussions above, it is noted that there are a series of projects, which have already been announced in the MENA and GCC region in particular and these will be requiring SAW Pipes/ Line Pipes. Based on the data available in the secondary domain, it is understood that series of other projects are also to be announced, in coming year and hence there will be ample demand for SAW Pipes in the region.

HSAW Pipes Scenario

Based on the discussion with industry experts, official of NPC and data available in secondary domain, the Consultants understand that the demand for the HSAW Pipes in the MENA/ GCC region is expected to grow at 4.20% per annum in medium term.

On the supply side, it is noted that no new project/ capacity addition/ capacity augmentation projects have been announced in the region and hence the installed capacity of the region is expected to remain at 1,555,000 TPA level for the foreseeable future.

In term of capacity utilisation level, the peak possible capacity utilisation level as expected will be in range of 60%-65% level, considering the variety of HSAW Pipes which will be required to be produced. The peak can be considered to be 65% level.

In term of imports, the Consultants understand that localisation or push for local manufacturing has gained momentum again from CY 2024. Hence the imports are expected to grow at subdued CAGR of 5.00% from the year 2025 for the foreseeable future.

Meanwhile the exports of HSAW Pipes is expected to continue to grow at 7.57%, in line with the historical growth rate.

Considering all these points into consideration, the expected demand supply scenario for the HSAW industry in region is presented in the exhibit below –

Year	Demand	Installed Capacity	Utilisation	Production	Imports	Exports	Supply	D-S Gap
2025	1,261,433	1,555,000	60%	929,654	397,191	65,412	1,261,433	-
2026	1,314,413	1,555,000	62%	967,729	417,050	70,366	1,314,413	-
2027	1,369,618	1,555,000	65%	1,007,411	437,903	75,695	1,369,618	-
2028	1,427,142	1,555,000	65%	1,010,750	459,798	81,428	1,389,119	38,023
2029	1,487,082	1,555,000	65%	1,010,750	482,788	87,596	1,405,942	81,140
2030	1,549,539	1,555,000	65%	1,010,750	506,927	94,230	1,423,447	126,092
2031	1,614,620	1,555,000	65%	1,010,750	532,273	101,367	1,441,656	172,964
2032	1,682,434	1,555,000	65%	1,010,750	558,887	109,044	1,460,593	221,841
2033	1,753,096	1,555,000	65%	1,010,750	586,831	117,303	1,480,278	272,818
2034	1,826,726	1,555,000	65%	1,010,750	616,173	126,187	1,500,735	325,991
2035	1,903,449	1,555,000	65%	1,010,750	646,982	135,745	1,521,987	381,462
2036	1,983,394	1,555,000	65%	1,010,750	679,331	146,026	1,544,055	439,339

Source: SCPL Estimates

The following is noted from the exhibit above –

1. The peak possible capacity utilisation for the industry will be 65%, which corresponds to 1,010,750 Tons of HSAW Pipe production at the regional level.

2. A demand supply gap of 38,023 Tons is expected to appear in the industry by the year 2027.
3. The demand supply gap level is expected to reach 439,339 Tons level by the year 2036.
4. At peak 59.50% capacity utilisation considered for NPC for financial projection, ample amount of market will be available for HSAW Pipes produced by NPC in the region, provided the Company is able to secure the contracts on time.

LSAW Pipes Scenario

As discussed in the section above, there are a series of oil and gas-based projects which have been announced in the MENA/ GCC region, which are expected to fuel the demand for LSAW Pipes in the region. Based on the data available in the secondary domain and the research undertaken by the Consultants, the demand for LSAW Pipes in the region is expected to grow at a CAGR of 3.50% in medium to long term.

Meanwhile, on account of the Global Pipe Company (KSA) plant being under semi closure state, the operational capacity of the region will remain pegged at 1,230,000 TPA till end of 2025. Subsequent to 2025, the operational capacity of the region will be 1,605,000 TPA.

In line with the HSAW Pipes industry, the LSAW Pipe industry can achieve maximum capacity utilisation level of around 65% and the same has been considered for the purpose of future demand supply projection.

Imports of LSAW Pipes will experience a degrowth, on account of focus towards localisation of production or push for local production. The imports are expected to decline at a CAGR of 5.00% per annum. Meanwhile the exports of LSAW Pipes will continue to grow at 6.50% per annum, which was the growth rate experienced during last 5 years.

Taking these points into consideration, the expected demand supply gap scenario for the GCC region has been presented in the exhibit below –

Year	Demand	Installed Capacity	Utilisation	Production	Imports	Exports	Supply	D-S Gap
2025	1,134,388	1,605,000	48%	772,180	430,290	65,412	1,137,058	-
2026	1,174,092	1,605,000	52%	834,982	408,775	69,665	1,174,092	-
2027	1,215,185	1,605,000	56%	901,044	388,336	74,195	1,215,185	-
2028	1,257,716	1,605,000	58%	928,983	407,753	79,020	1,257,716	-
2029	1,301,736	1,605,000	62%	998,529	387,366	84,158	1,301,736	-
2030	1,347,297	1,605,000	65%	1,043,250	367,997	89,630	1,321,617	25,680
2031	1,394,453	1,605,000	65%	1,043,250	349,598	95,458	1,297,389	97,064
2032	1,443,259	1,605,000	65%	1,043,250	332,118	101,666	1,273,702	169,556
2033	1,493,773	1,605,000	65%	1,043,250	315,512	108,276	1,250,485	243,287
2034	1,546,055	1,605,000	65%	1,043,250	299,736	115,317	1,227,669	318,385
2035	1,600,167	1,605,000	65%	1,043,250	284,749	122,815	1,205,184	394,982
2036	1,656,172	1,605,000	65%	1,043,250	270,512	130,801	1,182,961	473,212

Source: SCPL Estimates

The Consultants note the following from the exhibit above –

1. The peak possible capacity utilisation level for the regional LSAW Pipe industry is expected to be 65% level. The minimum capacity utilisation level will be 48%.

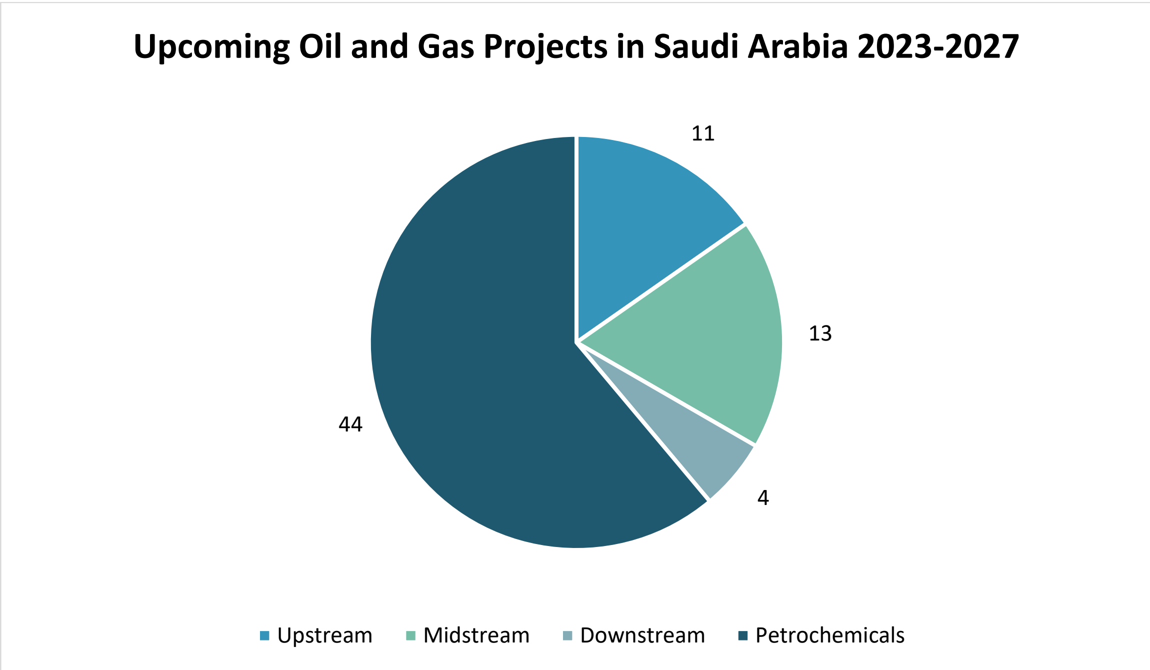
2. A demand supply gap of 28,400 Tons will appear in the regional market during the year 2024, the plant of Global Pipe Company is expected to become operational from 2026 and the D-S gap will disappear.
3. Demand supply gap will reappear in the markets from the year 2030 and will peak around 2034 at 473,212 Tons.
4. At peak 65% capacity utilisation considered for NPC for financial projection, ample amount of market will be available for LSAW Pipes produced by NPC in the region, provided the Company is able to secure the contracts on time.

Kingdom of Saudi Arabia

Demand Drivers

Oil, Gas and Petrochemical Industry

Petrochemical projects will constitute the majority of the upcoming oil and gas projects that are expected to commence operations in Saudi Arabia during the period 2023 to 2027, accounting for around 61% of the total projects, says GlobalData, a leading data, and analytics company. Their report reveals that petrochemical projects are expected to witness the highest project starts in Saudi Arabia (44), followed by midstream (13), upstream (fields) and refineries with 11 and four, respectively. Of the 44 petrochemical projects that are expected to start operations in the country by 2027, 36 are likely to be new builds and the remaining are expansion projects.



Among upcoming petrochemical projects in Saudi Arabia by 2027, Saudi Aramco Total Refining and Petrochemical Company Al-Jubail Ethylene Plant is a major project with a capacity of 1.50 mtpa and costing USD 2.7 billion. The plant will provide feedstock to other petchem and specialty companies in the Jubail industrial city. A more recent and detailed outlook of pipelines deployment in Saudi Arabia is illustrated in exhibit below –

Saudi Aramco - Master Gas System Expansion - Phase 3			
SR	**Item Description**	**Qty in (mtrs)**	**Qty. in MT**
Package 3	56" east-west gas pipelines 3 and 4 from EWPS-1 to Shedgum	56,000	27,750.72
Package 4	56" east-west gas pipelines 3 and 4 from EWPS-1 to EWPS-3	3,00,000	1,48,664.57
Package 5	56" east-west gas pipelines 3 and 4 from EWPS-3 to EWPS-4	1,66,000	82,261.06
Package 6	56" east-west gas pipelines 3 and 4 from EWPS-4 to EWPS-6	3,70,000	1,83,352.96
Package 7	56" east-west gas pipelines 3 and 4 from EWPS-6 to EWPS-8	3,40,000	1,68,486.51
Package 8	56" east-west gas pipelines 3 and 4 from EWPS-8 to EWPS-10	4,80,000	2,37,863.31
Package 9	56" EWJZG-1 from BGCS-10 to sector I STS 2	4,58,000	2,26,961.24
Package 10	56" EWJZG-1 from sector I STS 2 to sector II STS 1	3,10,000	1,53,620.05
Package 11	56" EWJZG-1 from sector II STS 1 to Shoqaiq	4,62,000	2,28,943.43
Package 12	16", 12", 4" lateral pipelines in Eastern Province and Qassim region clusters	1,20,000	7,074.38
Package 12	16" lateral pipelines	40,000	3,730.96
Package 12	12" lateral pipelines	40,000	2,775.82
Package 12	4" lateral pipelines	40,000	567.60
Package 13	16", 12", 10", 4" lateral pipelines in Riyadh clusters	4,80,000	27,936.94
Package 13	16" lateral pipelines	1,20,000	11,192.88
Package 13	12" lateral pipelines	1,20,000	8,327.47
Package 13	10" lateral pipelines	1,20,000	6,713.79
Package 13	4" lateral pipelines	1,20,000	1,702.80
Package 14	56", 16", 10" lateral pipelines in Shoaiba clusters	2,10,000	45,133.96
Package 14	56" lateral pipelines	70,000	34,688.40
Package 14	16" lateral pipelines	70,000	6,529.18
Package 14	10" lateral pipelines	70,000	3,916.38
Package 15	20", 16", 12", 4” lateral pipelines in Yanbu to Rabigh	2,00,000	14,700.60
Package 15	20" lateral pipelines	50,000	5,857.62
Package 15	16" lateral pipelines	50,000	4,663.70
Package 15	12" lateral pipelines	50,000	3,469.78
Package 15	4" lateral pipelines	50,000	709.50
Package 16	pipeline to serve customers in Jeddah	52,000	-
Package 17	30", 16" lateral pipelines in Jizan cluster	1,70,000	22,960.42
Package 17	30" lateral pipelines	85,000	15,032.13
Package 17	16" lateral pipelines	85,000	7,928.29
	Total Demand of Pipelines	**53,54,000 m or 5,354 kms**	**1,693 thousand Metric Ton**

Water/Desalination Projects in Saudi Arabia

It is expected that the population of the Kingdom of Saudi Arabia will increase from 36 million to nearly 40 million people by 2026, and this growth in the population will put pressure on the country’s water infrastructure.

According to industry reports, seawater desalination currently contributes to more than 90 percent of all daily water needs in the GCC region, and the GCC countries, desalination capacity is expected to increase by approximately 37 percent over the next five years. The global desalination market is expected to grow from USD 17.7 billion in 2020 to USD 32.1 billion by 2027.

Saudi Arabia has recently announced numerous water projects in year 2023 for desalinization, transmission, strategic reservoir, and wastewater treatment. The Kingdom continues a vigorous development of water infrastructure deploying public-private partnerships, continuing its position as the largest market for water desalination in the world. Last year, Saudi Arabia announced at the 3rd MENA Desalination Projects Forum some 60 water projects worth 35 billion Saudi riyals (9.33 billion US dollars). Once completed, these projects will increase the desalination capacity from 2.54 cubic meters per day in 2021 to 7.5 million cubic meters of water per day in 2027

According to regional news and business intelligence service ZAWYA, the projects will be owned by the Ministry of Environment, Water & Agriculture, procured under Build-Own-Operate (BOO) and Build-Own-Operate-Transfer (BOOT) contracts. The snapshots of the announced projects is provided in the exhibit below –

- **De-salinization Projects –** 7 projects which includes Ras Alkhair 2, Ras Alkhair 3, Tabuk, Alshuqaiq, Rabigh 5, Rayis 2, and Jazan.
- **Independent Water Transmission Pipeline Projects –** 4 projects which include Tabuk-Al Ula, Rabigh-Jeddah, Jazan, and Ras Al khair – Hafr Al Batin.
- **Independent Strategic Water Reservoir Projects –** 10 projects under PPP model which includes Jeddah, Asir, Makkah, Tabuk, Al Qassim, Najran, Jazan, Al Madinah, Riyadh, and Al Baha.
- **Wastewater Treatment Projects –** 5 Independent Sewage Treatment Plant (ISTP) projects and one Small Sewage Treatment Plant (SSTP) and Collection Network project. These will treat wastewater for reuse. These include Hadda, Uranah, South Najran, Abu Arish 3, North Jeddah 1, and the SSTP in Northern Province.

A brief snapshot of the upcoming SAW pipes demand from water transmission sector is provided in the exhibit below –

S. No.	Name of Project	Project Owner	Pipelines in km	Pipelines in MT
1.	Ras Al Khair to Riyadh	SWCC	800	4,20,000
2.	Riyadh City Southern Ring	SWCC	300	1,80,000
3.	NEOM Project	NEOM	120	51,500
4.	Jubail to Buraydah	SWPC	600	-
5.	Rayis - Rabigh	SWPC	150	-
6.	Ras Muhaisen - Baha - Makkah	SWPC	300	1,50,000
7.	Riyadh - Qassim	SWPC	1392	-
8.	Al Duwadimi to Afif Water Transmission System project	WTTCO	450	-
9.	Shuqaiq Jizan Transmission Line	WTTCO	575	-
10.	Aradha Water Transmission Project		110	31,500
	Total Pipeline Demand		**4,797 km**	**8,33,000 MT**

Source: SWCC and Water Transport Department (WTTCO)

Future Demand Supply Gap Scenario

From the discussions above, it is noted that there are a series of projects, which have already been announced in KSAW and these will be requiring SAW Pipes/ Line Pipes. Based on the data available in the secondary domain, it is understood that series of other projects are also to be announced, in coming year and hence there will be ample demand for SAW Pipes in the country.

HSAW Pipe Scenario

As discussed in the previous chapter, there are a total of 3 HSAW Pipe plants located in KSA, with total installed capacity standing at 1,270,000 Tons per Annum. Out of these, the NPC plant is currently not operational on account of working capital issues and lack of orders, while balance 2 plants i.e. Group 5 and East Pipes have received order from SWC for supply of HSAW Pipes and order book of these units are full till end of the year 2025. Based on discussion with official of NPC, industry experts and data available in secondary domain, SWCC is likely to announce new packages during the current year 2024 and this order is most likely going to be bagged by NPC, on account of the "Saudization" policy of KSA.

On account of the same Saudization Policy of KSA, the imports of HSAW pipes is expected to continue to reduce in medium to long term time period. The imports are expected to reduce at around 7.50% (levels prior to pandemic period) till CY 2030 and subsequently expected to grow on account of increasing demand-supply gap, with CAGR of 3.50%.

Meanwhile the exports of HSAW Pipes from the country are expected to grow at historical rate of 1.45%, in medium to long term.

Considering these points, the expected demand supply gap scenario for Kingdom of Saudi Arabia has been presented in the exhibit below –

Year	Demand	Installed Capacity	Utilisation	Production	Imports	Exports	Supply	D-S Gap
2026	848,153	1,270,000	57%	720,708	155,987	28,542	848,153	-
2027	895,820	1,270,000	61%	780,487	144,288	28,956	895,820	-
2028	946,165	1,270,000	65%	825,500	133,467	29,375	929,591	16,574
2029	999,339	1,270,000	65%	825,500	123,457	29,801	919,155	80,184
2030	1,034,316	1,270,000	65%	825,500	114,197	30,234	909,464	124,852
2031	1,070,517	1,270,000	65%	825,500	120,273	30,672	915,101	155,416
2032	1,107,985	1,270,000	65%	825,500	126,671	31,117	921,055	186,931
2033	1,146,765	1,270,000	65%	825,500	133,410	31,568	927,342	219,422
2034	1,186,902	1,270,000	65%	825,500	140,508	32,026	933,982	252,920
2035	1,228,443	1,270,000	65%	825,500	147,983	32,490	940,993	287,450
2036	1,271,439	1,270,000	65%	825,500	155,855	32,961	948,394	323,044

Source: SCPL Estimates

The following is noted from the exhibit above –

1. The peak capacity utilisation level for the local industry is expected to be 65%, taking into account the fact that multiple variety (in term of OD) of pipe will be manufactured by the companies.
2. The demand supply gap of 16,574 Tons will appear in the domestic markets by the year 2028.
3. The peak demand supply gap of 323,044 will appear by the year 2036.
4. The Consultants have considered peak utilisation level of 59.50% for NPC, for purpose of financial evaluation and the same will be achievable.

LSAW Pipes Scenario

As discussed in the sections above, it is noted that a series of oil and gas pipeline project have already been announced by Aramco. Orders for some of these projects are already placed with existing 4 LSAW Pipe manufacturing companies located in the Kingdom of Saudi Arabia, including with NPC. Balance of the order are yet to be placed and are expected to be placed during CY 2025 and CY 2026.

The total installed capacity of the 4 plants is 1,155,000 TPA. Further it is noted that currently the plant of Global Pipe Company is under semi operational state on account of one consignment being rejected by Aramco on quality issues. This plant is expected to become fully operational again by end of CY 2025. So effectively the current operational capacity of all plants together in KSA is 780,000 TPA, which will reach 1,155,000 TPA by CY 2026.

In line with the HSAW Pipe industry, the peak possible capacity utilisation for the LSAW Pipes industry will be around 65% and the same has been considered by the Consultants, while estimating the demand supply gap for future.

Further, on account of Saudization, the imports are expected to decline at a CAGR of around 7.50% level till CY 2028 and subsequently grow at 4.20% as demand-supply gap is expected to reappear.

Historically, exports of LSAW Pipes from KSA had seen a growth of around 1.00% and the same is expected to be maintained in near future as well.

Taking all these points into consideration, the future demand supply gap scenario for the LSAW Pipes industry in KSA has been presented in the exhibit below –

Year	Demand	Installed Capacity	Utilisation	Production	Imports	Exports	Supply	D-S Gap
2026	720,989	1,155,000	50%	577,131	175,901	32,043	720,989	-
2027	765,402	1,155,000	55%	635,057	162,708	32,363	765,402	-
2028	812,550	1,155,000	60%	694,732	150,505	32,687	812,550	-
2029	862,603	1,155,000	64%	738,791	156,826	33,014	862,603	-
2030	908,321	1,155,000	65%	750,750	163,413	33,344	880,819	27,502
2031	956,462	1,155,000	65%	750,750	170,276	33,677	887,349	69,113
2032	1,007,155	1,155,000	65%	750,750	177,428	34,014	894,164	112,991
2033	1,060,534	1,155,000	65%	750,750	184,880	34,354	901,276	159,258
2034	1,116,743	1,155,000	65%	750,750	192,645	34,698	908,697	208,045
2035	1,175,930	1,155,000	65%	750,750	200,736	35,045	916,441	259,489
2036	1,238,254	1,155,000	65%	750,750	209,167	35,395	924,522	313,732

Source: SCPL Estimates

The following is noted from the exhibit above –

1. The peak capacity utilisation level for the local industry is expected to be 65%, taking into account the fact that multiple variety (in term of OD) of pipe will be manufactured by the companies.
2. The demand supply gap of 27,502 Tons will appear in the domestic markets during the CY 2030.
3. The peak demand supply gap of 313,732 will appear by the year 2036.
4. The Consultants have considered peak utilisation level of 65% for NPC, for purpose of financial evaluation and the same will be achievable.

Business Model Assessment

SWOT Analysis

Strengths

- The Promoters of the Company have around 3 decades of experience in manufacturing, marketing and export of SAW pipes, both Longitudinal and Helical varieties. The Promoters have established track record of successfully operating LSAW and HSAW pipe plant, along with coating facilities and are considered pioneer in this segment of business in India.
- The Management of the Company has prior experience in implementing and successfully operating LSAW and HSAW pipe manufacturing facility. This experience will be handy while acquiring and operating the target Company.
- The Company has established international and national network of distributors/ agents, which will be leveraged by the proposed acquisition for marketing of finished products from the Project i.e. LSAW/ HSAW pipes. MIIL has its own 100% subsidiary operational in UAE, which is already marketing the products manufactured in India in the GCC countries. And proposed acquisition will be 100% subsidiary of Man Industries UAE.
- The location adjacent to the Port of Dammam Port (King Abdul Aziz Port) also is beneficial for exporting the finished products i.e. LSAW and HSAW pipes to Middle East and North Africa (MENA) region. The region is rich in oil and gas reserves and has been one of the most developing regions of the world in the recent times. There are variety of projects announced in the entire region like water supply projects, oil and gas projects etc., which are the primary demand centres for the finished products.
- MIIL has existing relationship with the customers in the Kingdom of Saudi Arabia. The Company has previously supplied large orders of API grade pipes to its customers in KSA including ARAMCO and others. This existing relationship will be leveraged by the target Company also the NPC has its own existing relationship in the Kingdom of Saudi Arabia which will leverage to locally market the products.
- Strong financial standing of the Company, which is listed on Stock Exchanges in India, Singapore and Dubai. The Company has successfully raised and repaid long term debt for execution of its other SAW Pipe and Coating plant projects within India.

Weakness

- The Target company is dependent of mostly on imports of basic raw materials i.e. HR Coils and Plates from companies like ArcelorMittal, Nippon Steel, POSCO, NISCO, TISCO etc., as well as it purchases HR coils from native suppliers i.e. Saudi Iron & Steel Company (Hadeed).

  The Company may consider trying to have long terms contract with suppliers of the basic raw material. Further an inventory of raw material needs to be maintained by the Company, to reduce the risk of non-availability of the raw material.

Opportunities

- The Government of Saudi Arabia has in recent years initiated a series of reforms for the industrial sector and Foreign Direct Investment (FDI) policy of the country has been relaxed to facilitate the industrial development, which is back by the infrastructure development in the country.

Threats

- The generic threat of recessionary trend in the global and the domestic economy of the country.
- The threat of substitution is prevalent for the pipes industry, where SAW Pipe plants face substitution from ERW Pipes and Concrete pipes for transportation of water.
- The threat of new entrant or threat of existing player in domestic market, expanding the capacity and thereby increasing the competition.

Risk Analysis and Mitigation Measures

The risk analysis and mitigation measures for the Project have been presented in the exhibit below –

Parameter	Carrier of Risk	Remark
Experience Risk	NPC/ MIIL	The Promoters and Management of MIIL have prior experience of successfully implementing and operating LSAW and HSAW pipes as well as coating plants in India, for almost 3 decades now. Further, the Management of the Company indicates that senior professional from the Company will be deputed to overseeing the acquisition and subsequently operations of the target Company in the KSA. Based on the discussions with the Executives of MIIL, it is understood that there will be a hand holding period, post acquisition, whereby the existing official of NPC will be retained for a period of 1 year post acquisition, so as to handhold and pass on operating knowledge to staff recruited/ appointed by MIIL. The Company will also be recruiting experienced professionals from the industry for taking care of day to day operations of the proposed acquisition. *Hence experience related risk is not associated with the Project.*
Funding Risk	NPC/ MIIL	The proposed acquisition of NPC + HSAW Pipe mill upgrade cost has been estimated at USD 103.00 Million is to be funded in a DER of 3:1 or 75:25. Meaning the Project will be funded by Equity of USD 26.25 Million and foreign currency term loan of USD 78.75 Million. The Company has not achieved financial closure as off the date of release of this report.

Parameter	Carrier of Risk	Remark
		Hence there is a funding risk associated with the Project, till the time the Project achieves financial closure.
Time Overrun Risk	NPC/ MIIL	Not applicable, as NPC is already operational company. Further no addition in terms of capex is proposed as part of the Proposal by MIIL.
Cost Overrun Risk	NPC/ MIIL	As per the discussion with the management of the company the acquisition cost of the NPC has been negotiated to a final bid of USD 102.00 Million. As the cost of acquisition has been finalised, the cost overrun risk is not associated with the project.
Statutory Approvals Risk	NPC/ MIIL	As discussed in previous sections of the report, the MIIL is acquiring the target Company which is already operation since 1980 and major approvals and clearances are in place with it. Hence approvals related risk is not associated with the said acquisition proposal.
Demand Risk	NPC/ MIIL	Based on the market assessment undertaken by the Consultants, there is ample demand for the finished products in the local KSA markets. The Government of Saudi Arabia has announced a series of infrastructure projects including water supply and oil and has projects have been announced, which will be the demand centres for the HSAW and LSAW pipes proposed to be manufactured by the Company. Further, the proposed Project will be in position to cater to the demand arising from the GCC/ MENA region, which is currently one of the fastest growing regions in the world. Hence, there is no demand related risk associated with the Project.
Pricing Risk	NPC/ MIIL	The Consultants note that the Company will be following the policy of "Mark to the Market", while pricing the products, proposed to be manufactured by the Project.

Parameter	Carrier of Risk	Remark
		Further, it is noted that the industry works on blocking the price, by placing large orders. NPC will be entering into back-to-back, similar contracts with raw material suppliers, so that not to allow the price fluctuation at raw material level impacting the price of the final product. The Consultants tested sensitivity, considering 2.50% decrease in selling price and 2.50% increase in raw material prices, to see the impact on the financial viability of the Unit. The said acquisition remains viable under both adverse scenarios. Since the Company will be marking the product price at market levels, hence pricing risk is not envisaged.
Raw Material Risk	NPC/ MIIL	The basic raw material for the manufacturing of LSAW and HSAW are HR Plats and HR Coils respectively. To fulfil the requirement, MIIL will need to import from China, India, Indonesia, Japan, Turkey etc., as the local supply in Saudi Arabia will not be sufficient enough to cater to the demand of HR coils and Plates from various industries/ demand centres. Hence raw material related risk is envisaged by for the Project and to mitigate the risk, the Company may consider entering long term contracts with suppliers, along with maintenance of sufficient inventory of atleast 2 months to reduce the raw material risk of the Project.
Technology Risk	NPC/ MIIL	The technology of manufacturing HSAW pipes from HR Coils and LSAW pipes from HR Plates through 2-Step and 3 Roll Bending process respectively, are established and economically reliable technology. There are a series of such plants operational across the globe and with MIIL already operating similar plants in India. Further, there are existing plants of LSAW & HSAW pipe manufacturing through these processes located in KSA as well. Hence technology related risk is not envisaged for the Project.
Force Majeure	MIIL / Insurer	Lenders may ask the Company to purchase adequate insurance cover to mitigate these risks.

Porter's Five Force Analysis

Threat of New Entry is LESS

As this industry generally requires significant capital investment in infrastructure, technology, and raw materials.
Also requires specific regulation and standard for setting up new business unit.
Additionally, the industry requires high level of expertise and years of experience to establish good relations with buyers.

Threat of Substitute is LESS

The steel industry's susceptibility to substitution depends on the specific application and industry requirements. However, the proposed product i.e. LSAW and HSAW requires a high degree of quality checks and standards.
Also, Emerging technologies that offer alternatives to traditional steel manufacturing methods can pose a threat.
But as of now the thread of substitute is low.

Competitive Rivalry is MEDIUM

At present, there are very few players in the market; however, they might rise in the future. The product is generally homogenous which makes substitution for buyers easy. Industry rivalry, at present, is therefore **MEDIUM**. Players with good relation with Aramco, SWCC and WTTCO, and high quality assurance are expected to prevail

Bargaining Power of Buyers is HIGH

The bargaining power of buyers is considered to be **HIGH** as there are only limited buyers purchasing product in the industry. In Saudi Arabia, there is only limited buyers like Aramco, SWCC and Water Transport Department. The product offering by the companies is also similar which provides great degree of substitution power to the buyer, resulting in high bargaining power.

Bargaining Power of Suppliers is MEDIUM

As, there are various suppliers of Raw Materials in the KSA as well as Supply can virtually be from anywhere across the globe. In addition, steel being a homogenous product, allows flexibility to the company to choose its supplier.

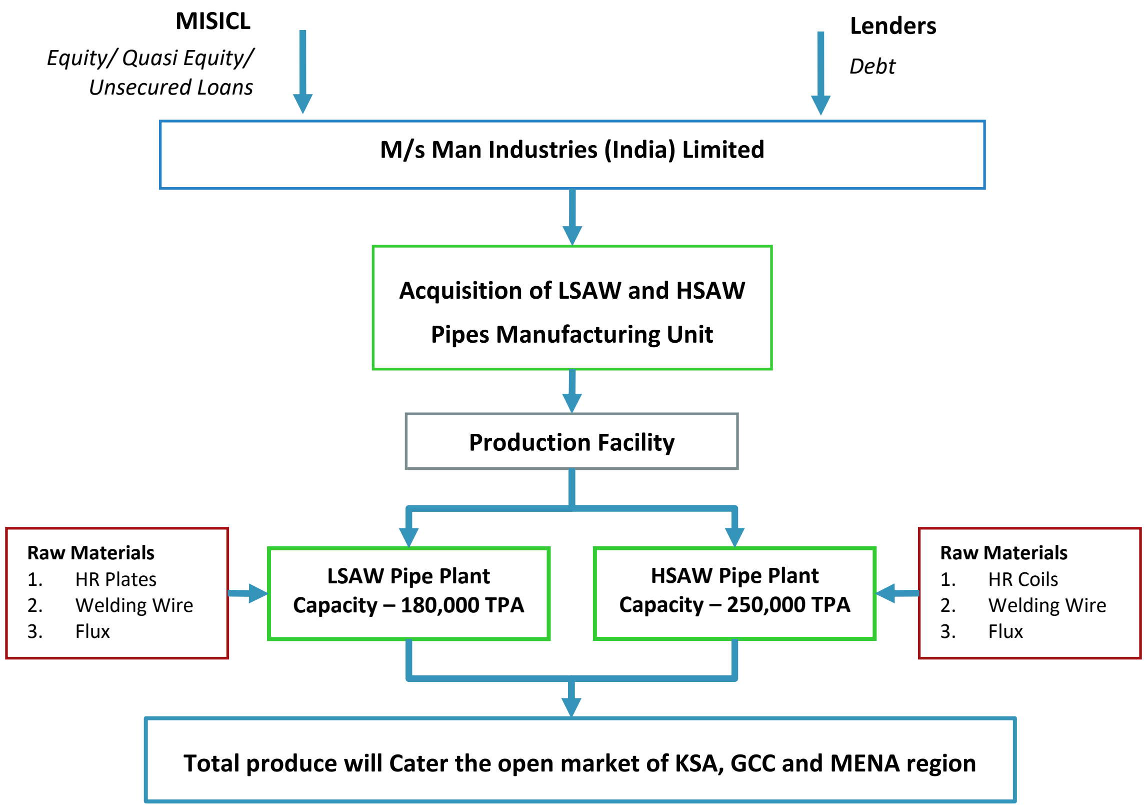
Technical Assessment

Introduction

As per the discussion with management of the NPC, the installed capacity of the company is 250,000 TPA for HSAW pipe Manufacturing and 180,000 TPA for LSAW pipe manufacturing which located at Al Khobar, Dammam, Saudi Arabia. In this chapter the consultants have assessed and discussed the Project configuration, Land location, Manufacturing Process, Raw material supply and all technical aspect of the project.

Project Configuration

The project configuration as envisaged by the Company is presented in the exhibit below –



Source: MIIL and SCPL Estimates

Land Details

According to the Company management and information shared by the Company it is noted that the total installed capacity of the target company is 430,000 TPA which includes 250,000 TPA for HSAW pipe Manufacturing and 180,000 TPA for LSAW pipe manufacturing. The project is implemented over a total land area of 400,000 Sq. M. and the land is freehold property of NPC.

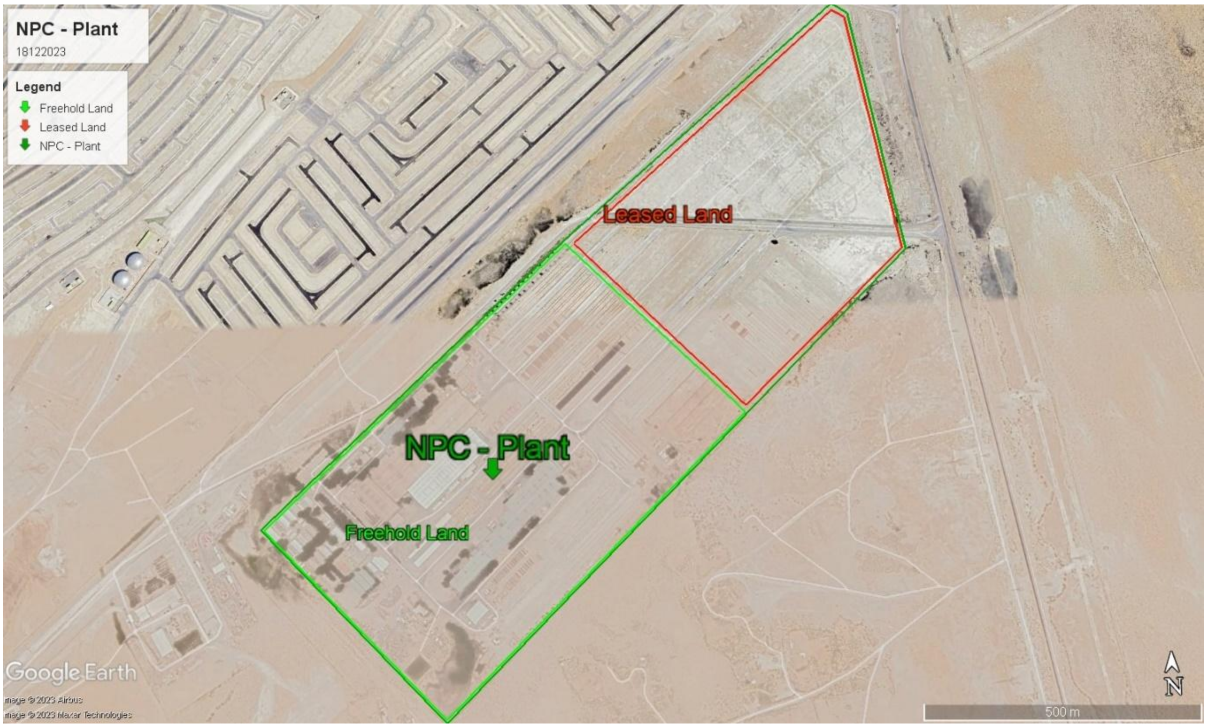
The Consultants during the site visit to the facility were shown the copies of the land papers, which were in Arabic. The Consultants had requested for a certified true copy of the land papers in English language and the same was duly provided to the Consultants for review purpose.

***SCPL Observation – The Consultants note that the Company has ample amount of land available with it for any possible future expansion on site like setting up a coating plant or expanding existing capacities of HSAW and LSAW Pipes*.**

Plant Location and Topography of Site

The plant is situated in Dammam, the capital and largest city of the Eastern Province of Saudi Arabia, renowned for its strategic location and economic significance, housing the major King Abdulaziz Seaport on the Arabian Gulf. More accurately the plant is located in Dhahran, is a part of the Dammam Metropolitan Area, features a predominantly flat and arid topography with a gentle slope toward the coast. Known for its pivotal role in the region's oil industry, Dhahran hosts the headquarters of Saudi Aramco amidst a landscape marked by urban development, industrial facilities, and residential areas. While the immediate surroundings lack significant natural elevations, the broader Eastern Province offers diverse landscapes, including deserts and mountains further inland from the coastal areas.

NPC - Plant is located at a satellite coordinates of Latitude 26°13'44.34"N and Longitude 50° 3'26.02"E And a satellite image has been exhibited below –



Source: Google Earth

NPC - Office is located at a satellite coordinates of Latitude 26°19'29.50"N and Longitude 50°11'38.67"E, by road, it is around 30.00 Km away from the NPC – Plant. Also, a satellite image has been exhibited below –

Source: Google Earth

A satellite image of NPC – Office and NPC – Plant and vis-à-vis nearby location from the plant and office has been exhibited below –

Source: Google Earth

The distance of the plant location from various key demand centres has been provided in table below –

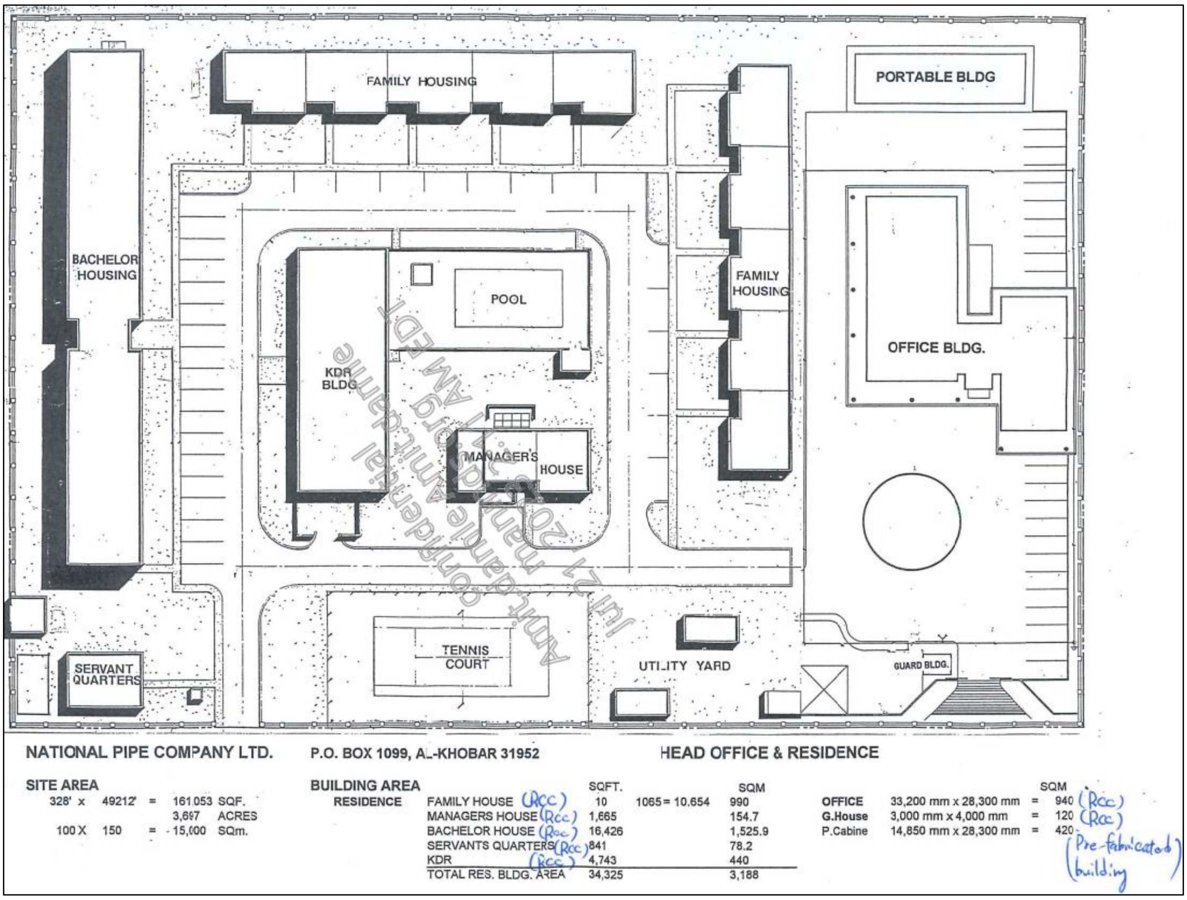
Demand Centre	Distance
From NPC - Plant	
Nearest Industrial City	Dammam 2nd Industrial City (10.00 Km)
Nearest Highway	Cooperative Council Road/ Route - 80 (2.00 Km)
Nearest Sea Port	Dammam Port (King Abdulaziz Seaport) (35.00 Km)
Nearest International Airport	King Fahd International Airport (50.00 Km)
Capital of KSA	Riyad (410.00 Km)
From NPC - Office	
Nearest Industrial City	Dammam 2nd Industrial City (30.00 Km)
Nearest Highway	King Fahd Road/ Route - 610 (2.00 Km)
Nearest Sea Port	Dammam Port (King Abdulaziz Seaport) (20.00 Km)
Nearest International Airport	King Fahd International Airport (55.00 Km)

Source: Google Earth / Google

Office and Plant Layout

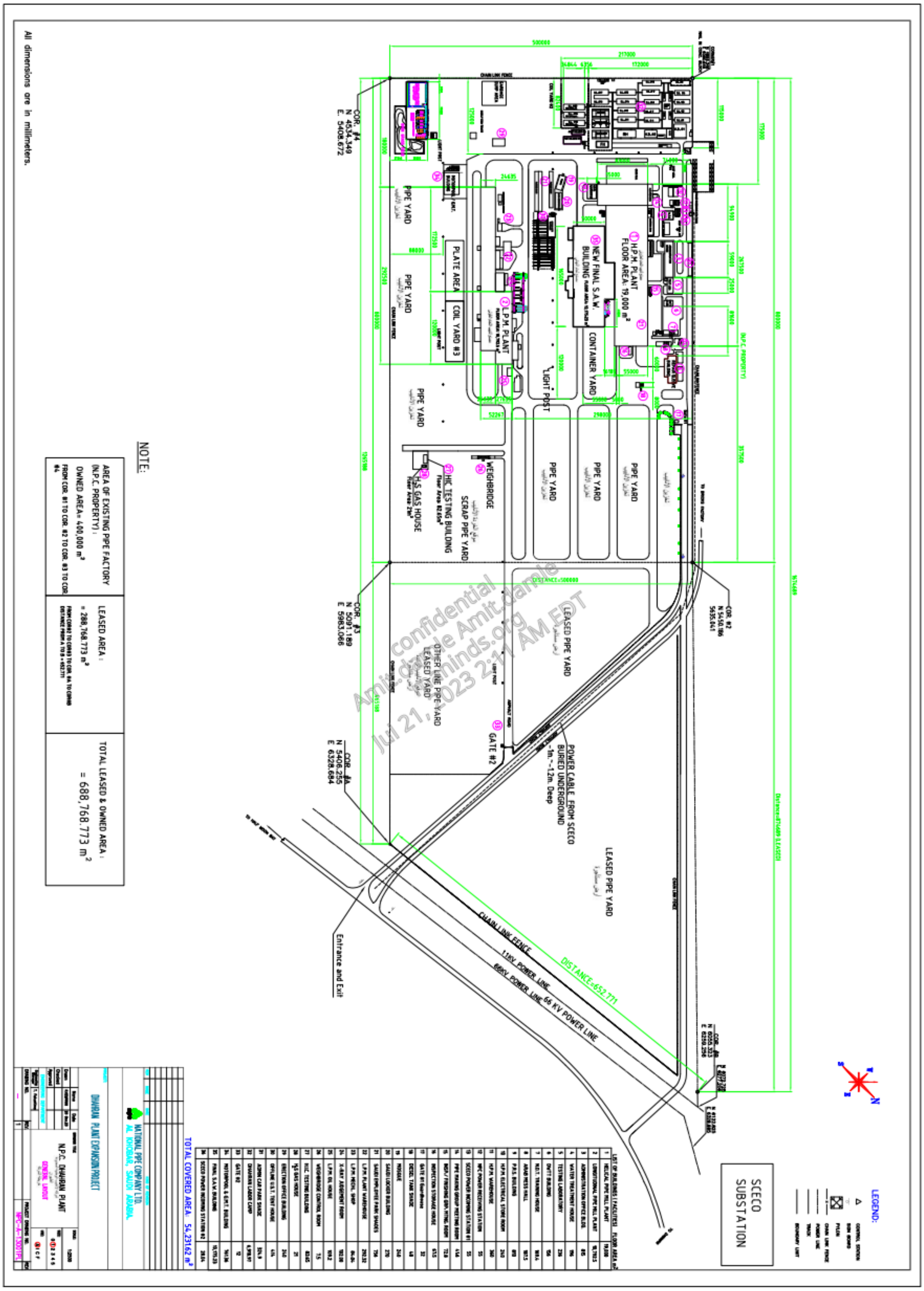
Consultants has sought the approved plant layout of the proposed project from the Company. The Company has duly provided proposed plant layout which has been exhibit below –

Office Layout



Source: MIIL

Main Plant Layout



Source: MIIL

Manufacturing Process – Line Pipes

Line Pipes are carbon steel pipes which are used in transmission of Oil, Gas and Petroleum as well as Water transmission. They are manufactured as per the standard/specification of American Petroleum Institute (API). Basically, the line pipes are manufactured by two welding process namely –

1. **Electric Resistance Welding**

   Resistance Welding Takes place by passing electric current through the specimen being joined to heat the interface point enough to causes melting of the materials at the Interface and a clamping force is applied during heating process which hold the specimen until the melt material is solidified. In this welding process no filler Material is used.

2. **Electric Fusion Welding**

   Fusion welding is just any welding process which inhibits the phase transition from solid to liquid and then back to solid state. Generally, two types of pipes are made through electric fusion welding method which are Longitudinal Submerged Arc Welding and Helical/ Spiral Submerged Arc Welding. A detailed explanation for both LSAW and HSAW pipe and pipe manufacturing process has been mentioned below –

1. Longitudinal Submerged Arc Welding (LSAW) Pipes

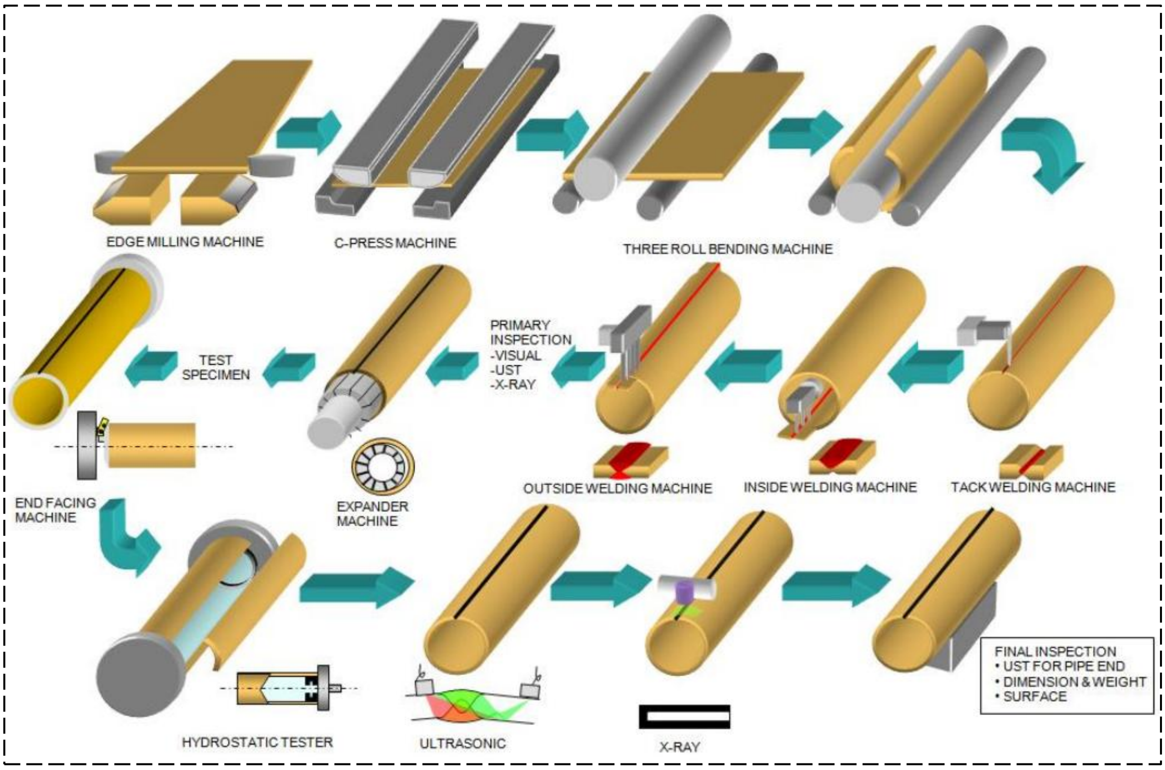
Longitudinal Submerged Arc Welding (LSAW) pipes use single plate steel as raw material, applied double-sided submerged arc welding. LSAW pipes have larger diameter and can withstand high pressure and low temperature, they play major role in oil and gas pipelines.

LSAW pipes are used where high-pressure transportation of large quantities of hydrocarbons is required. They are used for both onshore and offshore projects and are generally used for oil & gas transportation.

LSAW pipes play a major role in oil and gas pipelines with features of high strength, high quality and long distance. According to the standard of American Petroleum Institute (API), the LSAW pipe is the only designated pipe in the large, scaled oil and gas transportation, especially when the pipelines cross the densely populated urban areas and first- and second-class cities.

Longitudinally submerged arc welded pipes (LSAW) form the majority of our pipe production. Using steel plates as the key input factor, the plates are produced by suppliers according to the exact specifications of the EUROPIPE Group. They are formed into a pipe, welded by submerged arc welding and then mechanically expanded into their final shape. There are generally three most used process for LSAW pipe manufacturing like JCOE Process, UOE Process, 4RB Process and 3RB Process.

The NPC uses 3RB process to produce LSAW pipes with outside diameters of between 508 mm and 1,524 mm (20" and 60") and with a wall thickness from 7 mm to 45 mm (0.276" to 1.772"). The process flow chart for 3RB process has been exhibit below –



3RB (Three Roll Bending)

This process involves welding steel pipes using a continuously fed wire electrode and a submerged arc. Here are the key steps involved in the manufacturing of saw pipes, as below –

Material Preparation:

The process begins with the preparation of steel plates. These plates are passed through ultrasonic testing before it put into the production line.

Edge Milling:

The edges of the steel strip are milled to ensure a proper fit during the welding process. It uses a milling machine to shape both sides of the steel plate. So as achieve the required plate dimension, the parallelism of the plate edge.

3-Roll Bending Process

To pre-bend the plate edge to meet the required curvature using a pre-bending machine called 3-roll bending machine, used to make plate in circular shape or "O" shape.

Pre-Welding or Tack welding

The two edges of the steel strip are pre-heated using an induction coil or other means. This is followed by initial tack welding to hold the edges together. For double seam saw pipe, the first two halves joined by tack weld which called fit-up. Double seam saw pipe is having a two weld seam opposite to each other. Both the welding seams welded from the internal and external of the steel pipe. In the case of heavy thickness steel pipe, multiple pass welding done.

Submerged Arc Welding (SAW)

The edges of the steel strip are then welded together using the submerged arc welding process. In this process, a granular flux is used to protect the weld zone from atmospheric contamination. The arc is submerged beneath the flux, hence the name "submerged arc welding."

Here, the pipe is passed through a welding machine where a submerged arc welding process is used to create the longitudinal seam weld. This can be a single or double-sided welding process.

Weld Seam Inspection and Flux & Slag Removal

The welded seam is inspected to ensure quality. Various non-destructive testing methods may be employed, such as Hydrosonic testing, ultrasonic testing or X-ray inspection.

The welded pipe is then passed through a series of devices to remove excess flux and slag.

Pipe Formation and Pipe Expansion

The welded strip is then formed into a cylindrical shape. This can be done by a series of forming rolls. Also, the pipe may be expanded to achieve the desired diameter and wall thickness. This process is performed by a mechanical expander.

Sizing and Cutting

The pipe is sized to its final dimensions using a series of sizing rolls. The pipes are cut to the desired lengths.

End Facing and Bevelling

The ends of the pipes may be faced and bevelled for welding or other joining processes.

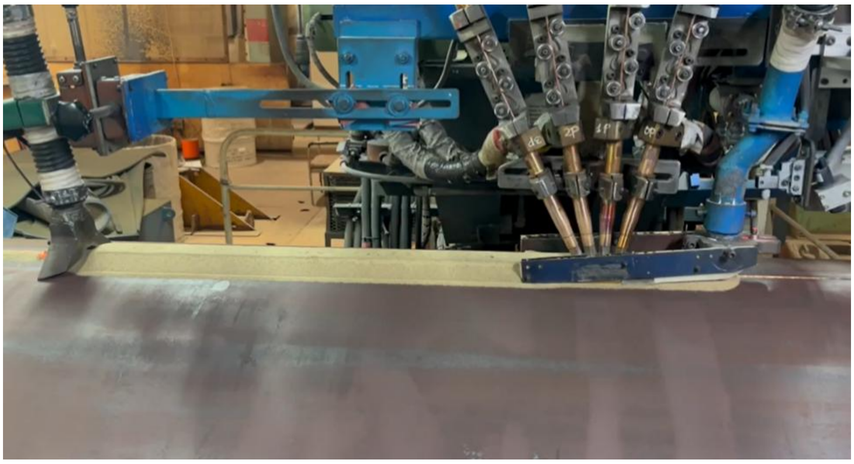
Final Inspection

The finished pipes undergo a final inspection to ensure they meet the required standards and specifications.

For more visual understanding. A pictorial presentation of the process has been illustrated in the exhibit below –

Plates Preparation	

Edge Milling	
3RB Forming Process	
	
Tack Welding/ Pre-welding	

Conveyance Channel for Pre Welded Pipes	
Outer and Inner Submerged Arc welding	 
X-Ray Inspection	

Stock Yard for the finished SAW pipes	

Source: Site Visit by SCPL and Site Photographs

Range and Grade Dimensions

At NPC, LSAW pipes are manufactured in sizes from 24" (610 mm) to 60" (1524 mm) with wall thickness from 6.4 mm to 30.5 mm. Pipe size and wall thickness depend mainly on the available plates but the processes that are used have limitations with respect to forming forces and these also limit pipe size and wall thickness.

In terms of grades the main trigger is given by the pre-material. Usually grades with strength classes up to L 485 (X70) are available, but also grade L555 (X80) are frequently used for pipelines worldwide as standard grades. Even higher grades, L690 (X100) and L830 (X120) have been developed, but the pipeline market is reluctant to use these very high strength pipe grades at this this time.

Higher grades allow reduction of pipe weight with smaller wall thickness at the same diameter and operational pressure, which is beneficial for pipe transport and laying activities.

SWOT Analysis of LSAW Pipes

Strength

- Established market around the world due to historically proven strength.
- Wall thickness of LSAW pipes makes them unique with respect to quality. Wall thickness allows pipelines to carry high pressure liquids or gas.
- Preferred for use on offshore drilling due to the ability to bear high pressure.

Weakness

- Expensive as compared to HSAW pipes because it uses steel plate as raw material which is considered as an expensive raw material.
- Wall thickness of LSAW pipes restricts its diameter and therefore decreases the amount of liquid/gas they can carry at one time.
- Cannot be used for larger diameter pipes as it is not efficient and is generally costly to produce
- production line takes more space as compared to HSAW and is more expensive.

Opportunity

- Emphasis on safety of pipelines by regulators around the world is expected to boost demand for LSAW pipes due its proven record of high quality
- Revival of overall energy market will generate opportunities for LSAW pipe producers as more LSAW pipes will be needed.
- Increase in offshore projects due to the implementation of large projects

Threats

- Cheap substitutes such as HSAW/SSAW are expected to replace the use of LSAW
- Slower production: the speed of producing LSAW is comparatively slower than HSAW therefore mills may not be able to generate enough return on investment and therefore scrap producing LSAW
- Technological improvements in HSAW; further improvement may result in decline in demand for LSAW pipes

Application of LSAW Pipes

- LSAW pipes are mostly used in the transmission of oil and gas. But currently it is being used in water transmission.
- LSAW pipes are used by National Association of Corrosion Engineers (NACE) in fighting Hydrogen Induced Cracking (HIC) and Sulphide Stress Cracking (SSC).
- LSAW are also used in Construction and Erection of Wind Turbine Structures.

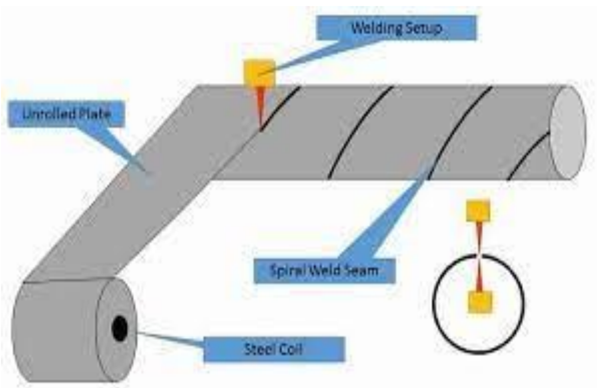
SCPL Observation

The Consultants note that the LSAW Pipe plant was operational at the time of the site visit undertaken by the Consultants. The team from NPC showed the entire plant along with associated utilities and support infrastructure to the team from SCPL. The plant and machinery were found to be well maintained and based on the discussions with the team from NPC, it is understood that the Company follows the preventive maintenance philosophy and for the same, regular shut down are undertaken on monthly basis.

Here its should be noted that the LSAW Pipe mill of NPC was operational and undertaking switch-over of pipe size, when the site visit was undertaken by team from SCPL on 2nd and 3rd December 2025.

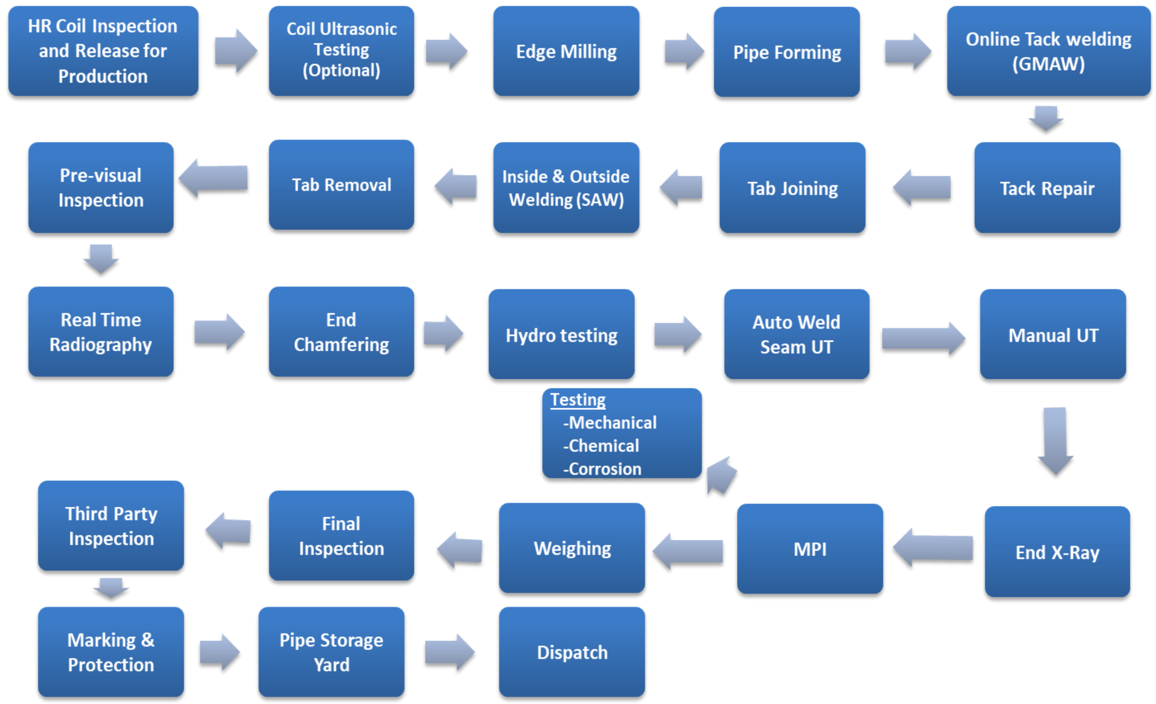
2. Helical/Spiral submerged arc welded (HSAW/SSAW) Pipes

Spiral welded pipe was first introduced on an industrial basis in the United States in 1880, but the technology used for pipe production today traces back to 1960s.



Nowadays many new developments have undertaken in Line pipe manufacturing with new technologies and equipment's being introduced but, in this heading, we shall discuss the basic process of Line pipe (spiral) manufacturing. First, a coil of hot rolled wide strip is mounted into the straightening machine. The strip is positioned to the correct helix angle. Smaller pipes require greater angles than larger pipes manufactured from the same width coil. Before passing the strip through, the leading and trailing ends of the incoming and already inserted coil strip respectively are welded together. Shears/Plasma cutting trim the edges that are then machined simultaneously on both sides by an edge miller. In the forming section, a 3-roller bending device along with a Jactuator (powered screw jack) bend the strip helically into a cylindrical pipe. Then the longitudinal edges of the incoming strip are brought into low point contact and the resulting butt joint is continuously welded using the Submerged Arc Welding process. Finally, it is cut off to the desired length with a plasma torch. The machine on which above process is carried out is referred as "Spiral Pipe Mill" and most of the processes are automatic or semiautomatic. Also, it is a called as continuous process.

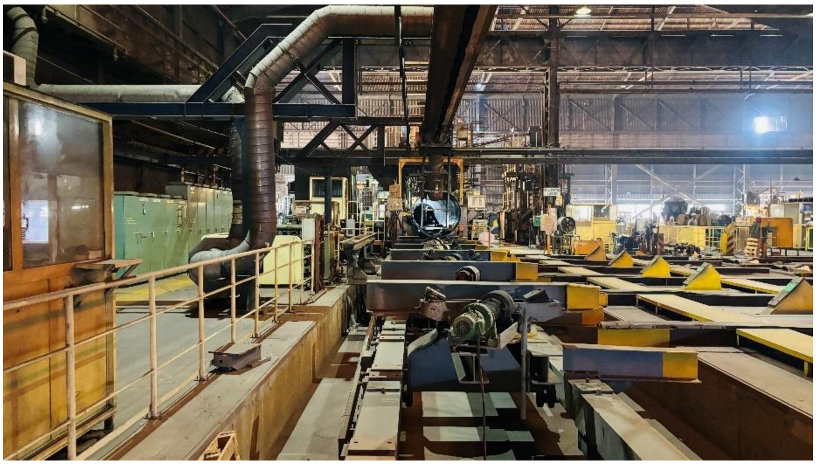
A step-by-step flow chart of manufacturing process of HSAW pipe is presented below –



Source: MIIL

Helical submerged arc welded (HSAW)/ Spiral submerged arc welded (SSAW) pipes has spiral welding joints and the pipes are welded and formed at the same time. Welding of HSAW pipes is generally considered of low quality as compared to LSAW pipes. But nowadays with the improvement in process through changes in technology has made HSAW pipes quality close to LSAW pipes. A pictorial presentation of the process has been illustrated in the exhibit below –

Coil Preparation/ Uncoiling	 
Tack Welding/ Pre-welding	

Conveyance Channel for Pre Weld Pipes	
Final Welding or Submerged Arc welding	
Hydrosonic Testing	

X-Ray Inspection	

Stock Yard for the finished SAW pipes	

Source: Site Visit and Site Photographs

HSAW pipes are used in various industries such as public waterworks, petrochemical industry, chemical industry, electric power industry, agricultural irrigation and urban construction. They are used for onshore projects only due to low pressure and are generally used for water pipelines.

Range and Grade Dimensions

Pipes are generally manufactured in outer diameter range from 508 mm to 2134 mm, in length from 4 m to 24 m (line pipes) and from 4 m to 30 m (pipes for construction purposes), with plain or bevelled ends. The manufacturing of Pipes also involves hydrostatic tests and depending on requests: automatic non-destructive ultrasonic tests and/or X-ray test (pipelines).

The dimensions of spiral weld pipes are continuously adjustable so that any diameter, as desired by the costumer can be produced from a base material of the same width. Unlike LSAW it is not necessary to use a large number of forming tools for various separate sizes. Spiral welded pipes are manufactured in steel grades from API 5L Grade B to X80 and to various other national and international standards.

SWOT Analysis of HSAW Pipes

Strength

- Established market around the world due to historically proven strength
- Lower cost alternative to LSAW pipes due to the use of HRC as raw material
- Lower wall thickness increases its diameter and therefore increases the practical capacity of carrying liquids/gases
- Production line cost less plus take less space

Weakness

- The lower wall thickness makes HSAW pipes vulnerable to ruptures and breaking of joints and therefore HSAW pipes can prove to be safety hazard
- Since HSAW pipes cannot be used for high pressure liquid delivery therefore they are not used for offshore drilling projects

Opportunity

- Emphasis on safety of pipelines by regulators around the world is expected to boost demand for LSAW pipes due its proven record of high quality
- Increasing demand of water transportation in KSA
- The increased competition and squeezing margins of oil and gas exploration companies requires them to minimize costs therefore they are more expected to use HSAW pipes in their projects as it saves them cost.
- Technology improvement to further strengthen the weld will be game changer

Threats

- Emphasis on safety regulations by regulators around the world can restrict the use of HSAW as due to its lower wall thickness it is often prone to damage
- Lower pressure in pipelines which can result in breaking of pipes and could be costly and time consuming.
- ERW can also be use as replacement for transportation projects due to same characteristics but higher electrical resistance

Application of HSAW Pipes

- HSAW pipes have multiple uses from liquid transportation to gas transmission apart from its use in transportation of liquids and gases it is also used in structures.
- HSAW steel pipe is mainly used in public waterworks, petrochemical industry, chemical industry, electric power industry, agricultural irrigation, and urban construction etc. The strength and flexible manufacturing of spiral welded pipe make it the product of choice for a variety of applications.
- They are also used in the design of multiple structures such as piling, bridge, wharf, road and architectural structures etc.

SCPL Observations – The HSAW Pipe plant of the Company is old and was established in the year 1980 and has been maintained in fairly good condition. The Consultants note that the instrumentation of the forming machine and the tack welding machine installed at site were analogue in nature and require upgradation to modern system, which can be undertaken by MIIL in 2-3 years' time period, post-acquisition. The Company has installed 3 new offline welding machines and new X-Ray machine. The workshop of the Company is located adjacent to the HSAW Pipe plant and the workshop also had upgrade machines and newly installed CNC Lathe as well.

Here it should be noted, that the HSAW Pipe mill was operational at the time of site visit undertaken by team SCPL on 2nd and 3rd December 2025.

Plant and Machinery

The Consultants had sought the copy of the fixed asset register of NPC for inspection purpose during the site visit and the same was duly provided by the Company. The list of plant and machinery as shared by the Company has been provided as Annexure 2 of the Report.

SCPL Observations – The Consultants note that both the HSAW and LSAW Pipe lines installed at the site are complete in all records and are capable of manufacturing the desired finished goods as required.

However, here it should be also noted that the HSAW Pipe mill installed at site is currently capable of manufacturing pipe with maximum diameter of 84 inches. However, in the current circumstances, the order being received from local customers are mainly for larger diameters upto 100 inches to 12 inches. Hence Man Group will be undertaking an upgrade of HSAW pipe mill, almost immediately after acquisition of the unit and the cost of this upgrade has been estimated at USD 3.00 Million.

Technology Provider

Going through the information received and based on the site visit observation, the Consultants have noted that for the plant is operational and the plant is well supported with details below –

Nippon Steel Welding & Engineering (Japan)

Nippon Steel Welding & Engineering Co. Ltd., are a subsidiary company of Nippon Steel Corporation which is one of the world's biggest steel maker.

Over 80 years, it supply the breakthrough welding consumables used for FCAW, SMAW, GMAW, GTAW and various welding equipment included plasma welding to match and enhance the welding skill.

It is the first and only Japanese manufacturer to mass produce the Seamless Flux Cored Wire since 1981, which has excellent performance and quality for low temperature and high-tensile strength use. Also, it promises to provide appropriate solutions and create a breakthrough for your welding situation.

Sumitomo Heavy Industries (Tuck Welding)

Known initially as Izumi-ya, Sumitomo began its operations in the fields of copper refining, trading, and mining. The company went through several name changes during the course of its history, first with the change to Kikaika (or "Machinery Division") in 1894, and then to Niihama Seisakusho (or "Niihama Works") in 1928. The company was established as an independent public corporation in 1934 under the name of Sumitomo Machinery Co., Ltd. In 1940, its name was again changed to Sumitomo Kikai Kogyo Co., Ltd. (or "Sumitomo Machinery Industries").

Sumitomo Heavy Industries has always strived for technological innovation and applies cutting-edge technologies wherever possible. The company operates in a broad range of business areas ranging from manufacturing equipment and infrastructure to fields making use of the latest advanced technology. Most recently, the company's research and development efforts have focused on fine, mechatronics, and systems technologies and by doing so it has been able to gain a foothold in the areas of precision control devices and key components. For example, the company has succeeded in developing equipment for the manufacturing of LCDs and semiconductors as well as digital home appliances. This new area of business is positioned to become the third core competency for the company after power transmission equipment and plastics machinery.

The company has its networks in Europe, East Asia, North America, Southeast Asia, Oceania and Latin America with total overseas manufacturing bases of 30. Also, it has multiple branches in its native country.

And has acquired 7,494 patents with a consolidated employees of 25,541 as of FY 2022.

SMS Germany (PWS) (Inside Pipe Cleaning, Over Head Cranes)

SMS group has always been a truly international company. Right from the very beginning, we have opened branches and subsidiaries in the main markets. Today, we are present in 31 countries representing more than 80 percent of global steel production.

To be closer to our customers, we have implemented a regional setup for sales and project execution, creating new market opportunities and driving growth based on a lean setup. Our four regions Americas, Europe, China and APAC & MEA (India, Asia Pacific, Middle East and Africa) team up with our worldwide Center of Excellence (CoEs) for Products and for Services as well as with the CoEs for Implementation and Supply Chain to make your project a success.

And it is located in various corners of the globe like in China, USA, Brazil, India, Italy, Japan, Luxembourg, Germany (The Home Base).

Mitsubishi Heavy Industries (Edge Milling Machine)

Mitsubishi Heavy Industries (MHI) Group is one of the world's leading industrial groups, spanning energy, smart infrastructure, industrial machinery, aerospace, and defence.

MHI Group combines cutting-edge technology with deep experience to deliver innovative, integrated solutions that help to realize a carbon neutral world, improve the quality of life and ensure a safer world.

With a perspective gained from 130 years of history and tradition on land, at sea, in the sky and in space, MHI address social issues and take on challenges for the future.

Haeusler AG (Roll Bending Machine)

Haeusler Group is a globally active company and continues to set new standards in the metal forming industry. Our inventive pioneering spirit has been growing for over 85 years and draws on a long family tradition. We are looking for new challenges – on which we can grow to ensure your success.

HAEUSLER has many years of experience in the development and production of production machines for on- and offshore applications. HAEUSLER is the world's leading manufacturer of roll bending pipe mills for large pipe production. HAEUSLER VRM and HDR machines are ideally suited for the production of components for drilling platforms and other steel constructions. Here are the key sector where it facilitates its services –

- Plate bending for Steel construction
- Plate bending for the production of wind turbines
- Plate bending for the aerospace
- Plate bending for Power plant construction
- Plate bending for the container construction
- Plate bending for shipbuilding
- Plate bending for the automotive industry
- Plate bending for the production of boilers and heat exchangers
- Plate bending for the tunnel & mining industry

Raw Material

The basic raw material required for manufacturing of LSAW and HSAW pipe manufacturing are HR Coils and Plates. However, the supplier needs to be approved supplier of Aramco, in case of supply of raw material for pipes to be delivered to Aramco.

The Consultants had sought copies of sample invoices/ purchase orders of some of the suppliers from NPC and the same were duly provided by NPC for review by the Consultants. The list of current raw material suppliers of the Company include –

1. Voest- Alpine (Austria)
2. SMI Steel LLC
3. Nippon Steel Company
4. POSCO (Korea)
5. Japan Future Enterprise (Japan)
6. NISCO-CHINA
7. JIANGYIN-CHINA
8. DILLINGER-Germany
9. Tata Steel Limited
10. Sinomaterial International Co., Limited
11. Saudi Iron and Steel Company
12. Golden Source Steel Company
13. Morouge Ltd.

The Consultants understand that the Company will continue to import the basic rate material from these established and approved players in future as well.

Acquisition/ Project Cost

The total cost of acquisition has been finalised at SAR 382,500,000 or USD 102.00 Million and will be funded by promoters' equity of USD 25.50 Million (Rs. 227.97 Crores) and Long-Term Debt of USD 76.50 Million (Rs. 683.92 Crores) and the same has been exhibited in the table below –

All Figures in USD Million*	
Description	**Total**
Acquisition Cost	102.00
Upgrade/ Modernisation of HSAW Mill	3.00
Total Means of Finance	**105.00**

Source: MIIL and SCPL Estimates

*** - Exchange rate considered is 1 USD = Rs. 89.4013 (exchange rate as on 29thj November 2025)**

Here it should be noted that the cost of acquisition of NPC has been considered at USD 102.00 Million and additional USD 3.00 Million has been considered for upgrade of HSAW Pipe mill, which has current capability to produce pipes of dimeter upto 84 inch. This is essential, as the newer orders from local customers in recent times indicate that the market trend for diameters upto 100 inches to 120 inches is gaining popularity. The upgrade project is expected to be completed within 12 months of acquisition of NPC by Man Group. This is the reason for considering the overall Project Cost for USD 105.00 Million.

The Consultants had sought the quotation for the proposed upgrade/ modernisation project of HSAW Pipe Mill from the Company, however the Consultants were informed that the estimate is based on experience of Man Group in undertaking implementation of SAW Pipe projects in past and that the Group will float request for proposals for upgrade/ modernisation project immediately after acquisition of NPC.

Utilities

For the successful operation of the project following utilities are installed and were operational at the time of site visit by team from SCPL.

Power

Based on the discussion with the management of the company and site visit undertaken by the consultants it is noted that the power requirement is being fulfil by a permanent incoming supply of 17.5 KV issued from Saudi Electricity Company (SECO).
The power supply as taken by the NPC is sufficient for operations of both the HSAW and LSAW pipe plants simultaneously.

Water

Water requirement is fulfilled through 2 Nos. of Borewell available within the premise of company. The depth of both the borewells is 150 meters. There are 3 RO systems installed at site for treatment of water for drinking purpose and housekeeping purposes. The installed capacities of these RO systems were 216 Cu. M./ Day, 177.6 Cu. M./ Day and 199.2 Cu. M./ Day.

All the 3 RO Plants were operational at the time of site visit undertaken by the Consultants.

Water Treatment Plant

The Consultants note that there is a water treatment plant with installed capacity of 100 Cu. M./ Day, which has been installed and was operational at the time of the site visit undertaken by the Consultants.

Testing Laboratory

The Company maintains a full fledged laboratory at the location to undertake destructive and non-destructive testing of raw material and finished goods. The testing machines located at site include the hyposonic tester, X-Ray machine, chemical analyser, universal testing machine, hardness testing machine, various sizes of micro-scopes etc.

The Consultants had sought the copies of various tests being undertaken by the Company and the same were provided at site for review purposes.

Staff Colony

The Consultants note that NPC has a full-fledged staff quarters located at the plant site itself. The staff quarters are sufficient to have a total of 350 employee (for foreign employees). The Arab (local) employees of the Company prefer staying at their respective homes.

Additionally, the Company also has staff quarters for middle and senior management of the Company, located at the head office at Al Khobar in the city. The senior employee of the Company including the General Manager of the Company, stay at the Al Khobar.

The Company has its own mess/ canteen for the food requirement of officers as well as employees of the factory, where Arabian and Continental food is served.

Residual Life of Plant

HSAW Pipe Plant

The HSAW Pipe plant of the Company was established in the year 1980 and has been upgraded a few times. The plant and machinery have been maintained in a fair condition. However, the instrumentation system of the Forming Machine, T-Cross Welding Machine and Hydrotesting machine located at site are analogue in nature and need an upgrade. Taking these points and site visit observation into consideration, the Consultant opine that the residual life of these sections of the HSAW Pipe plant in the prevailing circumstances will be 10-12 years, if proper maintenance philosophy is followed.

However, here it should be noted that Man Group proposes to undertake an upgrade and modernisation project on HSAW Pipe Mill, immediately after acquisition of NPC. This is on account of the fact that the current HSAW Pipe Mill is capable of manufacturing a maximum diameter size of 84 inches and the current set of orders being floated by customer in KSA are mostly for pipe diameters in range of 100 inches to 120 inches. The proposed upgrade/ modernisation of the line will result in the HSAW Pipe Mill to be able to manufacture pipes with maximum diameter upto 120 inches. With the proposed upgrade, the residual life of the existing HSAW Pipe mill is expected to extend by another 5-6 years and hence the extended residual life of the existing HSAW Pipe Mill after upgrade/ modernisation project is expected to be 15-17 years.

Further there have been new additions in the line which include Offline Final Welding Machine, X-Ray (Digital), Overhead Cranes, RFID System, Online Ultrasonic Testing Machines etc. during the last 5-6 years period. The Consultants opine that the residual life of these sections of plant and machinery will be 17-20 years period, if proper maintenance philosophy is followed.

With the proposed upgrade, the residual life of the Offline Final Welding Machine is expected to extend by another 5-6 years and hence the extended residual life of the existing HSAW Pipe Mill after upgrade/ modernisation project is expected to be 22-25 years.

LSAW Pipe Plant

The LSAW Pipe plant of the Company was established during the year 2001. The plant and machinery have been maintained in fair to good condition. The instrumentation system of the entire plant is digital and updated version. Taking these points and site visit observations into consideration, the Consultants opine that the residual life of LSAW Pipe plant will be in a range of 14-18 years.

The machine-wise residual life as estimated for the Project has been summarised in the table below –

Asset details	Purchase Year	Age of machine	Residual Life
HPM			
Pipe Mill 1 & 2 (Forming / Inside T-Cross / Tack Welding)	1980	45	10 to 12
T-Cross Welding	1980	45	10 to 12
Inside Pipe Cleaning	2015	10	10 to 12
Final Welding machine	2015	10	17 to 20
X-ray machine (Film type)	1980	45	10 to 12
X-ray machine (Digital)	2016	9	17 to 20
Hydrotesting machine	1980	45	10 to 12

Asset details	Purchase Year	Age of machine	Residual Life
Online UST	2013	12	17 to 20
Magnetic Particle Testing	2011	14	12 to 15
Over Head Crane (HPM)	1980	45	10 to 12
Over Head Crane (FWM)	2015	10	17 to 20
Tracing System (RFID)	2017	8	17 to 20
Other misc items	1980	45	10 to 12
Electrical	1980	45	10 to 12
LPM			
Edge milling machine	2001	23	14 to 18
Crimping press machine	2011	14	17 to 20
Roll bending machine	2001	24	14 to 18
Tack welding machine	2001	24	14 to 18
Inside SAW	2001	24	14 to 18
Outside SAW	2001	24	14 to 18
Expander machine	2001	24	14 to 18
End Facer machine	2001	24	14 to 18
Hydrostatic Testing machine	2001	24	14 to 18
Digital X-ray machine	2001	24	14 to 18
Auto Line UST	2001	24	14 to 18
Over head Crane (LPM)	2000	25	14 to 18
Gantry Crane	2010	15	14 to 18
Tracing System (RFID)	2017	8	17 to 20
Other misc items	2001	24	14 to 18
Electrical	2001	24	14 to 18
Workshop/ Laboratory			
Tensile and Bend Test Machine (2 units)	2012	13	10 to 12
Charpy Impact Test (1 unit)	2010	15	10 to 12
DWTT (1 unit)	2001	24	10 to 12
Hardness Tester 1	1979	46	Need Replaced
Hardness Tester 2	1979	46	Need Replaced
Hardness Tester 3	2011	14	10 to 12
Micro and Macroscopic Examination 1	1995	30	Need Replaced
Micro and Macroscopic Examination 2	2019	6	10 to 12
Chemical Analyser 1	1994	31	Need Replaced
Chemical Analyser 2	2015	10	10 to 12
Chemical Analyser 3	2015	10	10 to 12
Chemical Analyser 4	2015	10	10 to 12
Machinery of Test Preparation (Lathe)	1979	46	Need Replaced
Machinery of Test Preparation (Milling machine)	2010	16	10 to 12
Water treatment facility (Drinking facility)	2011	14	10 to 12
Water treatment facility (Waste water management)	2017	8	10 to 12
Yard Crawler crane - 250 T	1979	46	Need Replaced

Asset details	Purchase Year	Age of machine	Residual Life
Yard Crawler crane - 250 T	1979	46	Need Replaced
Yard Crawler crane - 150 T	2002	23	10 to 12
Yard Crawler crane - 150 T	2006	19	10 to 12
Yard Crawler crane - 150 T	2011	14	10 to 12
Pipe Transportation Trailers	2011	14	10 to 12

Source: SCPL Estimates

Manpower

The Consultants understand that since the entire plant is not operational at the moment, with HSAW Pipe mill being in stand-by condition, the manpower strength of the Company is limited to manpower requirement for 2 shift of operations or 340 personnel, which have been deployed on LSAW Pipe plant, since Aramco's order was being processed. With current manpower, the Company will be able to operate both the plants on single shift basis, in case order for both HSAW and LSAW Pipes are available. So, the Company will have to recruit additional manpower in case both lines are to be operated on optimal basis.

As per the information received from the company and discussion with executive of the company, the consultants have understood that there will be a total of 567 manpower for peak utilisation. The total manpower required and their respective salary has been presented below –

Description	Unit	Value	Unit	Value
Manpower Requirement				
General Manager	Nos.	1	USD/Month	7,420
Senior Managers	Nos.	4	USD/Month	4,857
Middle Management/ Engineers	Nos.	24	USD/Month	4,018
Administrative Support	Nos.	38	USD/Month	1,478
Supervisors	Nos.	50	USD/Month	1,925
Skilled	Nos.	150	USD/Month	1,231
Semi-Skilled	Nos.	300	USD/Month	895
Total Manpower	**Nos.**	**567**		

Source: MIIL and SCPL Estimates

Along with the salary per month, employees will be facilitated with an annual benefit of 15% of its base salary.

Here it should be noted that as per the Saudisation policy, every unit operational within KSA needs to have minimum 35% of the employee as Saudi Nationals, otherwise the unit is termed as Red Zone unit and is heavily penalised. Any Unit, which has Saudi National at 50% or above of the total employees, then the unit is categorised as Green Zone company and does not have to pay any penalty. Currently, the total Saudi Employees in NPC are 50% or above range and hence NPC falls in the Green Zone Unit category.

Statutory Approvals and Clearances

The Consultants had sought the copies of approvals and clearances from the Company, and the copies of available approvals and clearances was dully provided by the company. The details of approvals and clearances have been exhibited below –

Name of the Documents	Reference	Remark
(NPC) Company Registration Certificate	Certificate No. - 2051003936	Valid upto 09th August 2028
Certificate of Authority to use the Official API Monogram	License Number – 2B - 0073	Expiration Date – 7th June 2024 Entitled API Spec. Q1 and API-2B
Certificate of Authority to use the Official API Monogram	License Number – 5L - 0140	Expiration Date – 7th June 2024 Entitled API Spec. Q1 and API-5L
Certificate for Management System as per DIN EN ISO 9001 : 2015	Registration No. 04 100 950464 Certification Body at TUV NORD CERT GmbH	Valid until – 22nd May 2023
SAGIA License	License No. – 1210330215119	Valid till – 15th August 2024
NPC Khobar municipality License	License No. – 3909674800	Received
NPC Factory municipality License	License No. – 40082118825	Received
NOC from General Authority of Competition	App. No. – RN0004171	Received as on 20th October 2025
Industrial Facility License of MISIC	License No. 45229217	Valid until – 22nd May 2026
Environmental License	Applied for renewal.	

Source: NPC

SCPL Observations/ Comments

From the table above, it is noted that the copies of some of the approvals and clearances as shared are outdated/ expired and require renewal. While for some approvals, the current Management of NPC has already applied for. The Consultants believe that NPC would have these updated by now and the updated copies were not shared in data room with MIIL, for scrutiny purposes, as both the HSAW Pipe and LSAW Pipe Mills of NPC were operational at the time of site visit as conducted by SCPL on 2nd and 3rd December 2025.

The Lenders may request the Company to get the updated copies of approvals and clearances from NPC and validate the same.

Site Visit Observation

The team of consultants have visited the plant located at Dhahran, Dammam, Saudi Arabia as date 02nd and 3rd December 2025. Based on the site visit undertaken by consultants following observation has been made as listed below –

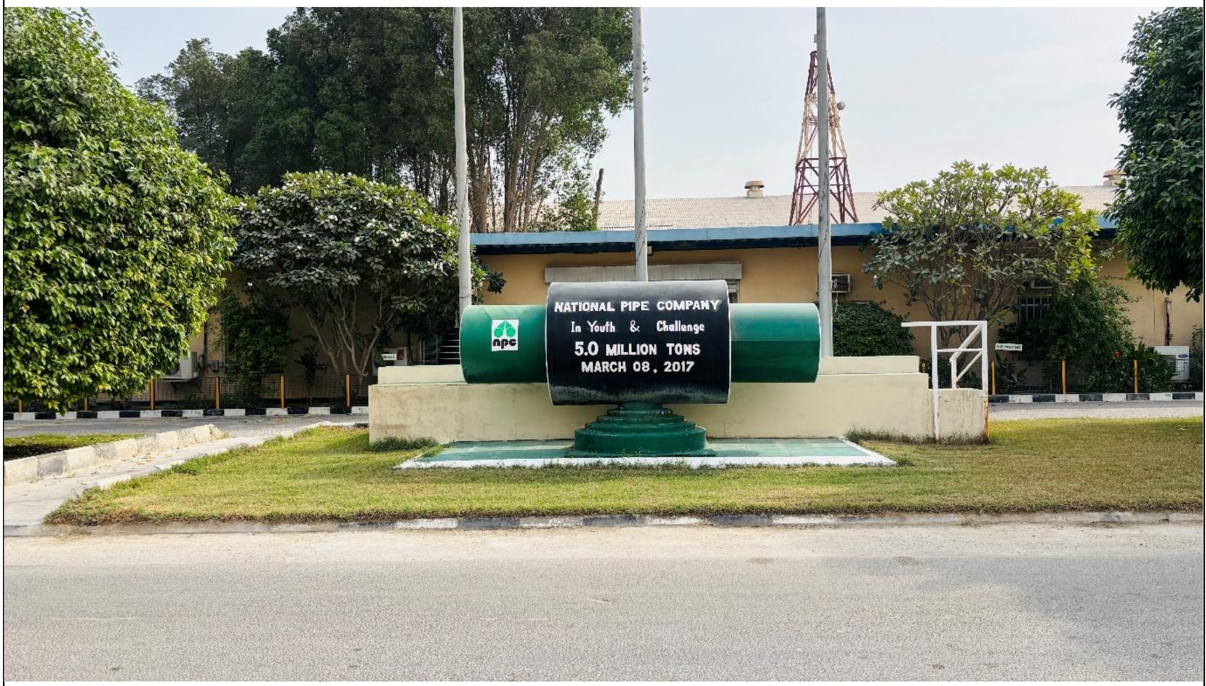
- The plant is well connected with road and completely surrounded by the metal fencing.
- There are mainly three plant sheds namely, First one is HSAW pipe manufacturing plant, Second one is, LSAW pipe manufacturing plant and Third one is Final Welding pipe manufacturing plant.

- Both HSAW & LSAW pipe manufacturing plant was running at time of site visit.
- There is a RO water plant within the premises of the NPC factory.
- The plant is well connected with internal road within the plant for smooth movement of Raw material and finished goods.
- There is well maintained testing laboratory located at the site for all testing’s to be undertaken in-house.
- Also, there are security cabins at every entry and exit gate of the factory.
- There is permanent power connection with an incoming supply of 17.5 KV which is further supported with step up transformer.
- There are separate canteens for Saudi resident and expats (like Indians, Bangladeshi, Philippines)
- There was a diesel storage tank within the boundary of factory for refuelling of diesel vehicle (Trucks, Cranes, etc)
- Lab Equipment’s were also physically verified & found all the machine in fair condition. AMC were available for critical machines like Spectro, UTS machines etc.
- Water requirement is fulfilled through 2 Nos. of Borewell available within the premise of company.
- Power is being supplied by SCECO at 15 KV supply voltage. Stepdown Transformers of suitable capacity i.e. 1500 KVA – 2 Nos., 2000 KVA – 1 No. & 2500 KVA – 2 Nos at 60 Hz is installed at site, which supply the power to entire facility.
- It was noted during the discussion with site engineers that dedicated earthing’s are provided at each machine, which is used as Power & Electronic Earth both.

Site Snaps

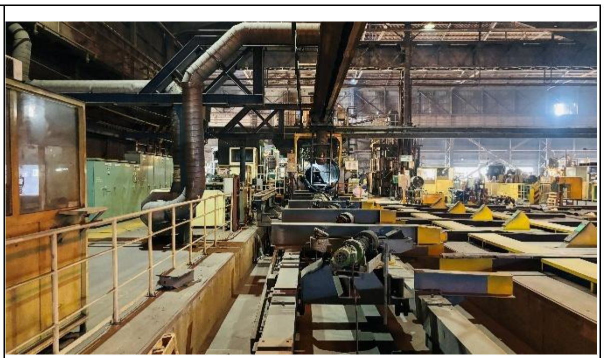
NPC – Manufacturing Plant

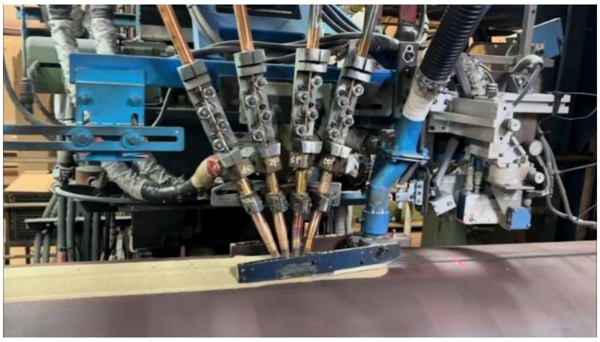
Plant snaps taken during the visit have been exhibited in the table below –

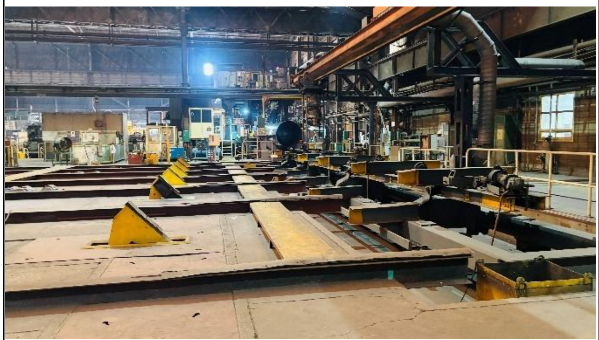


LSAW ↓	↓ HSAW
	
 Plates	 HR Coils
 Edge Milling	 Coil Flatter
 3-Roll Bending Process	 Tack Welding/ Pre-Welding


Tack Welding/ Pre-Welding


Conveyance Channel


Outer Submerged arc welding


Conveyance Channel

Inner Submerged arc welding


Final Welding


Mechanical Expander


Hydrosonic Testing

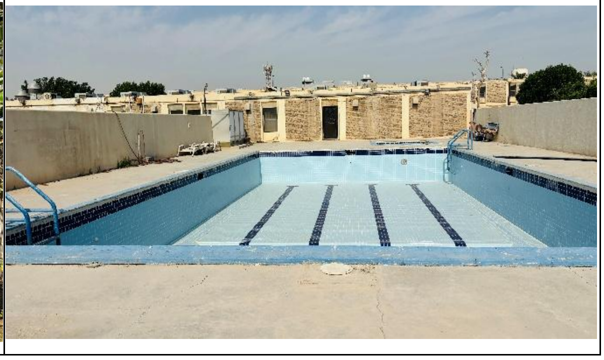
 Stock Yard	 Final SAW pipe
Utilities and Laboratory	
 RO Plant	 Universal Tensile Testing Machine
 NDT Training Center	 Spare Storage room
 Coils/ Plates Unloading	 Diesel Storage Tank

Source: SCPL Estimates

NPC – Office

Plant snaps taken during the visit have been exhibited in the table below –

Office Division ↓	↓ Residence Division
	
	

Conclusion

Based on the technical assessment undertaken, following points have been concluded as below –

- The plant is located at Al Khobar, Dammam, Saudi Arabia within an area of 400,000 Sq. M.
- The installed capacity of the target company is 250,000 TPA for HSAW pipe Manufacturing and 180,000 TPA for LSAW pipe manufacturing.
- Both the LSAW and HSAW plant was operation during the Site visit.
- NPC - Plant is located at a satellite coordinates of Latitude 26°13'44.34"N and Longitude 50° 3'26.02"E.
- NPC - Office is located at a satellite coordinates of Latitude 26°19'29.50"N and Longitude 50°11'38.67"E, by road, it is around 36.00 Km away from the NPC – Plant.
- The plant and machinery have been supplied and installed by renowned technology provider of the segment like (Nippon Steel Welding & Engineering, Sumitomo Heavy Industries, SMS Germany (PWS), Mitsubishi Heavy Industries, and Haeusler AG, Etc.)
- The Raw material will be procured from the open market from renowned supplier like (Nippon Steel Company, POSCO (Korea), Japan Future Enterprise, and NISCO – China, Etc.)
- The power requirement will be fulfilled by a permanent incoming supply of 17.5 KV issued from Saudi Electricity Company (SECO).
- Water requirement is fulfilled through 2 Nos. of Borewell available within the premise of company. Also, there is a RO water plant within the premises of the NPC factory.
- The Company maintains a full-fledged staff quarter and officers' quarters for employees at factory site and at head office site.
- The overall project cost has been estimated at USD 105.00 Million which includes the NPC acquisition cost of USD 102.00 Million and upgrade/ modernisation cost of HSAW Pipe Mill at USD 3.00 Million.
- There will be a total manpower of 597 for peak utilisation.
- Both the HSAW pipe mill and the LSAW pipe mill of the Company were operational at the time of site visit undertaken by the Consultants.
- The Project is found to be technically feasible preposition.

Financial Assessment

Introduction

The process of valuation of an entity, utilising the Discounted Cashflow Techniques requires projection of revenue and cost streams of the Company, considering the basic assumption that the Company will remain going concern. For purpose of estimating the Business Enterprise Value of National Pipe Company Limited, SCPL have undertaken a projection of 10 years, commencing from the current year 2024. This chapter lists out all the assumptions as considered by the Consultants while undertaking the projections and arriving at the Enterprise Business Value and Equity Value of the Company.

Capacity and Utilization

As discussed previously, the installed capacity of the HSAW pipe plant is 250,000 TPA and LSAW pipe plant is 180,000 TPA. The Consultants have considered the following while assessing the capacity utilisation levels for the Project –

- The HSAW Pipe plant is operational during CY 2026 but a lower capacity utilisation has been considered considering the current order book.
- The peak capacity utilisation level for the HSAW Pipe plant has been considered at 65%, taking into consideration the age and OD profile of HSAW Pipe expected in the orders. However during the projection period the same has reached a maximum of 59.50% by CY 2032.
- The peak utilisation level for LSAW Pipe plant has been considered at 60%, which is taking into consideration the variety of OD profile of LSAW Pipes expected in orders.
- The peak operational days for the Project in a year have been considered to be 330 days.

Considering the assumptions, the capacity and capacity utilisation for the Project has been presented in the table below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Days in Year		365	365	365	365	365	365	365
Operating Days		330	330	330	330	330	330	330
H-SAW Pipe								
Installed Capacity	TPA	250,000	250,000	250,000	250,000	250,000	250,000	250,000
Utilisation Level	%	38.50%	42.00%	45.50%	49.00%	52.50%	56.00%	59.50%
Production		96,250	105,000	113,750	122,500	131,250	140,000	148,750
L-SAW Pipe								
Installed Capacity	TPA	180,000	180,000	180,000	180,000	180,000	180,000	180,000
Utilisation Level	%	48.50%	52.00%	55.50%	59.00%	60.00%	60.00%	60.00%
Production		87,300	93,600	99,900	106,200	108,000	108,000	108,000

Source: SCPL Estimates

Stock Movement

The stock movement for the project, taking into consideration the inventory levels, has been presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe								
Opening Balance	Tons	4,500	12,130	13,233	14,336	15,438	16,541	17,644
Production	Tons	96,250	105,000	113,750	122,500	131,250	140,000	148,750
Closing Balance	Tons	12,130	13,233	14,336	15,438	16,541	17,644	18,747
Stock Available for Sales	Tons	88,620	103,897	112,647	121,398	130,147	138,897	147,647
L-SAW Pipe								
Opening Balance	Tons	7,500	11,002	11,796	12,590	13,384	13,611	13,611
Production	Tons	87,300	93,600	99,900	106,200	108,000	108,000	108,000
Closing Balance	Tons	11,002	11,796	12,590	13,384	13,611	13,611	13,611
Stock Available for Sales	Tons	83,798	92,806	99,106	105,406	107,773	108,000	108,000

Source: SCPL Estimates

Sales Quantity, Selling Price and Realization

Sales Quantity

The stock available for sales, after taking into consideration the stock movement has been summarised in the table below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe	Tons	88,620	103,897	112,647	121,398	130,147	138,897	147,647
L-SAW Pipe	Tons	83,798	92,806	99,106	105,406	107,773	108,000	108,000

Source: SCPL Estimates

Selling Price

The Consultants had sought copies of the invoices raised by NPC with Aramco, to understand the prevailing price of the contract.

Additionally, the Consultants had sought the selling price of the products from MIIL and had also individually sought quotation from other HSAW Pipe and LSAW Pipe manufacturers in India and internationally. The marine transport and insurance cost was added to the base FOB prices and the selling prices for the finished products was arrived at for purpose of financial evaluation. The selling prices as considered for the purpose of financial evaluation have been presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe	USD/ Ton	1,225	1,225	1,225	1,225	1,225	1,225	1,225
L-SAW Pipe	USD/ Ton	1,315	1,315	1,315	1,315	1,315	1,315	1,315

Source: SCPL Estimates

Sales Realisation

The sales realisation based on the sales quantity and selling prices as discussed above, has been presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe	USD Mn	108.56	127.27	137.99	148.71	159.43	170.15	180.87

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
L-SAW Pipe	USD Mn	110.19	122.04	130.32	138.61	141.72	142.02	142.02
Sales Realisation	**USD Mn**	**218.75**	**249.31**	**268.32**	**287.32**	**301.15**	**312.17**	**322.89**

Source: SCPL Estimates

Variable Cost

In this section of the chapter, the Consultants have discussed the various variable costs (i.e. Raw material cost, Stores and consumable cost, Power cost, Water cost and Repairs & maintenance cost etc.) associated with the Project, which are based on the industry norms and experience of the Consultants in undertaking similar assignments in recent past.

Raw Material Cost

The Consultants had sought the production and raw material consumption data of NPC for both the HSAW Pipe (when last operational) and LSAW Pipe plant. Additionally, to arrive at the raw material cost, the Consultant had also considered the industry norms prevailing in the international markets and based on the same, the raw material consumption norm has been considered for the financial evaluation of the Project.

For the purpose of raw material prices, the Consultants had called for quotation from international suppliers of HR Coil and HR Plates and had additionally sought copies of invoices for these raw materials from MIIL as well. Based on these inputs, the final raw material cost was arrived at after adding marine transport and insurance.

The raw material consumption norm and raw material purchase prices as considered for the Project have been presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe								
HR Coil Cost								
HR Coil	Ton/ Ton	1.030	1.030	1.030	1.030	1.030	1.030	1.030
Annual Requirement	Tons	99,138	108,150	117,163	126,175	135,188	144,200	153,213
HR Coil Purchase Price	USD/ Ton	865	865	865	865	865	865	865
HR Coil Cost	***USD Mn***	***85.75***	***93.55***	***101.35***	***109.14***	***116.94***	***124.73***	***132.53***
Welding Wire								
Welding Wire	Kg./ Ton	3.00	3.00	3.00	3.00	3.00	3.00	3.00
Annual Requirement	Tons	289	315	341	368	394	420	446
Welding Wire Price	USD/ Ton	1,051.44	1,051.44	1,051.44	1,051.44	1,051.44	1,051.44	1,051.44
Welding Wire Cost	***USD Mn***	***0.30***	***0.33***	***0.36***	***0.39***	***0.41***	***0.44***	***0.47***
Flux Cost								
Flux Required	Kg./ Ton	4.20	4.20	4.20	4.20	4.20	4.20	4.20
Annual Requirement	Tons	404	441	478	515	551	588	625
Welding Wire Price	USD/ Ton	1,387	1,387	1,387	1,387	1,387	1,387	1,387

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Welding Wire Cost	***USD Mn***	***0.56***	***0.61***	***0.66***	***0.71***	***0.76***	***0.82***	***0.87***
H-SAW Pipe Raw Material Cost	**USD Mn**	**86.62**	**94.49**	**102.37**	**110.24**	**118.12**	**125.99**	**133.86**
L-SAW Pipe								
HR Plate Cost								
HR Plate	Ton/ Ton	1.020	1.020	1.020	1.020	1.020	1.020	1.020
Annual Requirement	Tons	89,046	95,472	101,898	108,324	110,160	110,160	110,160
HR Plate Cost	USD/ Ton	925	925	925	925	925	925	925
HR Plate Cost	***USD Mn***	***82.37***	***88.31***	***94.26***	***100.20***	***101.90***	***101.90***	***101.90***
Welding Wire								
Welding Wire	Kg./ Ton	3.00	3.00	3.00	3.00	3.00	3.00	3.00
Annual Requirement	Tons	262	281	300	319	324	324	324
Welding Wire Price	USD/ Ton	1,051	1,051	1,051	1,051	1,051	1,051	1,051
Welding Wire Cost	***USD Mn***	***0.28***	***0.30***	***0.32***	***0.33***	***0.34***	***0.34***	***0.34***
Flux Cost								
Flux Required	Kg./ Ton	10.00	10.00	10.00	10.00	10.00	10.00	10.00
Annual Requirement	Tons	873	936	999	1,062	1,080	1,080	1,080
Welding Wire Price	USD/ Ton	1,387	1,387	1,387	1,387	1,387	1,387	1,387
Welding Wire Cost	***USD Mn***	***1.21***	***1.30***	***1.39***	***1.47***	***1.50***	***1.50***	***1.50***
L-SAW Pipe Raw Material Cost	**USD Mn**	**83.85**	**89.91**	**95.96**	**102.01**	**103.74**	**103.74**	**103.74**
Total Raw Material Cost	**USD Mn**	**170.47**	**184.40**	**198.32**	**212.25**	**221.85**	**229.73**	**237.60**

Source: SCPL Estimates

Power Cost

The power consumption norm for the Project has been considered in line with the power consumption levels already experienced by NPC for manufacturing of LSAW Pipes, while the power consumption norms for HSAW Pipes is based on the industry norms.

The power prices for the Project as considered are in line with the prevailing power prices, as experienced by NPC.

Considering the same, the power cost for the Project as estimated has been presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe								
Power Cost								
Power Consumption	Kwh/ Ton	215.00	215.00	215.00	215.00	215.00	215.00	215.00

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Annual Requirement	Mwh	20,694	22,575	24,456	26,338	28,219	30,100	31,981
Power Price	USD/ Kwh	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Power Cost	***USD Mn***	***0.99***	***1.08***	***1.17***	***1.26***	***1.35***	***1.44***	***1.54***
L-SAW Pipe		-	-	-	-	-	-	-
Power Cost								
Power Consumption	Kwh/ Ton	215.00	215.00	215.00	215.00	215.00	215.00	215.00
Annual Requirement	Mwh	18,770	20,124	21,479	22,833	23,220	23,220	23,220
Power Price	USD/ Kwh	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Power Cost	***USD Mn***	***0.90***	***0.97***	***1.03***	***1.10***	***1.11***	***1.11***	***1.11***
Total Power Cost	**USD Mn**	**1.89**	**2.05**	**2.20**	**2.36**	**2.47**	**2.56**	**2.65**

Source: SCPL Estimates

Material Handling and Transportation Cost

Material handling and transportation cost for the Project is the cost incurred by the Company in transporting the raw material and finished goods within the factory premises. The material handling and transportation cost as considered are based on the industry norm and experience of the Consultants in undertaking similar assignments in past. The material handling and transportation cost as considered for the Project for purpose of financial evaluation has been presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe								
Material Handling and Transport	USD/ Ton	2.65	2.65	2.65	2.65	2.65	2.65	2.65
Material Handling and Transport	***USD Mn***	***0.26***	***0.28***	***0.30***	***0.32***	***0.35***	***0.37***	***0.39***
L-SAW Pipe								
Material Handling and Transport	USD/ Ton	2.65	2.65	2.65	2.65	2.65	2.65	2.65
Material Handling and Transport	***USD Mn***	***0.23***	***0.25***	***0.26***	***0.28***	***0.29***	***0.29***	***0.29***
Total Material Handling and Transport	**USD Mn**	**0.49**	**0.53**	**0.57**	**0.61**	**0.63**	**0.66**	**0.68**

Source: SCPL Estimates

Repairs and Maintenance Cost

Repairs and maintenance cost for the Project have been considered in line with the experience of NPC in operating the plants. The repairs and maintenance cost for HSAW Pipe plant has been considered slightly higher on account of the fact that the line is aging. The repairs and maintenance cost for the Project as considered are presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe								
Repairs and Maintenance Cost	USD/ Ton	8.39	8.39	8.39	8.39	8.39	8.39	8.39
Repairs and Maintenance Cost	***USD Mn***	***0.81***	***0.88***	***0.95***	***1.03***	***1.10***	***1.17***	***1.25***
L-SAW Pipe								
Repairs and Maintenance Cost	USD/ Ton	8.39	8.39	8.39	8.39	8.39	8.39	8.39
Repairs and Maintenance Cost	***USD Mn***	***0.73***	***0.79***	***0.84***	***0.89***	***0.91***	***0.91***	***0.91***
Repairs and Maintenance Cost	**USD Mn**	**1.54**	**1.67**	**1.79**	**1.92**	**2.01**	**2.08**	**2.15**

Source: SCPL Estimates

Other Direct Expenses

The other direct expenses for the Project as considered include –

1. Inland transportation cost of raw material and finished goods
2. Cost of acquiring quality certifications like ISO, API etc.
3. Manpower training cost etc.

The other direct expenses as considered are based on the experience of NPC in handling the unit, industry norms and experience of the Consultants in undertaking similar assignment in past. The other direct expenses as considered for the Project have been presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
H-SAW Pipe								
Other Direct Expenses	USD/ Ton	27.96	27.96	27.96	27.96	27.96	27.96	27.96
Other Direct Expenses	***USD Mn***	***2.69***	***2.94***	***3.18***	***3.43***	***3.67***	***3.91***	***4.16***
L-SAW Pipe								
Stores and Consumable	Rs./Ton	27.96	27.96	27.96	27.96	27.96	27.96	27.96
Stores and Consumable	***USD Mn***	***2.44***	***2.62***	***2.79***	***2.97***	***3.02***	***3.02***	***3.02***

Source: SCPL Estimates

Manpower Cost

The overall manpower requirement for the project has been estimated at 441 personnel for the first two years and based on peak capacity it will be increased to 567 personnel including executives and non-executives. The manpower requirement of the project and their expected monthly salaries have been provided in the table below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Manpower Requirement								
General Manager	Nos.	1	1	1	1	1	1	1
Senior Managers	Nos.	4	4	4	4	4	4	4
Middle Management/ Engineers	Nos.	16	20	24	24	24	24	24
Administrative Support	Nos.	20	30	38	38	38	38	38
Supervisors	Nos.	40	45	50	50	50	50	50
Skilled	Nos.	120	135	150	150	150	150	150
Semi-Skilled	Nos.	240	270	300	300	300	300	300
Total Manpower	**Nos.**	**441**	**505**	**567**	**567**	**567**	**567**	**567**
Salaries								
General Manager	USD/ Month	7,420	7,420	7,420	7,420	7,420	7,420	7,420
Senior Managers	USD/ Month	4,857	4,857	4,857	4,857	4,857	4,857	4,857
Middle Management/ Engineers	USD/ Month	4,018	4,018	4,018	4,018	4,018	4,018	4,018
Administrative Support	USD/ Month	1,478	1,478	1,478	1,478	1,478	1,478	1,478
Supervisors	USD/ Month	1,925	1,925	1,925	1,925	1,925	1,925	1,925
Skilled	USD/ Month	1,231	1,231	1,231	1,231	1,231	1,231	1,231
Semi-Skilled	USD/ Month	895	895	895	895	895	895	895
Benefits	%	15%	15%	15%	15%	15%	15%	15%
Manpower Cost	**USD Mn**	**7.73**	**8.92**	**10.06**	**10.06**	**10.06**	**10.06**	**10.06**

Source: SCPL Estimates

Here it should be noted that the Consultants have considered benefits to the tune of 15% up and above per annum the base salaries as indicated in the table above.

Fixed Costs

Administrative Expenses

The administrative expenses for the Project have been considered at 2.50% of the sales realisation, subsequently the costs have been frozen at same level. The administrative expenses as considered include the following –

- The salaries and remuneration of the Directors
- Rental cost for land
- Audit cost
- Insurance cost
- Travelling, boarding and lodging cost of administrative staff
- Stationary cost etc.

Selling and Distribution Expenses

The selling and distribution cost of the Project has been considered at 0.75% of the sales realisation, subsequently the costs have been frozen at same level. The selling and distribution expenses as considered include –

- The discounts provided by Company on bulk purchases
- Losses due to pilferage
- Boarding, travelling and lodging cost of sales team

Based on the above, the fixed costs as estimated for the Project have been presented in the exhibit below –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Administrative Expenses	% of Sales	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%	2.50%
Selling and Distribution	% of Sales	0.75%	0.75%	0.75%	0.75%	0.75%	0.75%	0.75%
Administrative Expenses	USD Mn	5.48	6.25	6.25	6.25	6.25	6.25	6.25
Selling and Distribution	USD Mn	1.64	1.87	1.87	1.87	1.87	1.87	1.87

Source: SCPL Estimates

Working Capital Norms

The working capital norm of the Project as considered and based on the benchmarks of the industry and include the following –

Description	Unit	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Raw Material	Days	60	60	60	60	60	60	60
Work in Progress	Days	1	1	1	1	1	1	1
Finished Goods	Days	45	45	45	45	45	45	45
Debtors/ Receivables	Days	45	45	45	45	45	45	45
Trade Creditors	Days	15	15	15	15	15	15	0
Other Current Liabilities	Days	30	30	30	30	30	30	30
Margin Money	%	25%	25%	25%	25%	25%	25%	25%

Source: SCPL Estimates

Based on the holding norms as discussed above, the working capital requirement of the Project as estimated is presented in the exhibit below –

All Figures in USD Million							
Description	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Raw Material	28.02	30.31	32.60	34.89	36.47	37.76	39.06
Work in Progress	0.49	0.53	0.57	0.61	0.63	0.65	0.68
Finished Goods	23.63	25.68	27.60	29.38	30.60	31.59	32.58

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Debtors/ Receivables	26.97	30.74	33.08	35.42	37.13	38.49	39.81
Operating Current Assets	**79.11**	**87.26**	**93.85**	**100.30**	**104.83**	**108.49**	**112.13**
Trade Creditors	-	7.58	8.15	8.72	9.12	9.44	9.76
Other Current Liabilities	1.16	1.30	1.43	1.47	1.50	1.51	1.53
Operating Current Liabilities	**1.16**	**8.87**	**9.58**	**10.19**	**10.61**	**10.95**	**11.29**
Working Capital Gap	**77.95**	**78.38**	**84.27**	**90.10**	**94.21**	**97.54**	**100.83**
Margin Money	19.49	19.60	21.07	22.53	23.55	24.39	25.21
Eligible for Borrowing	**58.47**	**58.79**	**63.20**	**67.58**	**70.66**	**73.16**	**75.63**
Bank Borrowing	**-**	**-**	**-**	**-**	**-**	**-**	**-**
Utilisation	100%	100%	100%	100%	100%	100%	100%
Actual Borrowing	-	-	-	-	-	-	-
Interest Rate	8.25%	8.25%	8.25%	8.25%	8.25%	8.25%	8.25%
Interest for Period	-	-	-	-	-	-	-

Source: SCPL Estimates

Cost of Acquisition/ Means of Finance

The total cost of acquisition has been finalised at SAR 382,500,000 or USD 102.00 Million and will be funded by promoters' equity of USD 25.50 Million (Rs. 227.97 Crores) and Long-Term Debt of USD 76.50 Million (Rs. 683.92 Crores) and the same has been exhibited in the table below –

All Figures in USD Million*	
Description	**Total**
Promoters Equity	26.25
Term Loan from Lenders	78.75
Total Means of Finance	**105.00**

Source: MIIL and SCPL Estimates

*** - Exchange rate considered is 1 USD = Rs. 89.4013 (exchange rate as on 29thj November 2025)**

Here it should be noted that the cost of acquisition of NPC has been considered at USD 102.00 Million and additional USD 3.00 Million has been considered for upgrade of HSAW Pipe mill, which has current capability to produce pipes of dimeter upto 84 inch. This is essential, as the newer orders from local customers in recent times indicate that the market trend for diameters upto 100 inches to 120 inches is gaining popularity. The upgrade project is expected to be completed within 12 months of acquisition of NPC by Man Group. This is the reason for considering the overall means of finance for USD 105.00 Million.

Promoters Equity

Based on the discussions with the Management of MIIL, it is understood that the Promoters of MIIL will be infusing funds to the tune of Rs. 26.25 Crores from own sources of funds and the balance of the equity (Rs. 209.43 Crores) will be raised from proceeds of sales of land own by the Company near Vashi in Navi Mumbai.

Term Loan

The Consultants understand that the Lenders will be issuing Stand-By Letter of Credit ('SBLC') of USD 78.75 Mn (Rs. 704.04 Crores), which will be utilised by the Company to fund the acquisition. The broad covenants of the term loan have been presented in the table below –

Parameter	Details
Term Loan Amount	USD 78.75 Million
Interest Rate	8.25% Per annum (SOFR at 4.05% + Hedging and Risk Spread of 4.20%)
Door to Door Tenure	7 Years
First Disbursement	January 2026
Date of Acquisition	1st January 2026
Post Acquisition Moratorium	6 Months
Commercial Operations Date	1st January 2026
Repayment Duration	26 Quarters
Installments	26 Equal Quarterly Instalments
First Instalment	3rd Quarter of CY 2026
Last Installment	4th Quarter of CY 2032

Source: SCPL Assumptions

Other Assumptions

The other assumption related to Project, as considered by the Consultants have been presented in the table below –

Description	Unit	Value
Foreign Exchange Rates		
1 US$	Rs.	89.4013
1 SAR	Rs.	23.8123
1 USD	SAR	3.7700
Capital Structure		
Debt-Equity Ratio	Ratio	3.00
Cost of Equity	%	16.00%
Cost of Term Loan	%	8.25%
Cost of Working Capital Loan	%	8.25%
Post-Tax Cost of Capital	%	9.47%
Income Tax		20.00%

Source: SCPL Estimates

Key Financial Parameters

Note – The proposal is to purchase the equity shares of NPC through MISICL, which will act as the holding company in the KSA. Hence the term loan will not reflect in the books of NPC post-acquisition, albeit the same will reflect in the books of the Holding Company. Hence the Consultants have considered dividend equivalent to the Interest + Principal amount issued by NPC, so as to pay the debt obligation of the Holding Company. The projected financial statements have been prepared considering the same, however the key financial parameters/ ratios have been estimated with the view the debt is on books of the Company, so as to provide the actual debt repayment capacity of the Project.

The key financial parameters of the project have been presented in the table below –

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Revenue	219.27	249.83	268.84	287.84	301.67	312.69	323.41
Total Operating Costs	179.87	206.21	221.90	236.47	246.92	255.21	263.27
EBDITA	39.40	43.63	46.94	51.38	54.75	57.48	60.14
EBDITA Margin	17.97%	17.46%	17.46%	17.85%	18.15%	18.38%	18.60%
Contribution	46.53	51.75	55.06	59.50	62.87	65.60	68.26
Contribution Margin	21.22%	20.71%	20.48%	20.67%	20.84%	20.98%	21.11%
BEP Sales	77.78	84.86	72.27	71.61	71.02	70.55	70.13
BEP Capacity Utilisation	35.47%	33.97%	26.88%	24.88%	23.54%	22.56%	21.68%
Cash Break Even	37.92	44.03	44.53	44.12	43.76	43.47	43.21
Cash Break Even Margin	17.29%	17.62%	16.56%	15.33%	14.50%	13.90%	13.36%
Net Profit	30.03	34.17	33.31	35.76	38.46	40.64	42.77
Net Profit Margin	13.69%	13.68%	12.39%	12.42%	12.75%	13.00%	13.22%
Equity Share Capital	53.27	53.27	53.27	53.27	53.27	53.27	53.27
Statutory Reserves	26.64	26.64	26.64	26.64	26.64	26.64	26.64
Reserves and Surplus	75.22	91.41	107.79	127.67	151.31	178.17	208.21
Tangible Net Worth (TNW)	155.12	171.32	187.70	207.58	231.21	258.07	288.11
Term Loan	72.69	60.58	48.46	36.35	24.23	12.12	-
Debt Equity Ratio	0.47	0.35	0.26	0.18	0.10	0.05	-
Total Outside Liability (TOL)	7.11	14.83	15.53	16.15	16.56	16.90	17.24
TOL/ TNW	0.05	0.09	0.08	0.08	0.07	0.07	0.06
Closing Cash Balance	31.13	45.69	69.91	85.59	100.32	125.02	149.24
DSCR	3.71	2.80	2.69	2.97	3.32	3.68	3.99
Minimum DSCR	2.69						
Average DSCR	3.25						
NPV	121.88						
IRR	49.99%						
Cost of Capital	9.19%						

Source: SCPL Analysis

Sensitivity Analysis

The Consultants had undertaken sensitivity analysis to assess the impact of various scenarios on the financial parameters and the result of the same is presented in the table below –

Description	**NPV**	**IRR**	**WACC**	**Min. DSCR**	**Avg DSCR**
	USD Mn	**%**	**%**	**Ratio**	**Ratio**
Base Case	121.88	49.99%	9.19%	2.69	3.25
5% decrease in capacity utilisation	111.54	46.80%	9.21%	2.62	3.10
10% decrease in capacity utilisation	101.26	43.64%	9.23%	2.50	2.94
15% decrease in capacity utilisation	106.28	44.09%	9.22%	2.58	3.02
1.50% decrease in selling price	69.90	31.35%	9.32%	2.09	2.50
2.50% decrease in selling price	16.89	14.62%	9.62%	1.38	1.74

Description	NPV	IRR	WACC	Min. DSCR	Avg DSCR
	USD Mn	%	%	Ratio	Ratio
5% decrease in selling price	-46.68	-3.55%	10.19%	0.67	0.82
1.50% increase in raw material cost	81.90	35.14%	9.29%	2.25	2.67
2.50% increase in raw material cost	41.46	21.77%	9.45%	1.70	2.09
5% increase in raw material cost	-1.18	9.45%	9.78%	1.16	1.50
1% increase in interest rates	118.72	50.91%	9.82%	2.64	3.20
2% increase in interest rates	115.71	51.85%	10.45%	2.59	3.15

Source: SCPL Estimates

It is noted from the exhibit above, that the Project will be susceptible to 5% decrease in selling price and 5% increase in raw material cost. However, it is also noted that the Project remains viable under all other adverse scenarios as NPV remains positive and IRR, remains above WACC under other all scenarios.

The average DSCR of the Project is 3.25, while the minimum DSCR of the Project 2.69 indicating the Project has fair repayment capacity.

Conclusions

Based on the study undertaken by the Consultants, the following is concluded about the proposed acquisition of MIIL –

- The manufacturing unit of NPC is located on 40-hectares, freehold land owned by the Company at Dhahran at the outskirts of city of Dammam in KSA.
- As per the confirmation provided by the Management/ Executives of NPC, the present installed capacity of NPC stands at –
  - 250,000 Tons of HSAW Pipe facility based on 2 step technology.
  - 180,000 Tons of LSAW Pipe facility based on 3 RB technology.
- The plant and machinery have been supplied and installed by renowned technology provider of the segment like (Nippon Steel Welding & Engineering, Sumitomo Heavy Industries, SMS Germany (PWS), Mitsubishi Heavy Industries, and Haeusler AG, Etc.)
- The Raw material will be procured from the open market from renowned supplier like (Nippon Steel Company, POSCO (Korea), Japan Future Enterprise, and NISCO – China, Etc.)
- The power requirement will be fulfilled by a permanent incoming supply of 17.5 KV issued from Saudi Electricity Company (SECO).
- Water requirement is fulfilled through 2 Nos. of Borewell available within the premise of company. Also, there is a RO water plant within the premises of the NPC factory.
- The company is operation and has received requisite approvals and clearances.
- The overall acquisition cost has been estimated at SAR 382,500,000 or USD 102.00 Million. In addition, the Group after acquisition of NPC will immediately undertake upgrade of HSAW Pipe mill, with an expected cost of USD 3.00 Million. Hence the overall project cost is estimated USD 105.00 Million.
- The proposed project is proposed to be funded in a Debt-Equity ratio of 3:1. The Promoters Contribution for the Project will be USD 26.25 Million, while the loan will be USD 78.75 Million.
- The Net Present Value (NPV) of the Project has been estimated at USD 121.88 Million, while Internal Rate of Return at 49.99% is higher than Post Tax Cost of Capita at 9.19%, indicating the Project is financially viable.

- The average DSCR of the Project has been estimated at 3.25, indicating fair capability of the Project in repaying the long-term debts.
- The peak manpower required for the proposed project is ascertained at 567 personnel for peak capacity utilisation.
- Based on the market assessment, technological assessment, SWOT and risk analysis and the financial evaluation as undertaken by the Consultants, the proposal of acquisition of NPC by MIIL is found to be techno-economically viable preposition.

Annexure 1 – Financial Statements of the Proposed Acquisition

Profit and Loss Account

All Figures in USD Million											
Description	Audited			Est	Projected						
	31-Dec-22	31-Dec-23	31-Dec-24	31-Dec-25	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Sales Realisation	34.48	29.32	119.90	186.77	218.75	249.31	268.32	287.32	301.15	312.17	322.89
Other Income	0.64	0.87	0.81	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
Total Revenue	**35.12**	**30.20**	**120.71**	**187.29**	**219.27**	**249.83**	**268.84**	**287.84**	**301.67**	**312.69**	**323.41**
Variable Cost											
Raw Material	32.24	21.73	80.09	156.55	170.47	184.40	198.32	212.25	221.85	229.73	237.60
Power Cost	0.35	0.46	0.74	1.74	1.89	2.05	2.20	2.36	2.47	2.56	2.65
Material Handling and Transport				0.45	0.49	0.53	0.57	0.61	0.63	0.66	0.68
Repairs and Maintenance	0.57	0.91	1.61	1.41	1.54	1.67	1.79	1.92	2.01	2.08	2.15
Other Direct Expenses	0.42	0.26	0.55	2.27	2.44	2.62	2.79	2.97	3.02	3.02	3.02
Manpower Cost	8.04	8.21	10.05	7.73	7.73	8.92	10.06	10.06	10.06	10.06	10.06
Total Variable Cost	**41.63**	**31.57**	**93.04**	**170.14**	**184.57**	**200.17**	**215.74**	**230.16**	**240.04**	**248.10**	**256.16**
Opening Balance - WIP	-	-	-	-	0.32	0.49	0.53	0.57	0.61	0.63	0.65
Sub-Total	**41.63**	**31.57**	**93.04**	**170.14**	**184.89**	**200.66**	**216.27**	**230.73**	**240.65**	**248.74**	**256.82**
Closing Balance - WIP	-	-	-	0.45	0.49	0.53	0.57	0.61	0.63	0.65	0.68
Opening Balance - FG	-	-	-	-	11.98	23.63	25.68	27.60	29.38	30.60	31.59
Sub-Total	**41.63**	**31.57**	**93.04**	**169.70**	**196.38**	**223.77**	**241.38**	**257.72**	**269.39**	**278.68**	**287.73**
Closing Balance - FG	-	-	-	21.73	23.63	25.68	27.60	29.38	30.60	31.59	32.58
Cost of Production	**41.63**	**31.57**	**93.04**	**147.97**	**172.74**	**198.09**	**213.78**	**228.35**	**238.80**	**247.09**	**255.15**
Fixed Costs											

All Figures in USD Million											
Description	Audited			Est	Projected						
	31-Dec-22	31-Dec-23	31-Dec-24	31-Dec-25	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Administrative Expenses	3.69	4.49	5.08	4.68	5.48	6.25	6.25	6.25	6.25	6.25	6.25
Selling and Distribution	1.14	1.09	0.96	1.40	1.64	1.87	1.87	1.87	1.87	1.87	1.87
Total Fixed Costs	**4.83**	**5.58**	**6.04**	**6.09**	**7.13**	**8.12**	**8.12**	**8.12**	**8.12**	**8.12**	**8.12**
Total Operating Expenses	**46.46**	**37.15**	**99.08**	**154.06**	**179.87**	**206.21**	**221.90**	**236.47**	**246.92**	**255.21**	**263.27**
EBITDA	**-11.35**	**-6.95**	**21.63**	**33.24**	**39.40**	**43.63**	**46.94**	**51.38**	**54.75**	**57.48**	**60.14**
EBITDA Margin	**-32.31%**	**-23.02%**	**17.92%**	**17.75%**	**17.97%**	**17.46%**	**17.46%**	**17.85%**	**18.15%**	**18.38%**	**18.60%**
Other Expenses											
Depreciation	8.76	8.60	8.35	8.46	8.46	8.46	5.68	5.68	5.68	5.68	5.68
Interest on Term Loan	-	-	-								
Interest on Working Capital Loan	0.05	0.56	2.53	2.58	0.92	1.00	1.00	1.00	1.00	1.00	1.00
Write Offs	1.92	1.64	-	-	-	-	-	-	-	-	-
Non-Operating Expenses (Loss)	- 0.26	-	- 0.13	-	-	-	-	-	-	-	-
Total Other Expenses	**10.47**	**10.80**	**10.74**	**11.03**	**9.38**	**9.46**	**6.68**	**6.68**	**6.68**	**6.68**	**6.68**
Total Expenditure	**56.93**	**47.95**	**109.82**	**165.09**	**189.25**	**215.66**	**228.58**	**243.15**	**253.60**	**261.89**	**269.95**
Profit Before Tax	**- 21.82**	**- 17.76**	**10.88**	**22.20**	**30.03**	**34.17**	**40.26**	**44.69**	**48.07**	**50.80**	**53.46**
Applicable Tax and Adjustments	- 0.49	- 0.35	2.10	-	-	-	6.95	8.94	9.61	10.16	10.69
Profit After Tax	**- 22.31**	**- 18.11**	**8.78**	**22.20**	**30.03**	**34.17**	**33.31**	**35.76**	**38.46**	**40.64**	**42.77**
PAT Margin	-63.53%	-59.96%	7.28%	11.86%	13.69%	13.68%	12.39%	12.42%	12.75%	13.00%	13.22%
Dividend				-	12.79	17.98	16.93	15.88	14.83	13.78	12.73
Retained Earning	**- 22.31**	**- 18.11**	**8.78**	**22.20**	**17.23**	**16.19**	**16.38**	**19.88**	**23.63**	**26.86**	**30.04**

Balance Sheet

Description	All Figures in USD Million										
	Audited			Est	Projected						
	31-Dec-22	31-Dec-23	31-Dec-24	31-Dec-25	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Liabilities											
Shareholders Funds											
Equity Capital	53.27	53.27	53.27	53.27	53.27	53.27	53.27	53.27	53.27	53.27	53.27
Statutory Reserves	26.64	26.64	26.64	26.64	26.64	26.64	26.64	26.64	26.64	26.64	26.64
Reserves and Surplus	45.10	27.00	35.78	57.99	75.22	91.41	107.79	127.67	151.31	178.17	208.21
Total Shareholder Funds	**125.01**	**106.90**	**115.69**	**137.89**	**155.12**	**171.32**	**187.70**	**207.58**	**231.21**	**258.07**	**288.11**
Term Liabilities											
Term Loan	-	-	-	-	-	-	-	-	-	-	-
Unsecured Loan	-	-	-	-	-	-	-	-	-	-	-
Employee Terminal Benefits	5.77	5.95	5.89	5.95	5.95	5.95	5.95	5.95	5.95	5.95	5.95
Deferred Tax Liability (Net)	-	-	-	-	-	-	-	-	-	-	-
Total Term Liabilities	**5.77**	**5.95**	**5.89**	**5.95**	**5.95**	**5.95**	**5.95**	**5.95**	**5.95**	**5.95**	**5.95**
Current Liabilities											
Trade Payables	2.54	0.85	-	-	-	7.58	8.15	8.72	9.12	9.44	9.76
Other Current Liabilities	-	51.36	11.63	1.12	1.16	1.30	1.43	1.47	1.50	1.51	1.53
Working Capital Loan	-	25.40	-	-	-	-	-	-	-	-	-
Provision for Zakat	0.59	0.50	1.91	-	-	-	-	-	-	-	-
Total Current Liabilities	**3.13**	**78.11**	**13.53**	**1.12**	**1.16**	**8.87**	**9.58**	**10.19**	**10.61**	**10.95**	**11.29**
Total Liabilities	**133.91**	**190.96**	**135.12**	**144.96**	**162.23**	**186.15**	**203.23**	**223.73**	**247.77**	**274.98**	**305.35**
Assets											
Gross Block	264.87	264.62	265.09	266.06	269.06	269.06	269.56	274.56	275.06	280.06	280.56
Cumulative Depreciation	197.80	206.26	214.72	223.18	231.63	240.09	245.77	251.45	257.14	262.82	268.50
Net Block	**67.07**	**58.36**	**50.37**	**42.89**	**37.43**	**28.97**	**23.79**	**23.11**	**17.93**	**17.25**	**12.07**
Current Assets											

All Figures in USD Million											
Description	Audited			Est	Projected						
	31-Dec-22	31-Dec-23	31-Dec-24	31-Dec-25	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Inventories	21.77	72.37	61.47	47.91	52.14	56.52	60.77	64.87	67.70	70.01	72.32
Debtors/ Receivables	25.73	36.01	7.07	23.03	26.97	30.74	33.08	35.42	37.13	38.49	39.81
Other Current Assets	-	-	-	-	-	-	-	-	-	-	-
Cash and Cash Equivalent	19.34	24.21	16.20	31.13	45.69	69.91	85.59	100.32	125.02	149.24	181.16
Total Current Assets	**66.84**	**132.60**	**84.74**	**102.07**	**124.80**	**157.17**	**179.44**	**200.61**	**229.85**	**257.73**	**293.29**
Total Assets	**133.91**	**190.96**	**135.12**	**144.96**	**162.23**	**186.14**	**203.23**	**223.73**	**247.77**	**274.98**	**305.35**

Cash Flow Statement

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Sources of Funds							
Profit After Tax	30.03	34.17	33.31	35.76	38.46	40.64	42.77
Depreciation	8.46	8.46	5.68	5.68	5.68	5.68	5.68
Expenses Written Off							
Increase in Equity	-	-	-	-	-	-	-
Increase in Statutory Reserves	-	-	-	-	-	-	-
Increase in Term Loan	-	-	-	-	-	-	-
Increase in Unsecured Loan	-	-	-	-	-	-	-
Increase in Employee Terminal Funds	-	-	-	-	-	-	-
Increase in Deferred Tax Liability (Net)	-	-	-	-	-	-	-
Increase in Trade Creditors	-	7.58	0.57	0.57	0.39	0.32	0.32
Increase in Expense Creditors	0.04	0.14	0.13	0.04	0.02	0.02	0.02
Increase in Working Capital Loan	-	-	-	-	-	-	-
Increase in Provision for Zakat	-	-	-	-	-	-	-
Decrease on Gross Fixed Assets	-	-	-	-	-	-	-
Decrease in Non-Current Assets	-	-	-	-	-	-	-
Decrease in Inventories	-	-	-	-	-	-	-
Decrease in Debtors	-	-	-	-	-	-	-
Decrease in Other Current Assets	-	-	-	-	-	-	-
Total Sources of Funds	**38.53**	**50.34**	**39.70**	**42.05**	**44.56**	**46.66**	**48.79**
Application of Funds							
Dividend Announced	12.79	17.98	16.93	15.88	14.83	13.78	12.73
Decrease in Equity	-	-	-	-	-	-	-
Decrease in Statutory Reserves	-	-	-	-	-	-	-
Decrease in Term Loan	-	-	-	-	-	-	-
Decrease in Unsecured Loan	-	-	-	-	-	-	-
Decrease in Employee Terminal Reserves	-	-	-	-	-	-	-
Decrease in Deferred Tax Liability (Net)	-	-	-	-	-	-	-
Decrease in Trade Creditors	-	-	-	-	-	-	-
Decrease in Expense Creditors	-	-	-	-	-	-	-
Decrease in Working Capital Loan	-	-	-	-	-	-	-
Decrease in Provision for Zakat	-	-	-	-	-	-	-
Increase in Gross Fixed Assets	3.00	-	0.50	5.00	0.50	5.00	0.50
Increase in Non-Current Assets	-	-	-	-	-	-	-
Increase in Inventories	4.23	4.38	4.25	4.11	2.82	2.31	2.31
Increase in Debtors	3.94	3.77	2.34	2.34	1.71	1.36	1.32
Increase in Other Current Assets	-	-	-	-	-	-	-
Total Application of Funds	**23.97**	**26.12**	**24.02**	**27.33**	**19.86**	**22.45**	**16.86**

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Net Cashflow from Operations	**14.56**	**24.22**	**15.68**	**14.72**	**24.70**	**24.22**	**31.93**
Opening Balance	31.13	45.69	69.91	85.59	100.32	125.02	149.24
Closing Balance	**45.69**	**69.91**	**85.59**	**100.32**	**125.02**	**149.24**	**181.16**

Depreciation Schedule

Depreciation Schedule – The Company Act - SLM

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Opening Gross Block	**266.06**	**269.06**	**269.06**	**269.56**	**274.56**	**275.06**	**280.06**
Freehold Land	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	46.21	46.21	46.21	46.21	46.21	46.21	46.21
Motor Vehicles, Furniture and Road	7.30	10.30	10.30	10.80	10.80	11.30	11.30
Plant and Machinery	211.56	211.56	211.56	211.56	216.56	216.56	221.56
Addition (Deletion)	**3.00**	**-**	**0.50**	**5.00**	**0.50**	**5.00**	**0.50**
Freehold Land	-	-	-	-	-	-	-
Building and Civil Works	-	-	-	-	-	-	-
Motor Vehicles, Furniture and Road	3.00	-	0.50	-	0.50	-	0.50
Plant and Machinery	-	-	-	5.00	-	5.00	-
Closing Gross Block	**269.06**	**269.06**	**269.56**	**274.56**	**275.06**	**280.06**	**280.56**
Freehold Land	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	46.21	46.21	46.21	46.21	46.21	46.21	46.21
Motor Vehicles, Furniture and Road	10.30	10.30	10.80	10.80	11.30	11.30	11.80
Plant and Machinery	211.56	211.56	211.56	216.56	216.56	221.56	221.56
Depreciation	**8.46**	**8.46**	**5.68**	**5.68**	**5.68**	**5.68**	**5.68**
Freehold Land	-	-	-	-	-	-	-
Building and Civil Works	0.51	0.51	0.51	0.51	0.51	0.51	0.51
Motor Vehicles, Furniture and Road	0.17	0.17	0.17	0.17	0.17	0.17	0.17
Plant and Machinery	7.78	7.78	5.00	5.00	5.00	5.00	5.00
Cumulative Depreciation	**231.63**	**240.09**	**245.77**	**251.45**	**257.14**	**262.82**	**268.50**
Freehold Land	-	-	-	-	-	-	-
Building and Civil Works	38.96	39.48	39.99	40.51	41.02	41.54	42.05
Motor Vehicles, Furniture and Road	6.95	7.11	7.28	7.45	7.62	7.78	7.95
Plant and Machinery	185.72	193.50	198.50	203.50	208.50	213.50	218.50
Net Fixed Assets	**37.43**	**28.97**	**23.79**	**23.11**	**17.93**	**17.25**	**12.07**
Freehold Land	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Building and Civil Works	7.24	6.73	6.21	5.70	5.18	4.67	4.15
Motor Vehicles, Furniture and Road	3.35	3.19	3.52	3.35	3.69	3.52	3.85
Plant and Machinery	25.84	18.06	13.06	13.06	8.06	8.06	3.06

Term Loan Repayment Schedule

All Figures in USD Million								
Description	**31-Dec-25**	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Interest Rate	8.25%	8.25%	8.25%	8.25%	8.25%	8.25%	8.25%	8.25%
Repayment Schedule								
Annual Summary								
Opening Balance	-	-	72.69	60.58	48.46	36.35	24.23	12.12
Addition during the Period	-	78.75	-	-	-	-	-	-
Repayment during the Period	-	6.06	12.12	12.12	12.12	12.12	12.12	12.12
Closing Balance	-	72.69	60.58	48.46	36.35	24.23	12.12	-
Interest for the Period	-	6.41	5.58	4.58	3.58	2.58	1.58	0.58
January								
Opening Balance		-	72.69	60.58	48.46	36.35	24.23	12.12
Addition during the Period		78.75						
Repayment during the Period								
Closing Balance	-	78.75	72.69	60.58	48.46	36.35	24.23	12.12
Interest for the Period	-	0.54	0.50	0.42	0.33	0.25	0.17	0.08
February								
Opening Balance	-	78.75	72.69	60.58	48.46	36.35	24.23	12.12
Addition during the Period								
Repayment during the Period								
Closing Balance	-	78.75	72.69	60.58	48.46	36.35	24.23	12.12
Interest for the Period	-	0.54	0.50	0.42	0.33	0.25	0.17	0.08
March								
Opening Balance	-	78.75	72.69	60.58	48.46	36.35	24.23	12.12
Addition during the Period			-					
Repayment during the Period			3.03	3.03	3.03	3.03	3.03	3.03
Closing Balance	-	78.75	69.66	57.55	45.43	33.32	21.20	9.09
Interest for the Period	-	0.54	0.49	0.41	0.32	0.24	0.16	0.07
April								
Opening Balance	-	78.75	69.66	57.55	45.43	33.32	21.20	9.09
Addition during the Period								
Repayment during the Period		-						
Closing Balance	-	78.75	69.66	57.55	45.43	33.32	21.20	9.09
Interest for the Period	-	0.54	0.48	0.40	0.31	0.23	0.15	0.06
May								
Opening Balance	-	78.75	69.66	57.55	45.43	33.32	21.20	9.09
Addition during the Period								
Repayment during the Period								

All Figures in USD Million								
Description	31-Dec-25	31-Dec-26	31-Dec-27	31-Dec-28	31-Dec-29	31-Dec-30	31-Dec-31	31-Dec-32
Closing Balance	-	78.75	69.66	57.55	45.43	33.32	21.20	9.09
Interest for the Period	-	0.54	0.48	0.40	0.31	0.23	0.15	0.06
June								
Opening Balance	-	78.75	69.66	57.55	45.43	33.32	21.20	9.09
Addition during the Period								
Repayment during the Period			3.03	3.03	3.03	3.03	3.03	3.03
Closing Balance	-	78.75	66.63	54.52	42.40	30.29	18.17	6.06
Interest for the Period	-	0.54	0.47	0.39	0.30	0.22	0.14	0.05
July								
Opening Balance	-	78.75	66.63	54.52	42.40	30.29	18.17	6.06
Addition during the Period		-						
Repayment during the Period								
Closing Balance	-	78.75	66.63	54.52	42.40	30.29	18.17	6.06
Interest for the Period	-	0.54	0.46	0.37	0.29	0.21	0.12	0.04
August								
Opening Balance	-	78.75	66.63	54.52	42.40	30.29	18.17	6.06
Addition during the Period		-						
Repayment during the Period								
Closing Balance	-	78.75	66.63	54.52	42.40	30.29	18.17	6.06
Interest for the Period	-	0.54	0.46	0.37	0.29	0.21	0.12	0.04
September								
Opening Balance	-	78.75	66.63	54.52	42.40	30.29	18.17	6.06
Addition during the Period		-						
Repayment during the Period		3.03	3.03	3.03	3.03	3.03	3.03	3.03
Closing Balance	-	75.72	63.61	51.49	39.38	27.26	15.14	3.03
Interest for the Period	-	0.53	0.45	0.36	0.28	0.20	0.11	0.03
October								
Opening Balance	-	75.72	63.61	51.49	39.38	27.26	15.14	3.03
Addition during the Period								
Repayment during the Period								
Closing Balance	-	75.72	63.61	51.49	39.38	27.26	15.14	3.03
Interest for the Period	-	0.52	0.44	0.35	0.27	0.19	0.10	0.02
November								
Opening Balance	-	75.72	63.61	51.49	39.38	27.26	15.14	3.03
Addition during the Period	-							
Repayment during the Period								

All Figures in USD Million								
Description	**31-Dec-25**	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Closing Balance	-	75.72	63.61	51.49	39.38	27.26	15.14	3.03
Interest for the Period	-	0.52	0.44	0.35	0.27	0.19	0.10	0.02
December								
Opening Balance	-	75.72	63.61	51.49	39.38	27.26	15.14	3.03
Addition during the Period	-							
Repayment during the Period		3.03	3.03	3.03	3.03	3.03	3.03	3.03
Closing Balance	-	72.69	60.58	48.46	36.35	24.23	12.12	-
Interest for the Period	-	0.51	0.43	0.34	0.26	0.18	0.09	0.01

Dividend Amount (Interest Part)	-	6.41	5.58	4.58	3.58	2.58	1.58	0.58
Withholding Tax at UAE	-	0.32	0.28	0.23	0.18	0.13	0.08	0.03
Dividend Amount (Principal Part)	-	6.06	12.12	12.12	12.12	12.12	12.12	12.12
Total Dividend	-	12.79	17.98	16.93	15.88	14.83	13.78	12.73

Financial Ratios

NPV and IRR

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Initial Cash Outflow							
Capital Investment	-105.00	-	-0.50	-5.00	-0.50	-5.00	-0.50
Margin Money	-19.49	-0.43	-5.88	-5.84	-4.11	-3.33	-3.29
Total Initial Cash Outflow	**-124.49**	**-0.43**	**-6.38**	**-10.84**	**-4.61**	**-8.33**	**-3.79**
Operating Cash Flow							
PAT	30.03	34.17	33.31	35.76	38.46	40.64	42.77
Depreciation	8.46	8.46	5.68	5.68	5.68	5.68	5.68
Expenses Written Off	-	-	-	-	-	-	-
Interest Coverage	6.58	4.62	3.67	2.87	2.07	1.27	0.80
Total Operating Cash Flow	**45.07**	**47.25**	**42.65**	**44.30**	**46.21**	**47.59**	**49.25**
Terminal Cash Flow							
Salvage Value							12.07
Working Capital							42.37
Total Terminal Cash Flow	**-**	**-**	**-**	**-**	**-**	**-**	**54.43**
Net Cashflow	**-79.42**	**46.82**	**36.27**	**33.47**	**41.60**	**39.26**	**99.89**
NPV	121.88						
IRR	49.99%						
Post-tax Cost of Capital	9.19%						

WACC

Loans	Amount	ROI	Tax Rate	Post tax cost of capital (%)	Proportion	WACC
Equity	26.25	16.00%	0.00%	16.00%	25.00%	4.00%
Unsecured Loan from Promoters	-	8.25%	16.11%	6.92%	0.00%	0.00%
Term Loan New	78.75	8.25%	16.11%	6.92%	75.00%	5.19%
Total	**105.00**					**9.19%**

DSCR

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
PAT	30.03	34.17	33.31	35.76	38.46	40.64	42.77
Depreciation	8.46	8.46	5.68	5.68	5.68	5.68	5.68
Expenses Written Off	-	-	-	-	-	-	-
Interest on Term Loan	5.90	4.86	3.81	2.76	1.71	0.66	0.03
Total	**44.39**	**47.49**	**42.80**	**44.20**	**45.85**	**46.98**	**48.48**
Interest on Term Loan	5.90	4.86	3.81	2.76	1.71	0.66	0.03
Repayment for the Period	6.06	12.12	12.12	12.12	12.12	12.12	12.12
Total	**11.96**	**16.98**	**15.93**	**14.88**	**13.83**	**12.78**	**12.14**
DSCR	3.71	2.80	2.69	2.97	3.32	3.68	3.99
Min DSCR	2.69						
Average DSCR	**3.25**						

FACR

All Figures in USD Million							
Description	**31-Dec-26**	**31-Dec-27**	**31-Dec-28**	**31-Dec-29**	**31-Dec-30**	**31-Dec-31**	**31-Dec-32**
Net Fixed Assets (excl Capex)	37.43	28.97	23.79	23.11	17.93	17.25	12.07
Term Loan Outstanding	72.69	60.58	48.46	36.35	24.23	12.12	-
Margin	**194.20%**	**209.07%**	**203.69%**	**157.27%**	**135.15%**	**70.25%**	**0.00%**
Fixed Asset Coverage Ratio (A-B)/C	**0.51**	**0.48**	**0.49**	**0.64**	**0.74**	**1.42**	**-**

Annexure 2 – List of Plant and Machinery

Helical Pipe Mill (HPM): List of Main Facilities

Machine Name	Manufacturer	Date Built	Date Refurbish	Annual machine
Pipe Mill#1 &#2 (Forming/Inside T-Cross/Tack Welding)	SUMITOMO HEAVY INDUSTRIES	1980	2015	250KT
T-Cross Welding	SMS Germany (PWS)	1980	2015	
Inside Pipe Cleaning Machine	SMS Germany (PWS)	2015	-	
Final Welding Machine	SMS Germany (PWS)	2015	-	
X-Ray Machine (Film Type)	YXLON/PHILIPS	1980	-	
X-Ray Machine (Digital)	YXLON	2016	-	
Hydrostatic Testing Machine	HYPREX	1980	-	
End Facing Machine	Daido Machinery	1980	-	
Online UST	Nippon Steel Technology	2013	-	
Magnetic Particle Testing	Eishin Kagaku Co. Ltd.	2011	-	
Over Head Cranes (HPM)	SUMITOMO HEAVY INDUSTRIES	1980	Crane No1 : 2019	
Over Head Cranes (FWM)	CESAN	2015	-	

Longitudinal Pipe Mill (LPM): List of Main Facilities

Machine Name	Manufacturer	Date Built	Date Refurbish	Annual machine
Edge Milling Machine	Mitsubishi Heavy Industires	2001	-	180KT
Crimping Press Machine	Sumitomo Heavy Industries	2011	-	
Roll Bending Machine	Hauesler AG	2001	-	
Tack Welding Machine	Hauesler AG	2001	-	
Inside SAW	NSTEX Eng/Daiden	2001	-	
Outside SAW	NSTEX Eng/Daiden	2001	-	
Expander Machine	SMS Germany	2001	-	
End Facer Machine	Daido Machinery	2001	-	
Hydrostatic Testing Machine	HYPREX	2001	-	
Digital X-Ray Machine	Yxlon	2001	2018	
Auto Line UST	Nippon Steel Technology	2001	2018	
Over Head Cranes (LPM)	SAUDI CRANES	2000	-	
Gantry Crane	KAMIUCHI ELECTRIC WORK (Gantry Crane)	2010	2018	

General: List of Main Facilities

Machine Name	Manufacturer	Date Built	Date Refurbish
Mechanical Testing Machine a. Tensile and Bend Test (2 units)	i.Tokyo Koki Seizosho ii.Shimadzu	i. 1979 ii. 2012	-
b. Charpy Impact Test (1 unit)	MPM Technology	2010	-
c. DWTT (1 unit)	JT Toshi Inc	2001	-
d. Hardness Test (3 units)	i. ZwickRoell ii. Akashi Seisakusho	i. 2011 ii. 1979	-
e. Micro and Macroscopic Examination (2 units)	i. Olympus + BUEHLER (software) ii. Olympus	i. 1995 ii. 2019	i.2010 (Software & stage only) ii. -
f. Chemical Analyzer (4 units)	i.&ii.ARL Fisons Instrument iii.&iv.Leco	i. 1994 ii. 2015 iii. 2015 iv. 2015	-
Machinery of Test Piece preparation (Lathe)	TAKISAWA IKEGAI DAINICHI DAINICHI	- - 1979 1979	-
Machinery of Test Piece preparation (Milling Machine)	NIIGATA (Japan, x3 machines) JAROCIN (Poland) CHESTER HERCULES (UK) ACRA (Taiwan) LAGUN (Spain)	1970 - 2010 - 2017	-
Water Treatment Facility (Drinking Water Production)	AL-KAWTHER	1980 2000 2011 2012	- 2020 - -
Water Treatment Facility (Waste Water Treatment)	Enviromental Equipment C	2017	-
Yard Crawler Cranes	Crane No 1 : LING BELT Crane No 2 : LING BELT Crane No 3 : SUMITOMO Crane No 4 : HITACHI SUMITOMO Crane No 5 : HITACHI SUMITOMO	1979 1979 2002 2006 2011	-
Pipe Transportation Trailers	MAN	2002 2011 2011	-

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